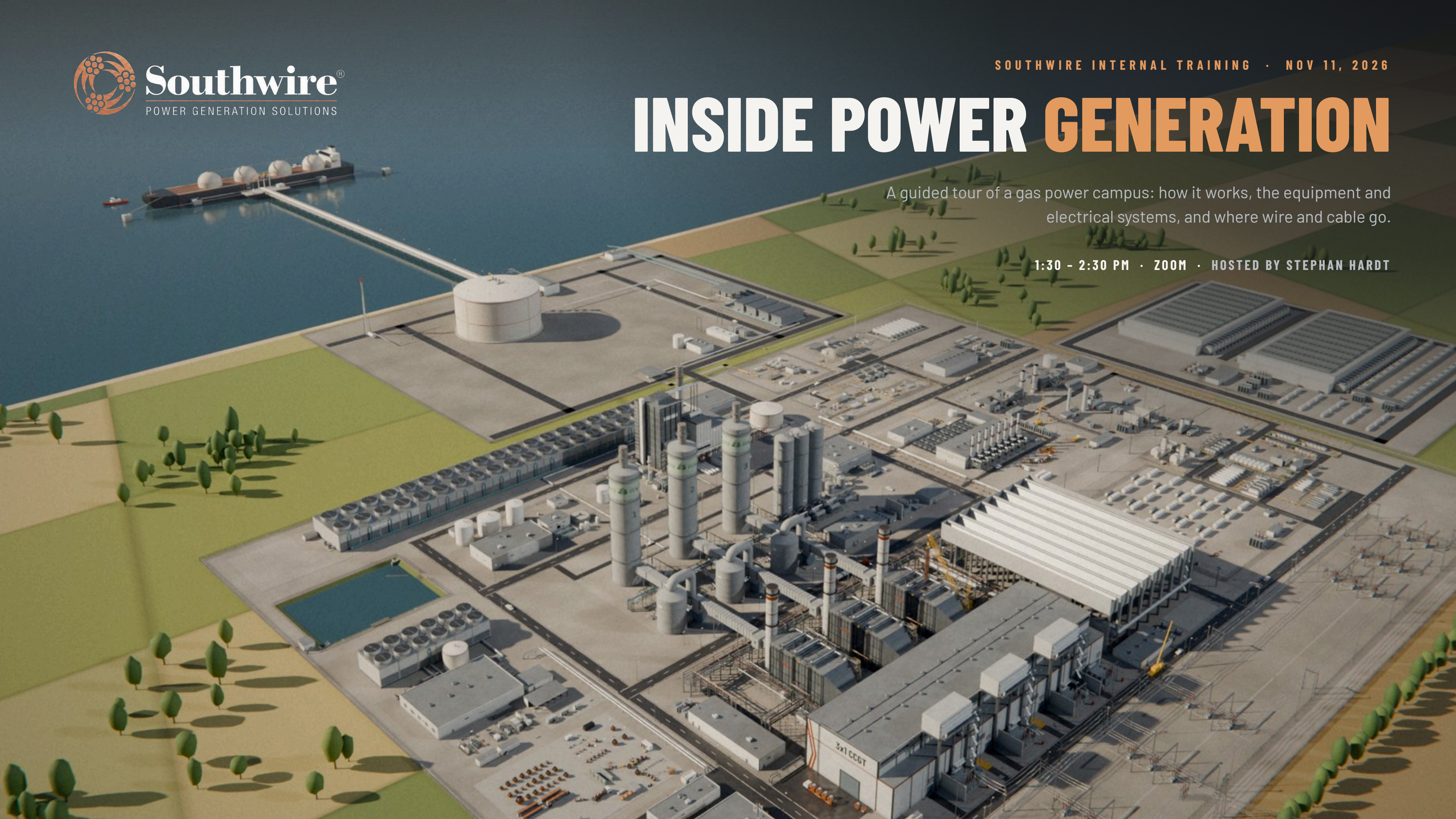
Southwire®
POWER GENERATION SOLUTIONS

SOUTHWIRE INTERNAL TRAINING · NOV 11, 2026

INSIDE POWER GENERATION

A guided tour of a gas power campus: how it works, the equipment and electrical systems, and where wire and cable go.

1:30 – 2:30 PM · ZOOM · HOSTED BY STEPHAN HARDT



TODAY · 60 MINUTES

THE HOUR

1:30	**WELCOME** why this matters	5 MIN
1:35	**MARKET AND FUNDAMENTALS** how power is made, the gas build; the machines, the grid, cable, who buys	25 MIN
2:00	**THE 3D TOUR** seven chapters, live in the model: what it does, its electrical system, its cable	18 MIN
2:18	**WIRE AND CABLE TAKEAWAYS** the whole plant in one number, products, pain points	6 MIN
2:24	**Q&A** questions from the chat	6 MIN

No engineering background needed. At every stop you get three things: **what it does**, its **electrical system**, and **the cable**.

Questions in the chat; we answer them at the end.

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WELCOME · WHERE TODAY'S TOUR STARTS

WELCOME TO THE SW-3X1 GAS POWER CAMPUS

A natural-gas generation campus, built as a 3D model with every cable routed.

~1,690 MW · 107 ACRES · 230 KV · ~$6B CAPEX

stephan.hardt@southwire.com · INSIDE POWER GENERATION 3/89

WHY BUILD A WHOLE PLANT? TO TELL THE CABLE STORY

1 NOTHING LIKE IT EXISTED

NO COMPETITOR MATERIAL, NO PUBLIC REFERENCE

OEM brochures stop at the turbine; EPC drawings are confidential. We found nothing that shows a whole gas plant with its cable, so we built it.

2 A WAY TO TALK WITH CUSTOMERS

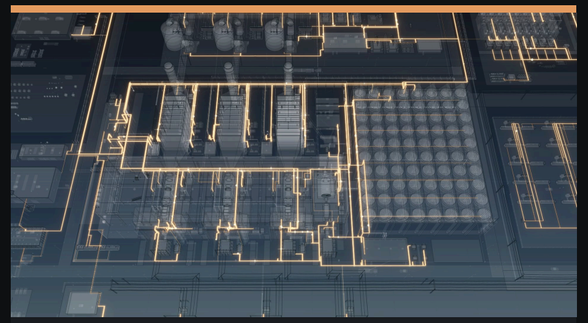
ONE PICTURE EVERYONE CAN POINT AT

“This is your plant, this is where our cable goes.” Sales, an EPC and an owner looking at the same view, in their language.

3 A SEAT AT THE IPP TABLE

EARLY, BEFORE THE EPC BUYS

IPPs and developers plan campuses: power block, batteries, data centre, behind-the-meter. The model matches how they think, so we join the conversation early.



4 CABLE IS THE STORY

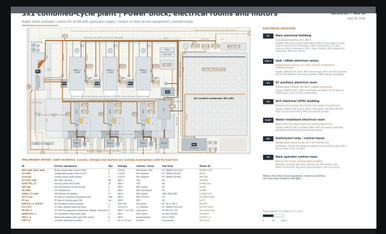
THE PLANT IS THE BACKDROP

Every render, brochure and slide shows our products applied where they go: MV-105 to the motors, TC-ER in the trays, ARMOR-X at the burners.

Not engineered for one customer: built so we can talk about theirs. One model behind the brochures, the decks, the 3D viewer and this training.

CHECKED AGAINST REAL PLANTS, THEN MEASURED

REAL PLANT DRAWINGS



1 RESEARCH

700+

pages of real combined-cycle drawings reviewed: layouts, one-lines, equipment and load lists. They confirmed the equipment, voltages and electrical systems.

2 DRAFT

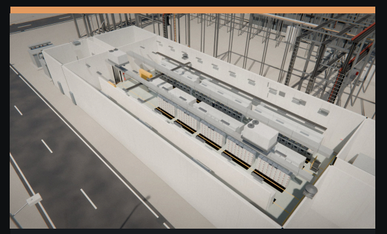
16

sheets: the SK-3X1 drawing set, with the power-block electrical layout and a preliminary motor and load schedule.

3 BUILD

1,177

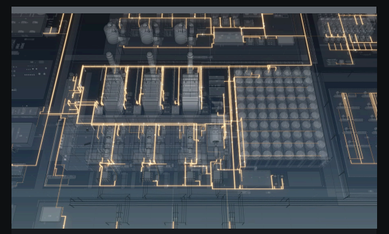
items, 186,362 parts and 1007 cable and pipe routes in the 3D model.



4 CHECK

211/211

footprints match the drawings; 21 heights checked; 1.6 M part pairs tested for clashes: none.



5 MEASURE

22,012

cables in 6,492 runs, each routed on the model's trays and duct banks: about 1,174 miles.

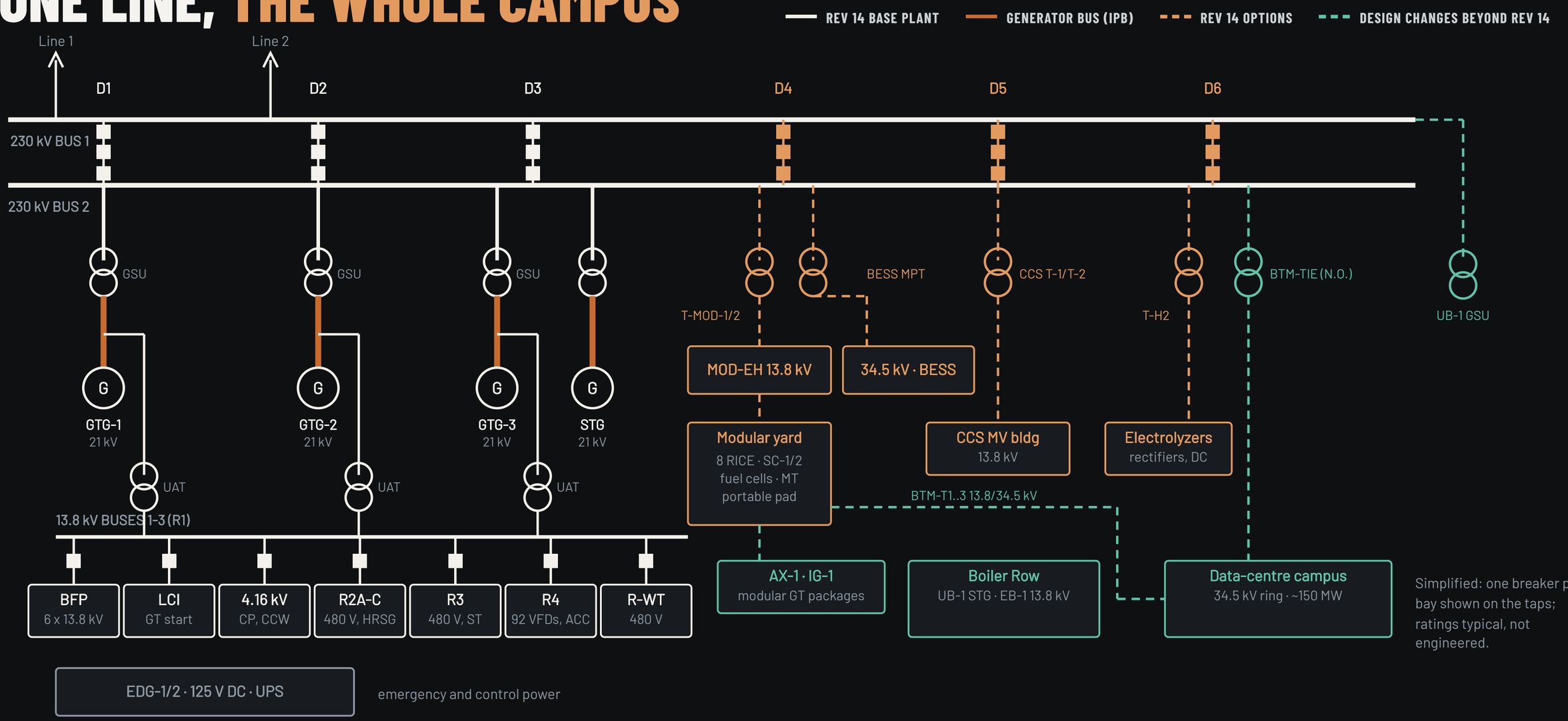
6 SHARE

1

model behind every render, brochure, deck, the 3D viewer, the guided tour and this training.

Conceptual, not engineered, but not invented. Every system in the model is one a real combined-cycle plant has, at the voltages and sizes real plants use; the checks run every time the model changes.

ONE LINE, THE WHOLE CAMPUS



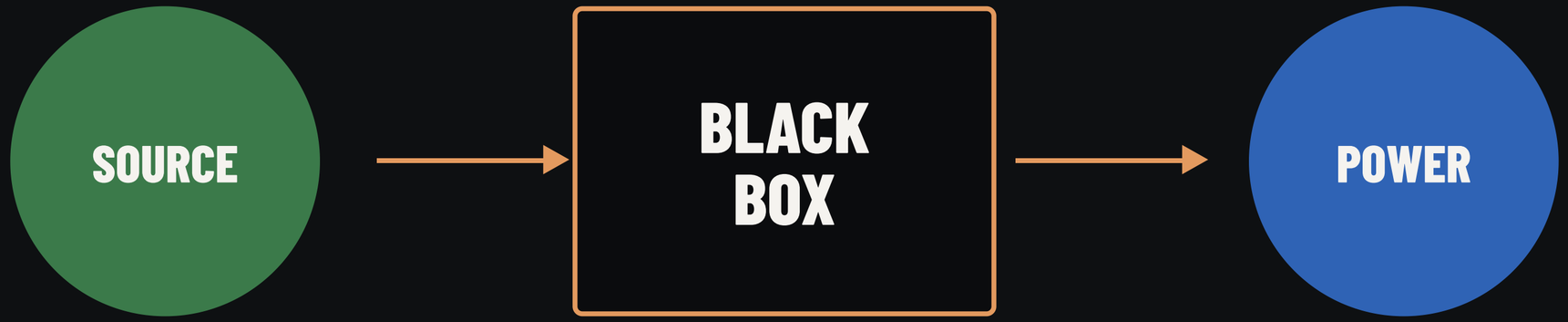
01

HOW POWER IS MADE

Every plant is a black box: what goes in, what comes out, and why we zoom in on gas.

EVERY POWER PLANT IS THIS

Input → conversion → output. The complexity is what's inside the box, and what's around it.



INPUT

The energy that goes in: fuel (gas, coal, uranium, biomass) or nature (sun, wind, water).

CONVERSION

The machine that turns the source into electricity: a turbine and generator, an engine, a solar panel, a fuel cell.

OUTPUT

Electricity out: to the grid through a switchyard, or straight to a load behind the meter.

Every plant, from a gas turbine to a solar farm, fits this picture. **What changes is the source and the machine in the box.** Next: the four families of power generation, then we open the box of a gas plant.

HOW ELECTRICITY IS MADE: FOUR FAMILIES

THERMAL

Heat spins a turbine and a generator.

- **NATURAL GAS** — FIRM
  simple cycle · combined cycle · engines · aero ← today
- **COAL** — FIRM
- **NUCLEAR** — FIRM
- **BIOMASS AND WASTE** — FIRM
- **GEOTHERMAL** — FIRM
- **OIL / DIESEL** — FIRM

WATER AND WIND

Moving water or air spins the turbine.

- **HYDRO** — FIRM*
- **WIND, ONSHORE AND OFFSHORE** — VARIABLE

Not on our campus; Southwire cables them too: hydro, onshore and offshore wind.

DIRECT

No turbine: sunlight or chemistry makes DC.

- **SOLAR PV** — VARIABLE
- **FUEL CELLS** — FIRM

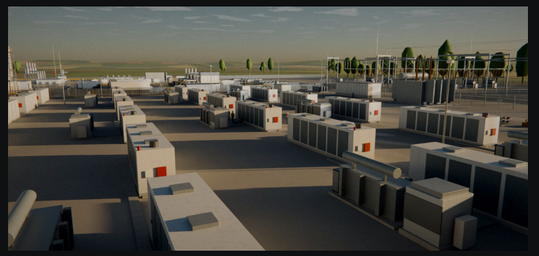
SOLAR CARPORT AND EV CHARGERS

In SW-3X1: the fuel-cell bank in the modular yard.

STORAGE

Shifts power in time; it does not make it.

- **BATTERIES (BESS)** — HOURS
- **PUMPED HYDRO** — HOURS



THE BATTERY YARD

In SW-3X1: the battery (BESS) yard.

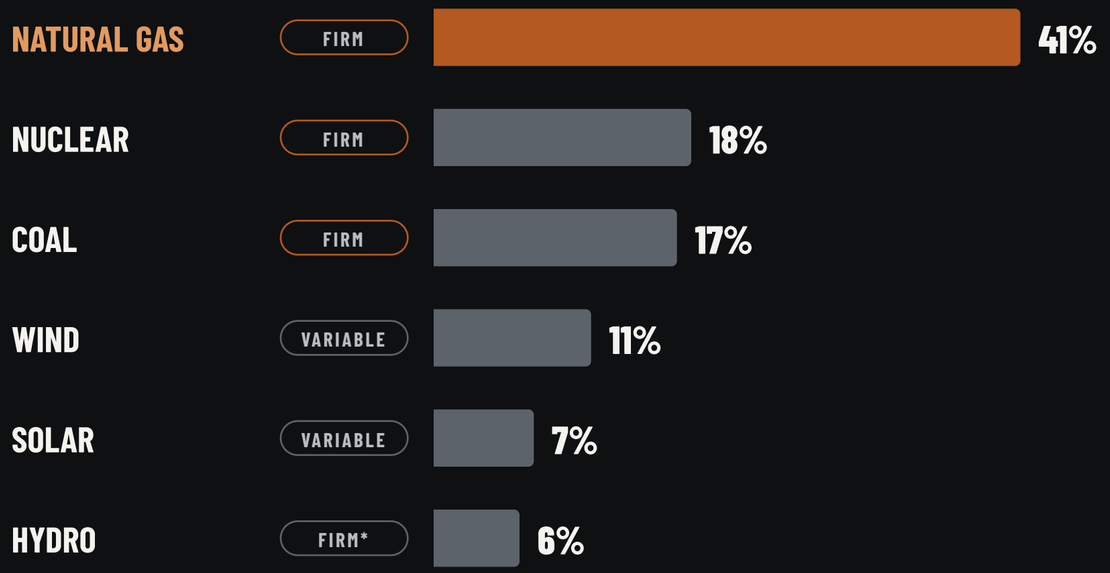
GAS, ENGINES, AERODERIVATIVES AND PACKAGE BOILERS: ALL THERMAL, ALL ON OUR CAMPUS

> **Almost all of them spin a magnet inside a coil: a generator.** Solar PV, fuel cells and batteries make DC and need inverters. Firm sources run on demand; variable ones follow the weather. Every one of them ends in cable.

Firm = can run on demand. *Hydro depends on water and season. Storage runs for hours, then recharges.

THE MIX TODAY, AND WHY WE ZOOM IN ON GAS

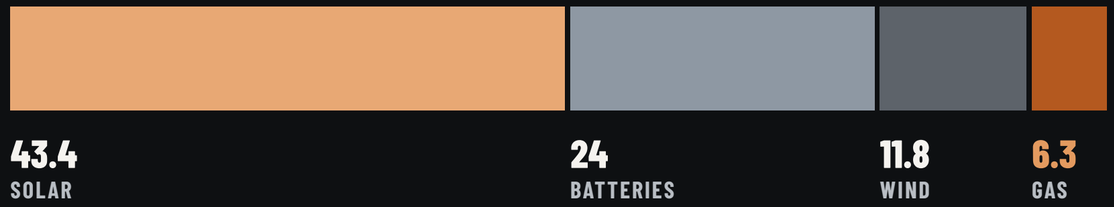
SHARE OF US ELECTRICITY, 2025



Firm = can run on demand. *Hydro depends on water and season. Batteries store power; they do not make it.

NEW CAPACITY PLANNED FOR 2026 · 86 GW

Most of what gets added is solar and batteries. Gas is a small share of the GW, and the one that runs on demand.



FIRM

Gas runs when it is called: day or night, any weather. Wind and solar follow the weather; batteries cover hours, not days.

THE BACKBONE

41% of US electricity today, and the backup that lets more wind and solar onto the grid.

THE NEXT WAVE

Data centres need firm power around the clock: 378 GW of gas is in development, much of it built later this decade.

Southwire cables every one of these: solar, wind, storage, transmission and distribution. **Today we focus on gas**: firm power, a boom-scale build, and some of the most cable-dense plants on the grid.

EIA, Electricity explained (2025 utility-scale generation, rounded) and Today in Energy, Feb 2026 (planned 2026 additions; developers' plans, not all will be built).

02

THE MARKET

A boom-scale gas build, and cable is on the critical path.

A BOOM-SCALE GAS BUILD, AND CABLE IS ON THE CRITICAL PATH

THE MARKET

378 GW

US gas power in development; 52 GW already under construction

THE CABLE

~$1.2B

core wire & cable, 2026–2035, at $8–13M per GW

THE PIPELINE

386

gas projects we track in 40 states, every owner type

THE TRACTION

$50M+

booked in Q3; $30M in September alone

TODAY'S GOAL: KNOW THE PLANT, SO YOU CAN FOLLOW THE CABLE THROUGH IT

Global Energy Monitor, Aug 2026; Southwire PowerGen planning model and internal figures, 2026.

MARKET CONTEXT

LOAD GROWTH IS PULLING GAS POWER FORWARD

166 GW

US peak load growth forecast by 2030

about 90 GW of it from data centres

378 GW

US gas power in development, mid-2026

189 GW tied to data centres

52 GW

under construction, up 76% in H1 2026

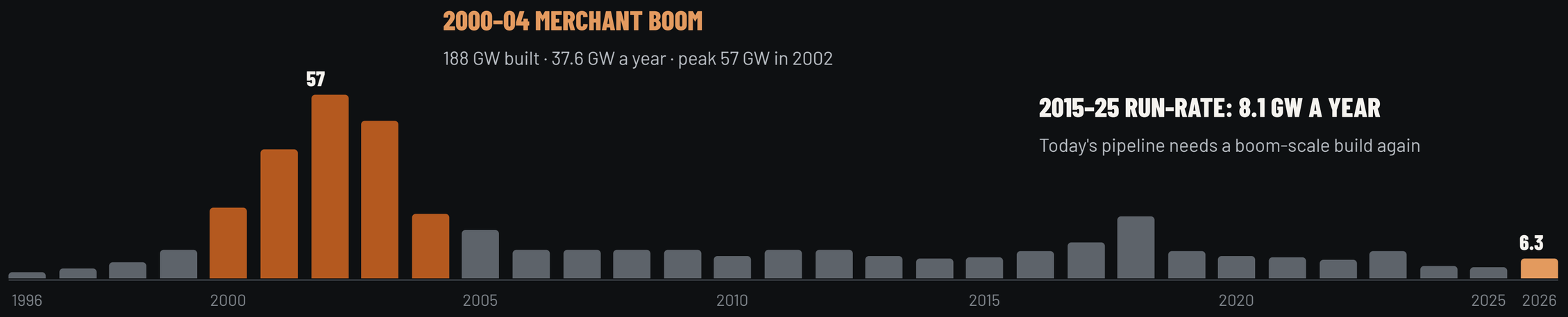
the largest gas build in the world

Up to 7 yrs

quoted lead time for some large gas turbines

procurement decisions are being made now

US GAS CAPACITY ADDED PER YEAR, GW · PROVISIONAL



Grid Strategies, 2025; Global Energy Monitor, Aug 2026; S&P Global, May 2025. History: EIA and Southwire provisional series, to be verified.

SIZING THE OPPORTUNITY · CONSERVATIVE PLANNING CASE

~$1.2B OF CORE GAS CABLE, 2026–2035

CAPACITY

93.7 GW

modeled additions through 2036, from the 378 GW pipeline: 90% of the 52 GW building, 13.5% of the rest

PLANT INVESTMENT

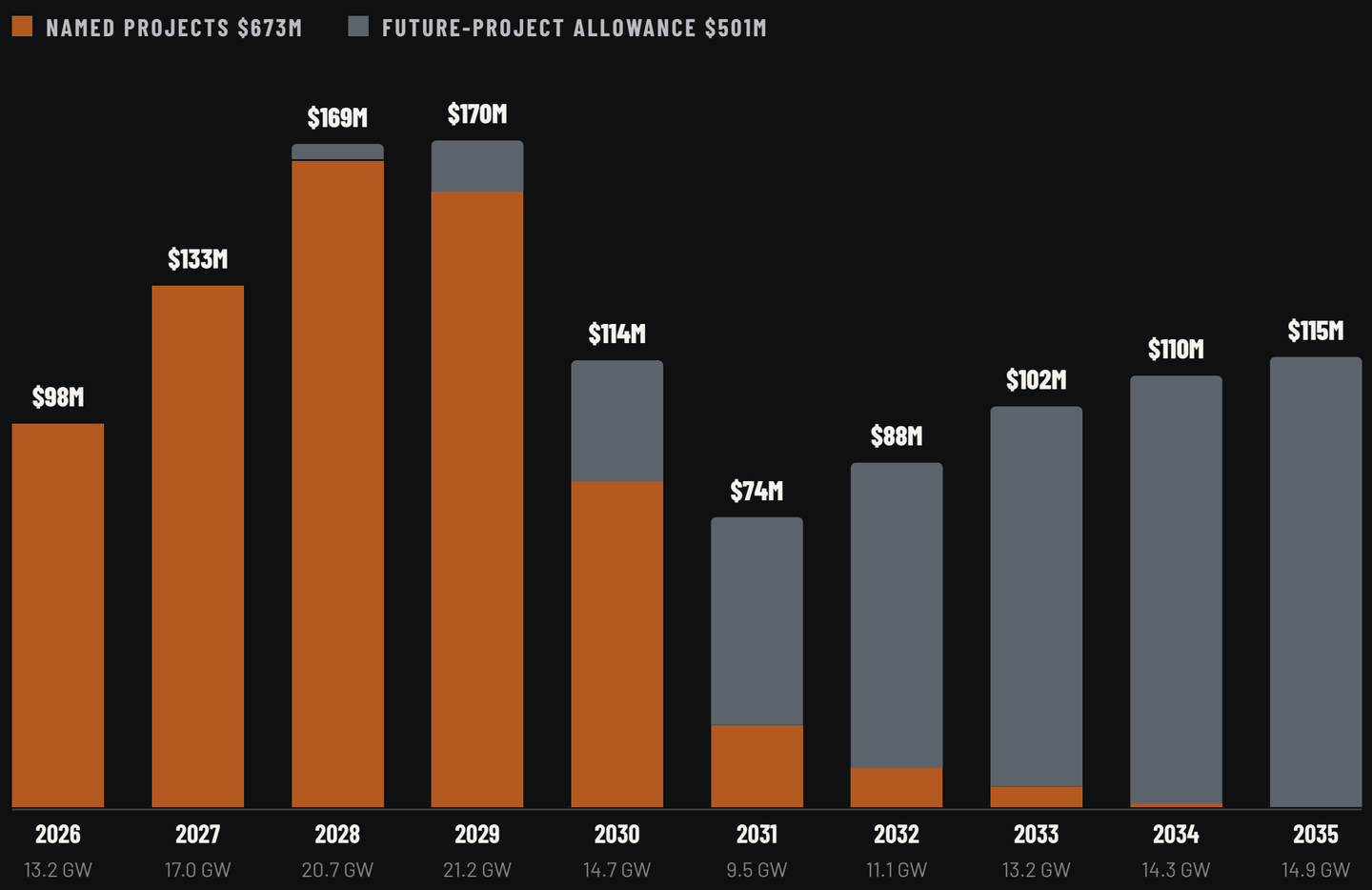
~$180B

93.7 GW × $1,923/kW blended (combined cycle, peakers, engines)

WIRE & CABLE

$0.75–1.2B

$8–13M per GW: under 1% of plant capex, on the critical path to first power



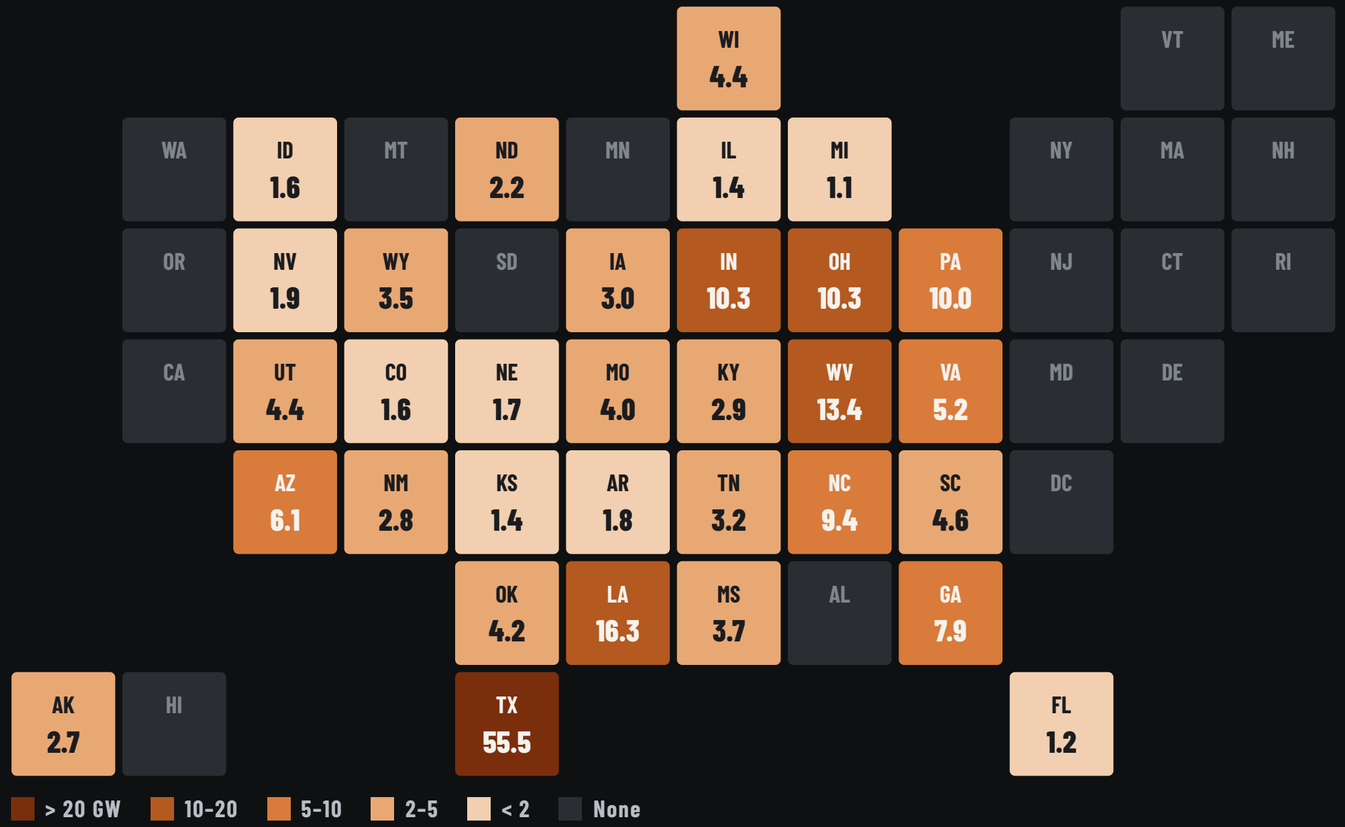
Core cable by purchase-order year; below each year, modeled PO-year GW (not installed GW). GW × $/kW × addressable share × win probability. OEM internal wiring and HV excluded.

Global Energy Monitor, Aug 2026; completion rates, blended cost and cable rates are Southwire planning assumptions.

PROJECT CONCENTRATION

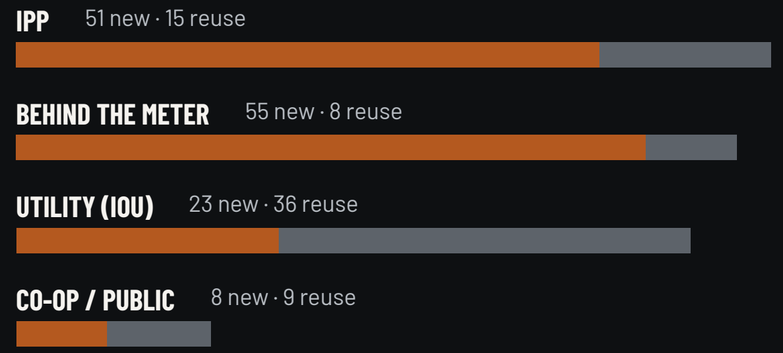
WHERE THE BUILD IS, WHO OWNS IT, WHAT KIND OF SITE

GROSS GW BY STATE · 386 PROJECTS, COD 2026-2035



OWNER × SITE · GROSS GW

GREENFIELD 137 · BROWNFIELD 69



UTILITIES REUSE OLD SITES; IPPS AND BTM BUILD NEW

Greenfield, grid-connected	77.0
Data-centre / BTM campus	54.5
Coal-site repower	41.3
Expansion at an operating plant	21.4
Industrial captive and site reuse	11.6

Southwire PowerGen register, Sep 11, 2026. Gross GW, unweighted. Site classes provisional (Rev01).

03

INSIDE A GAS PLANT

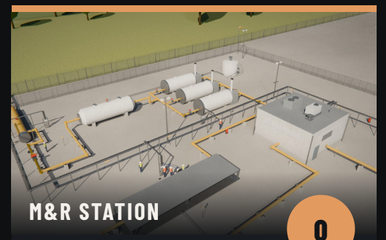
The machines, the grid, and the cable that ties them together.

OPEN THE BOX: THE PHYSICAL ANATOMY

INPUT
fuel / energy source

BLACK BOX
thermal conversion

OUTPUT
dispatchable power, delivered



FUEL INFRASTRUCTURE

- Gas production, pipelines, compression
- Storage and LNG logistics
- Plant-gate fuel delivery

FUEL INPUT

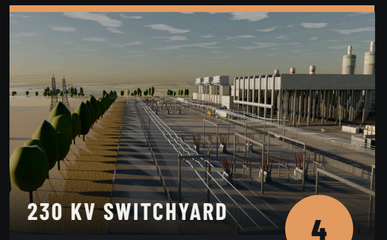
- Natural gas, coal, oil / diesel
- Hydrogen blends
- Biomass where it applies

THERMAL CONVERSION

- Frame and aero gas turbines, engines (RICE)
- Combined cycle: GT + HRSG + ST
- Modular and H2-ready packages

ELECTRICITY OUTPUT

- Dispatchable AC power
- Stable, reliable load
- Baseload, mid-merit or peaking



DELIVERY & FLEXIBILITY

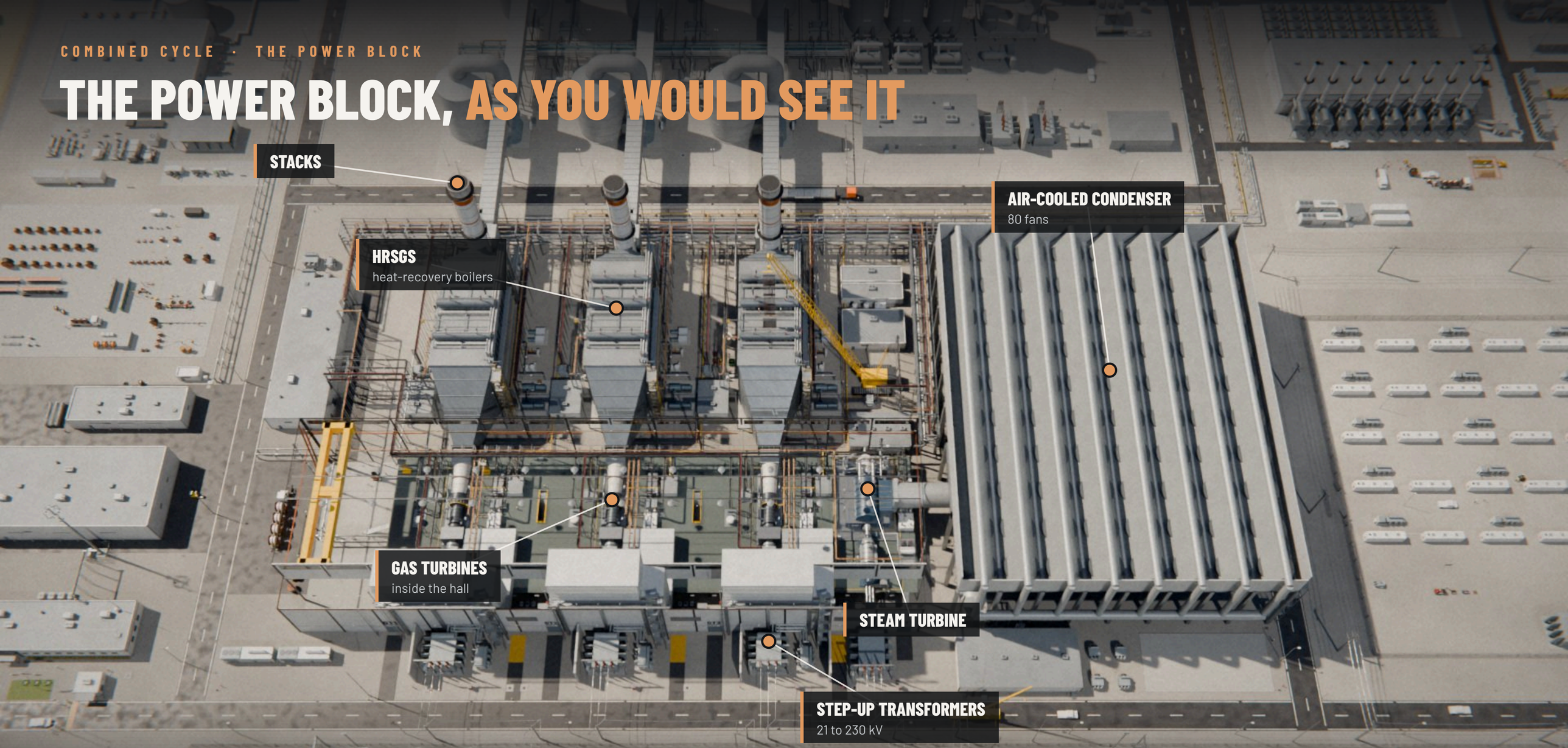
- Transmission and distribution
- MV / LV delivery, behind-the-meter on site
- Storage and flexibility support

END USE / OFF-TAKE

- Utilities, IPPs, industrial loads
- Data centres and other off-takers
- Critical infrastructure

Back to the black box, now for gas. Gas turbine, combined cycle or engines: **the same physics on every project.** What changes is the machine in the box, and who owns it.

THE POWER BLOCK, AS YOU WOULD SEE IT



Steel, concrete and machines: the part everyone photographs. **Next: how it makes power, one stage at a time.**

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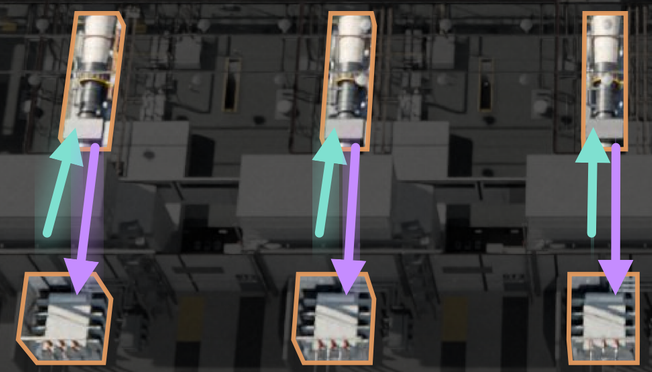
AIR POWER

THE GAS TURBINE: A JET ENGINE ON THE GROUND

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



It burns gas in compressed air; the hot gas spins the turbine and its generator. That makes **about two-thirds** of the plant's power, and leaves exhaust at about 1,100 °F.

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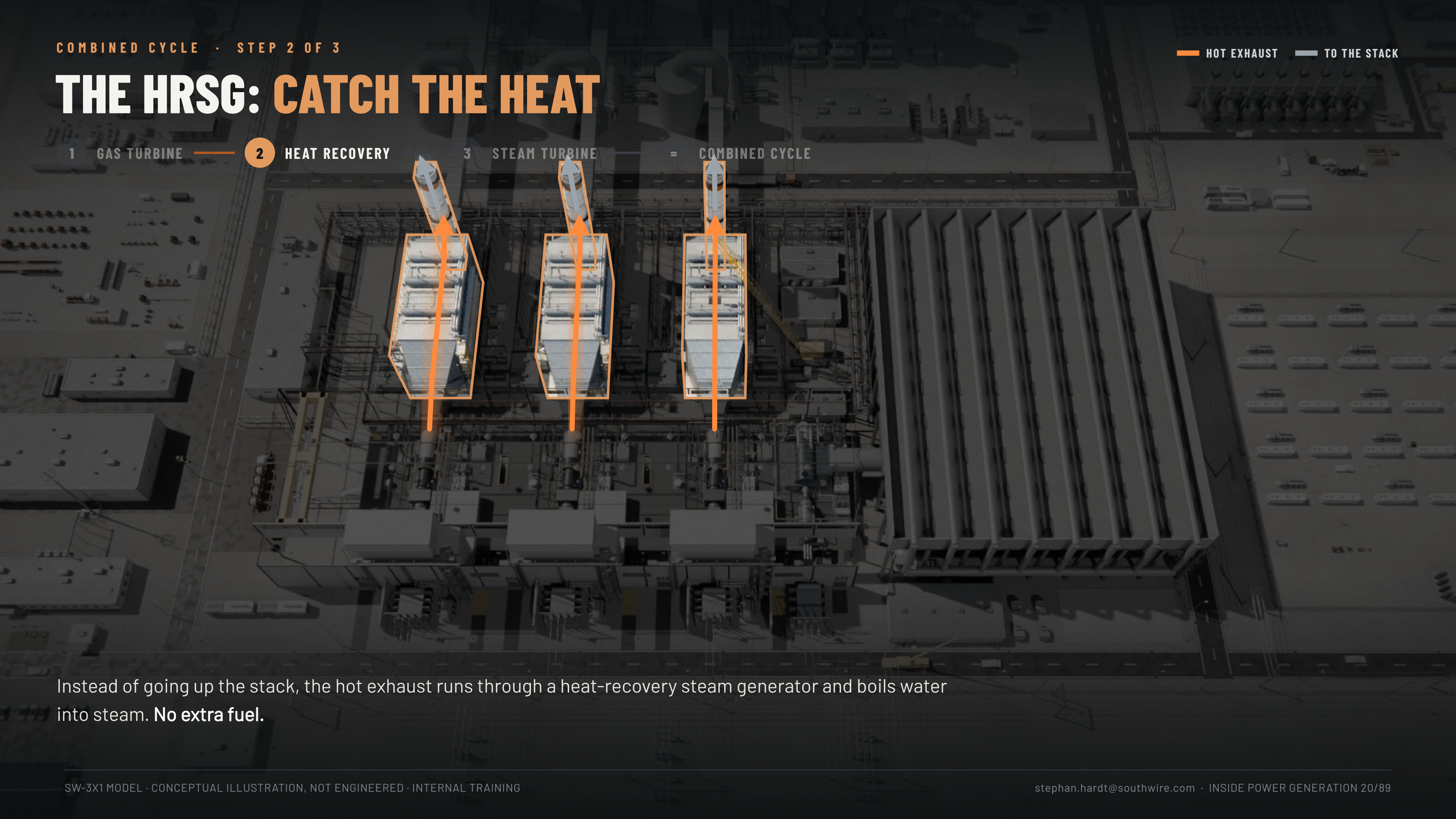
THE HRSG: CATCH THE HEAT

HOT EXHAUST — TO THE STACK

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



Instead of going up the stack, the hot exhaust runs through a heat-recovery steam generator and boils water into steam. **No extra fuel.**

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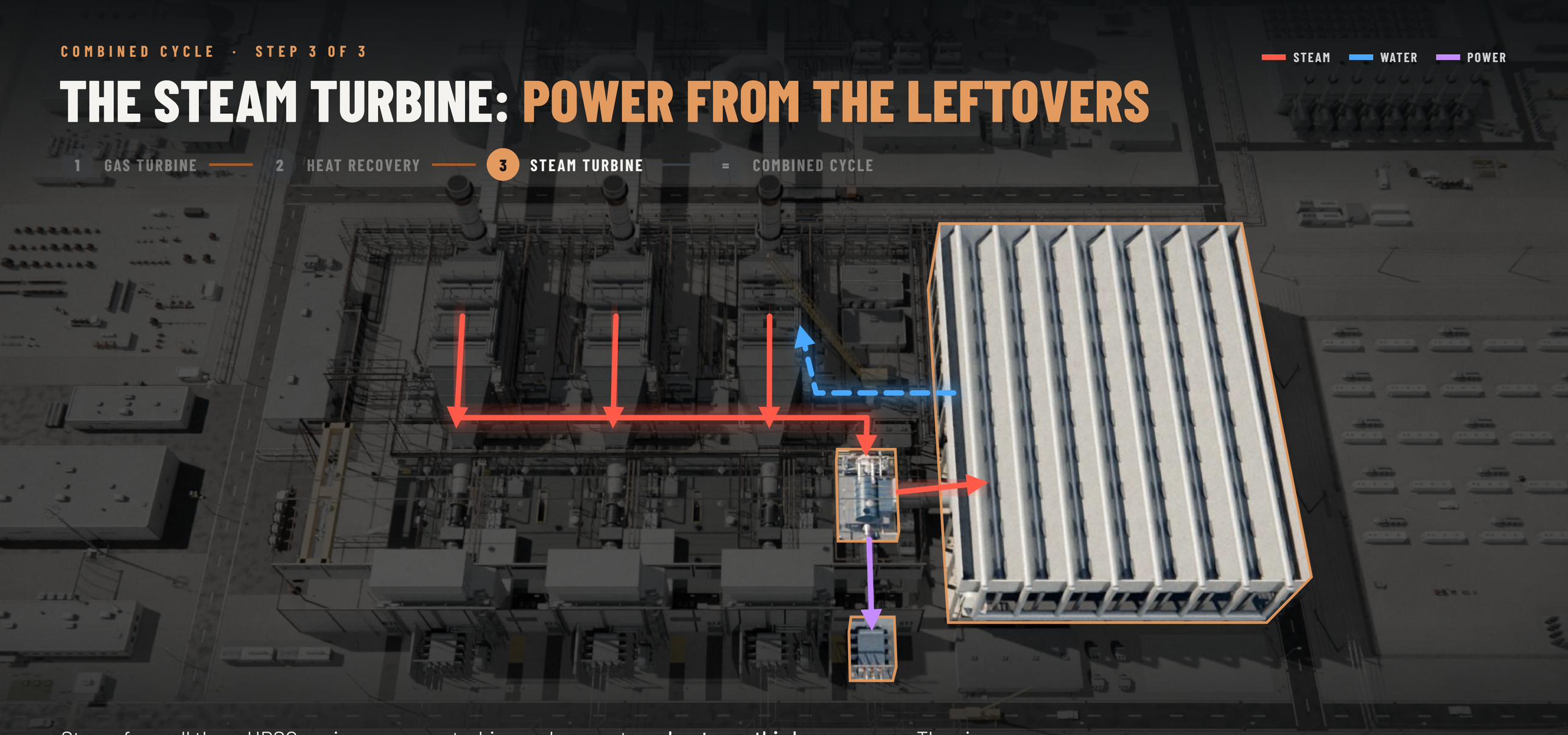
STEAM WATER POWER

THE STEAM TURBINE: POWER FROM THE LEFTOVERS

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



Steam from all three HRSGs spins one more turbine and generator: **about one-third more power.** The air-cooled condenser turns it back into water, and round it goes.

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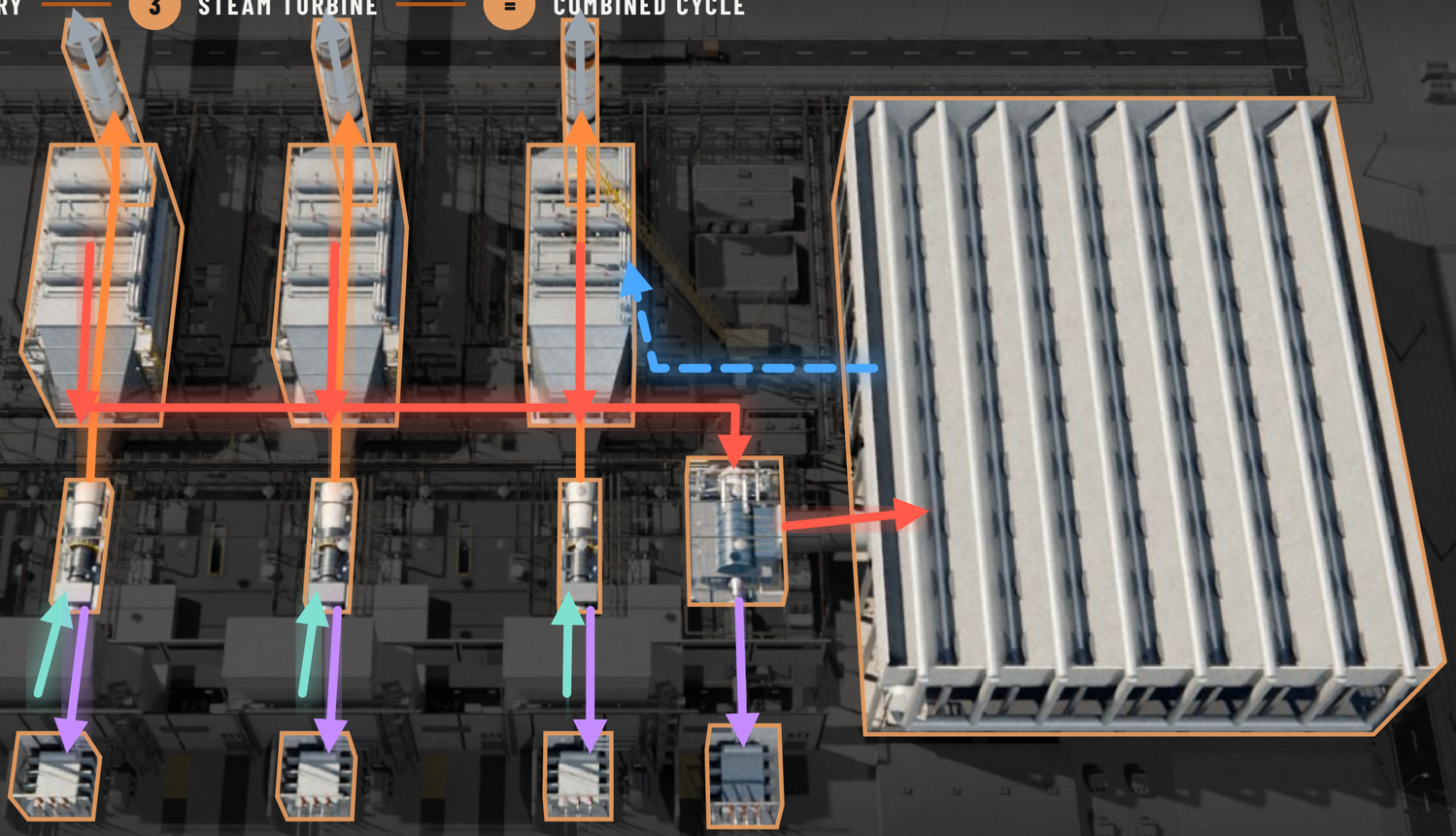
COMBINED CYCLE

BURN THE GAS ONCE, USE THE HEAT TWICE

EXHAUST · STEAM · WATER · POWER

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE — = COMBINED CYCLE



~60%
of the fuel becomes power, vs ~40% for a gas turbine alone

3 + 1
three gas turbines, three HRSGs, one steam turbine: a "3x1"

8,338
cables in this power block and its condenser, in our model

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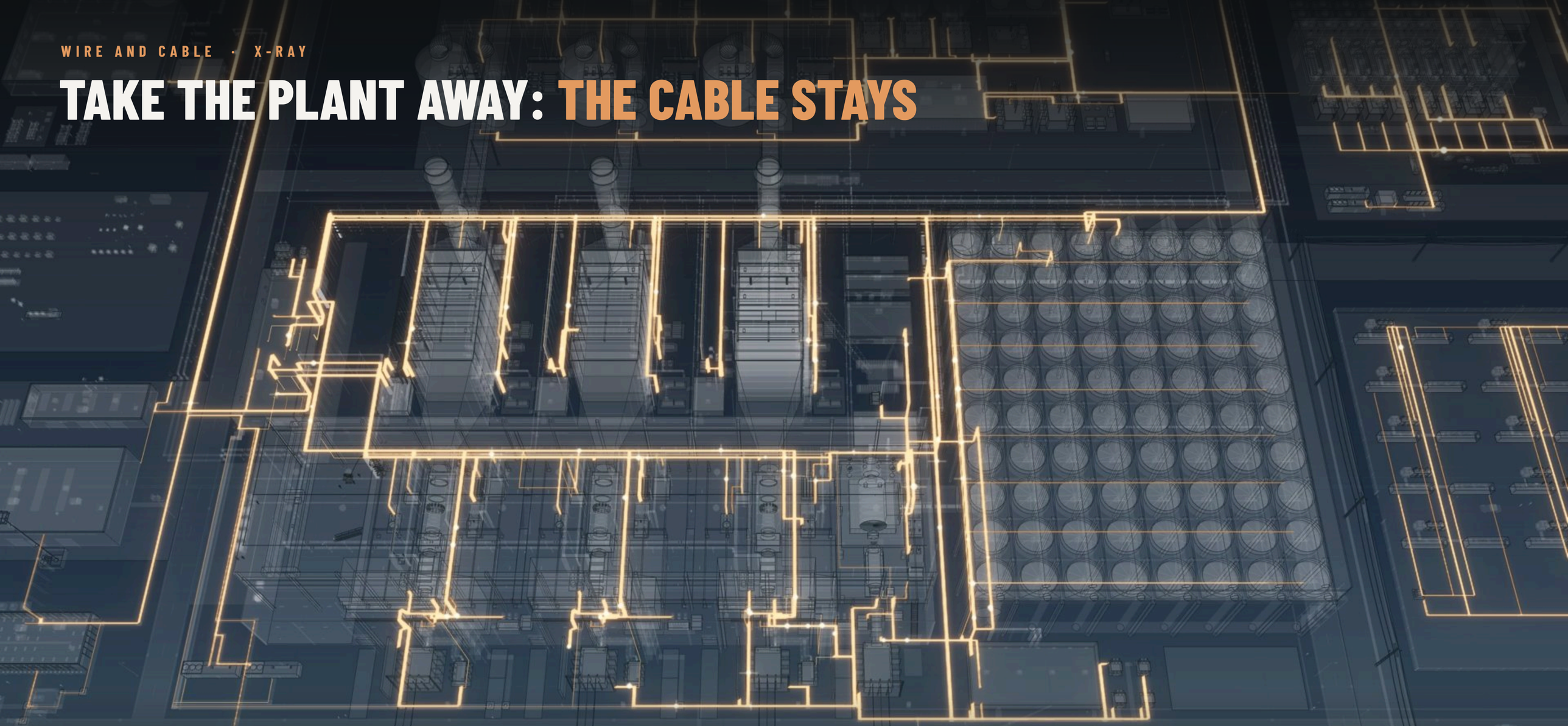
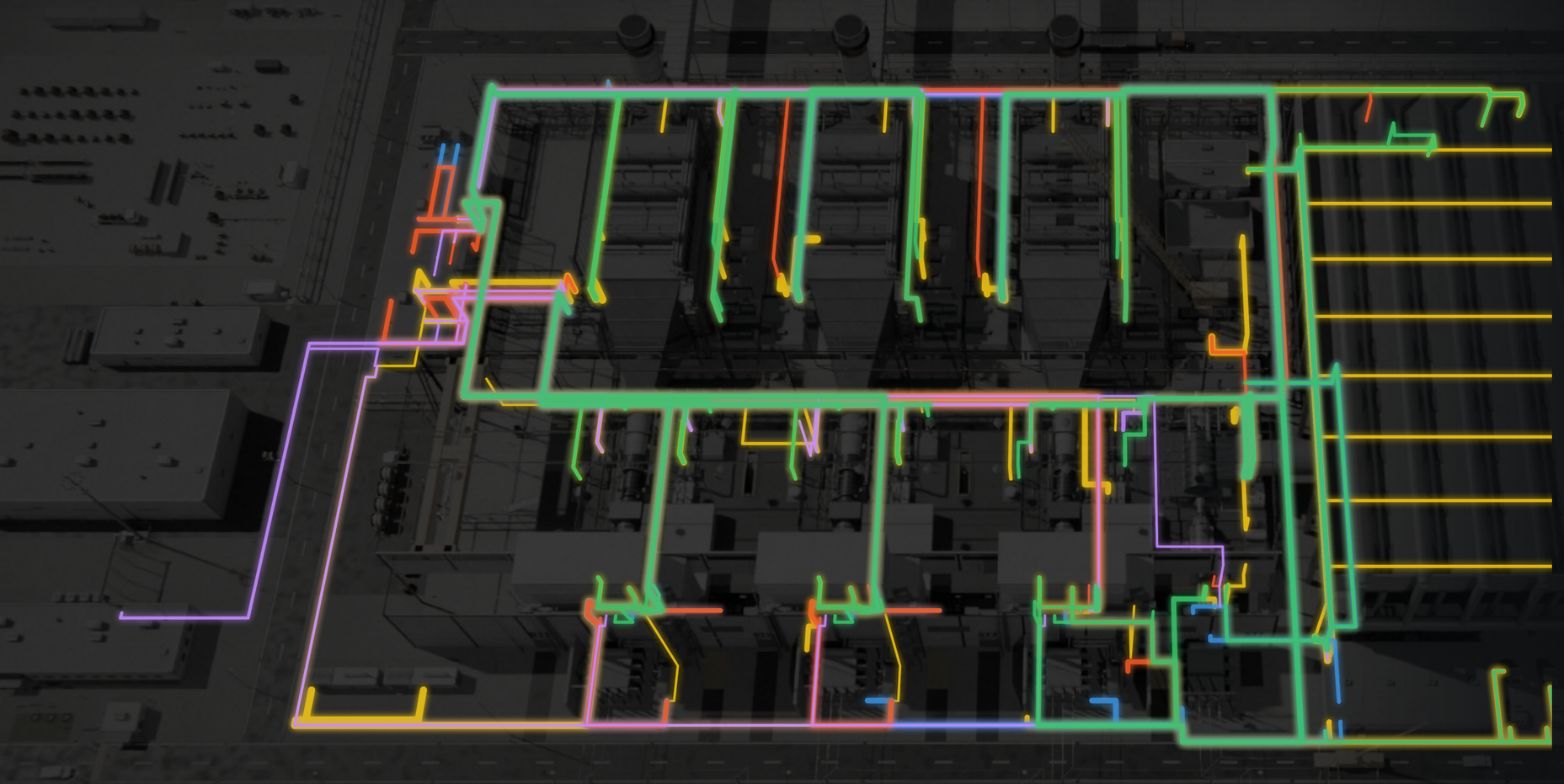
TAKE THE PLANT AWAY: THE CABLE STAYS



The same view, with the steel made glass. **About 8,300 cables and 445 miles** in the power block and its condenser, every one routed in our model.

NOW SORT IT BY JOB: FIVE KINDS OF CABLE



Every line is a route from the model's cable schedule, in trays, conduit and duct banks; thicker lines carry more cables. **The power block is a web of copper.**

THE CABLE IN THIS PICTURE

8,300 cables · ~444 miles

MV / HV POWER — 232 · 28 mi
The big feeders: generators, transformers, the largest motors. MV-105, HV XLPE.

LV POWER — 1,174 · 98 mi
600 V to motors, valves, drives, lighting. TC-ER, ARMOR-X MC-HL, VFD cable.

CONTROL — 1,253 · 94 mi
Starts, stops and trips: breakers, interlocks, MCC starters. Multi-conductor TC-ER.

INSTRUMENTATION — 5,538 · 203 mi
The plant's nerves: pressure, flow, temperature. Shielded pairs, thermocouple extension.

FIBRE AND FIRE ALARM — 103 · 21 mi
The control network, protection and life safety.

The power block and its air-cooled condenser only, from the SW-3X1 model cable schedule (building wire, grounding and overhead not counted). Conceptual estimate, not engineered.

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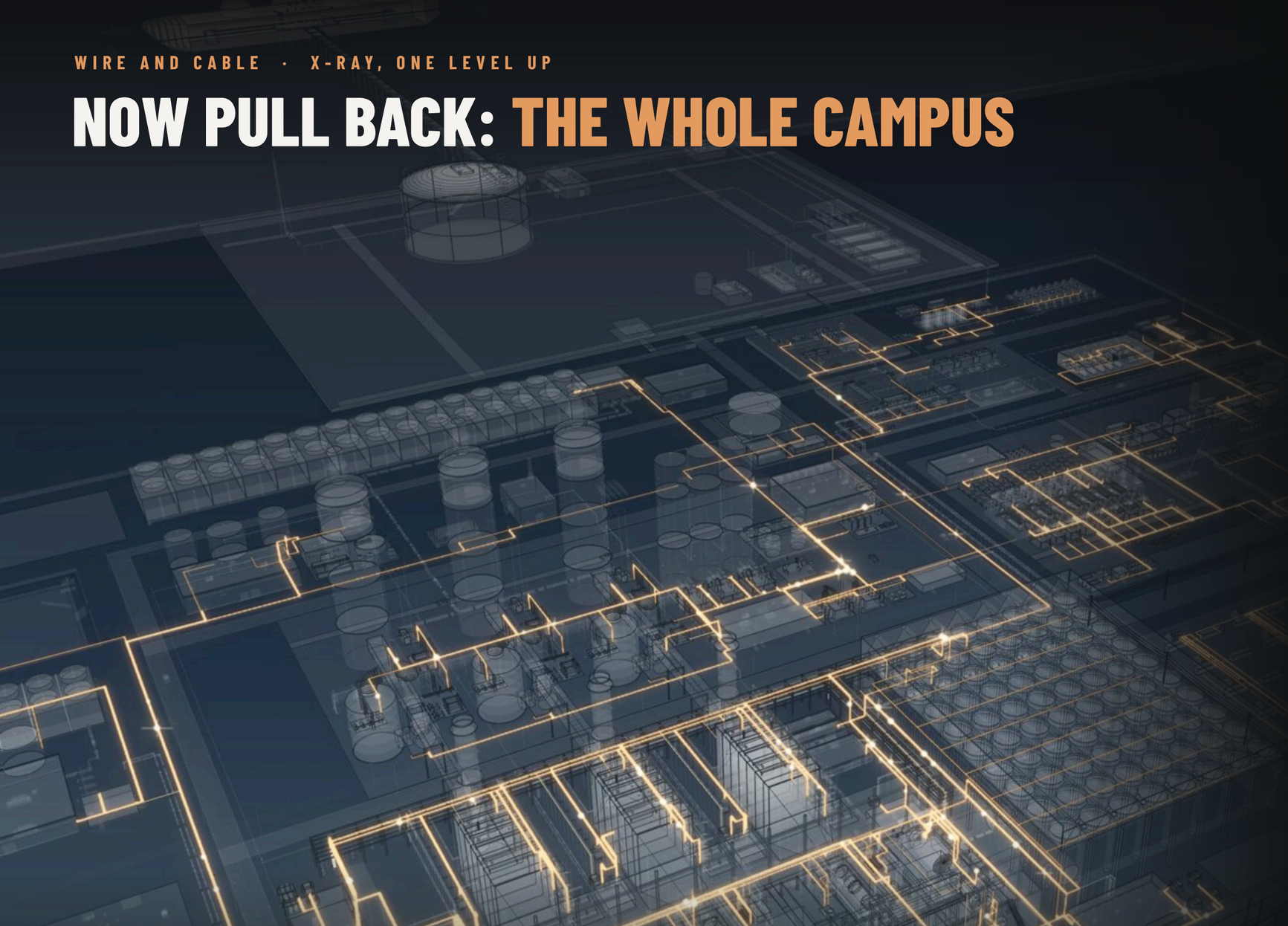
NOW PULL BACK: THE WHOLE CAMPUS

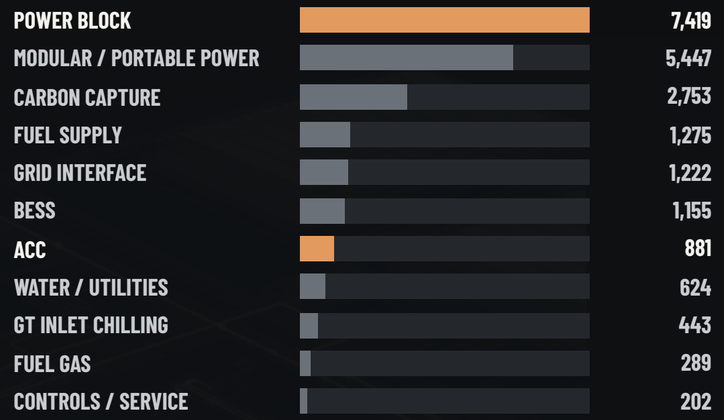


The power block and its condenser were the densest piece, **8,300 cables**. Add the grid, carbon capture, the batteries, the modular yard and the utilities, and **it is 2.6 times as much.**

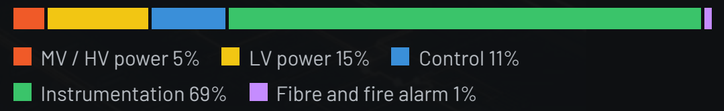
THE CABLE ON THE WHOLE CAMPUS

21,710 cables · ~1,095 miles

BY AREA ■ LAST SLIDE



BY TYPE



From the SW-3X1 model cable schedule (building wire, grounding and overhead not counted). Conceptual estimate, not engineered.

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CATCH THE CO_2 AFTER THE STACK

It bolts onto the combined cycle and **costs it steam and power**: steam leaves the turbine for the strippers, and the fans, pumps and compressors draw MW. A second plant's worth of motors.

1 EXHAUST LEAVES THE HRSGS
A damper diverts it to a booster fan, about 16 MW per train, that pushes it through the capture plant.

2 COOL IT DOWN
A direct-contact cooler, 60 ft across and 90 ft tall, brings the gas to about 40 °C.

3 ABSORB THE CO_2
In the absorber an amine solvent falls through the rising gas and takes up about 90 to 95 % of the CO_2. Clean gas leaves at the top.

4 STRIP IT BACK OUT
The rich solvent is heated in the stripper by low-pressure steam taken from the steam turbine. The CO_2 comes off; the lean solvent goes back to the absorber.

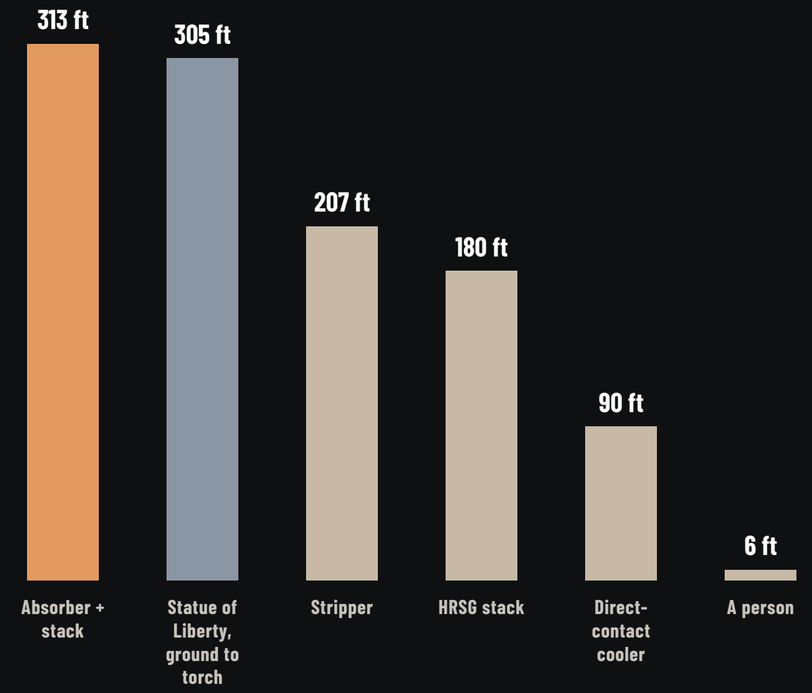
5 COMPRESS AND SEND
The CO_2 is dried and compressed for a pipeline to storage or use.

CABLE: about 2,750 cables in the capture plant: MV to the VFD-fed booster fans and CO_2 compressors, power and control to the solvent pumps, thousands of instrument loops.

REAL TOWERS, REAL SCALE

313 ft

Each absorber is 62 ft across and 262 ft tall, 313 ft with its stack: three of them, one per gas turbine. Sizes from the public permit drawings of a real capture project.



$85 / TONNE

45Q federal tax credit for CO_2 captured at a plant like this and stored (2025 law)

3 X 16 MW

booster fans on drives: VFD cable, MV feeders, a new MV lineup

THE CABLE FAMILIES IN A POWER PLANT

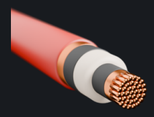
From the biggest to the smallest. The whole SW-3X1 campus: 22,012 cables, 1,174 miles.

HV — 69–345 kV

XLPE: switchyard and gen-tie

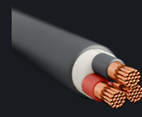
MV — 5–35 kV

MV-105: big motors, transformers, feeders



LV POWER — 600 V

TC-ER, MC, ARMOR-X: motors, panels, lighting



VFD — 600 V–2 kV

drive-rated: fans and pumps on drives

CONTROL — 600 V

TC-ER multi-conductor: breakers, valves, interlocks

INSTRUMENTATION — 300 V

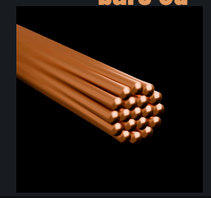
PLTC / ITC shielded pairs, thermocouple extension

FIBRE — –

controls network, protection

GROUNDING — bare Cu

the ground grid under everything



By length: Instrumentation 44% · LV 600 V power 19% · Control 18% · Fibre, fire alarm, other 12% · MV / HV 7%. The big cables get the attention; the small ones make up most of the miles.

Shares by footage from the SW-3X1 model cable schedule; voltage ratings typical.

ONE COMPANY SPECIFIES THE CABLE, ANOTHER ONE BUYS IT

DEVELOP

site, permits, interconnection request, fuel and water

WHO
Owner / developer

YEARS: THE INTERCONNECTION QUEUE IS OFTEN THE LONG POLE

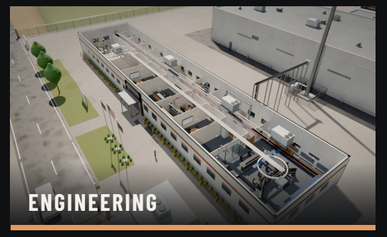
FINANCE

PPA or rate case, turbine slot, financing decision (FID)

WHO
Owner, lenders, OEM

TURBINES ARE RESERVED YEARS AHEAD

ENGINEER



owner's engineer writes the specs; the EPC does detailed design

WHO
Owner's engineer, EPC

CABLE IS SPECIFIED HERE

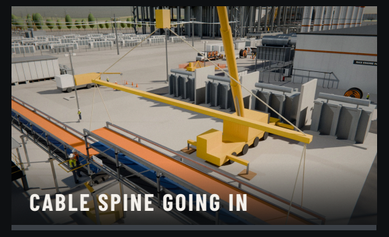
PROCURE

EPC and package builders buy equipment and bulk materials

WHO
EPC, OEMs, package builders

CABLE IS BOUGHT HERE

BUILD



civil, mechanical, then electrical: trays, pulls, terminations

WHO
EPC and electrical contractor

CABLE PULLS SIT ON THE CRITICAL PATH

OPERATE

commercial operation (COD), outages, upgrades, repowering

WHO
Owner / operator

REPLACEMENT AND BROWNFIELD WORK

The owner pays, the owner's engineer writes the spec, the EPC or package builder places the order. Be in the spec early, and be ready when the EPC buys: by then the schedule is tight and cable is on the critical path.

THREE BUCKETS OF CABLE, THREE DIFFERENT BUYERS

1 INSIDE THE PACKAGES

WHO BUYS

OEMS, E-HOUSE AND PACKAGE BUILDERS

E-house internals (switchgear and MCC interconnections, control and panel wire, short ties), turbine skids, modular gensets

WHEN

Every month, across many projects, months before site

OUR PRODUCTS

MV-105 ties, TC-ER, control and panel wire, pre-terminated harness kits

NOT IN OUR 445 MILES: A SEPARATE MARKET ON TOP

2 BETWEEN THE PACKAGES

WHO BUYS

THE EPC OR ITS ELECTRICAL CONTRACTOR

Field cable from the e-houses out to every motor, valve, instrument and transformer: the X-ray

WHEN

Once per plant, when detailed design fixes the cable schedule

OUR PRODUCTS

MV-105, TC-ER, ARMOR-X MC-HL, VFD, PLTC / ITC, thermocouple; SIMpull, cut-to-length

MOST OF THE MILES: ~440 IN THE POWER BLOCK

3 TO THE GRID

WHO BUYS

THE EPC'S HV CONTRACTOR OR THE UTILITY

GSU to switchyard, the gen-tie to the POI, grounding

WHEN

Early: HV cable has the longest lead time

OUR PRODUCTS

HV XLPE (69-345 kV), bare copper grounding

FEW CABLES, HIGH VALUE

Starting in the e-house is not the same as being bought by the e-house builder. Cover both: EPCs for the field cable; package and e-house builders for the cable inside. The builders buy every month and shape the spec at the terminations.

WHERE EVERY CABLE STARTS

The R1 switchgear room from above, roof off. **Power leaves the switchgear and MCCs; signals come home to the control room.** Every device in between is wired.

SWITCHGEAR

13.8 and 4.16 kV lineups: every MV feeder starts here

MCCS AND DRIVES

480 V starters and VFDs: motors, fans and pumps

CONTROL ROOM

where the instrument and control cable comes home

04

THE TOUR

Seven chapters on the SW-3X1 campus: what it does, its electrical system, its cable.

ONE CAMPUS, SEVEN CHAPTERS

1 ARRIVAL AND THE BUILD

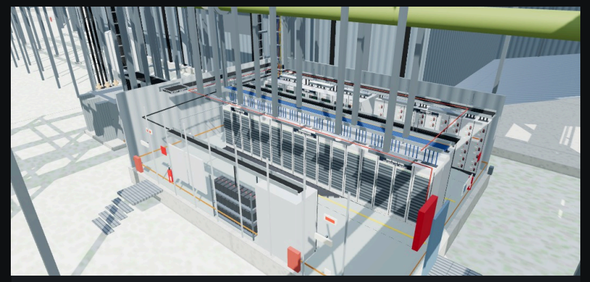
Admin and control room, the office being wired, reels, pullers, prefab

7 VIEWS

2 THE COMBINED CYCLE

Air and water in; the turbine hall: gas turbine, steam turbine, the switchgear

8 VIEWS



3 THE HRSG AND ITS E-HOUSE

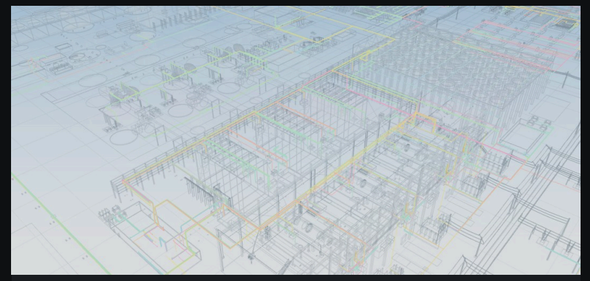
The boiler's own electrical house, the stack, 92 VFD cables to the ACC

3 VIEWS

4 TO THE GRID

Step-up transformer, the 230 kV switchyard, inside the relay house

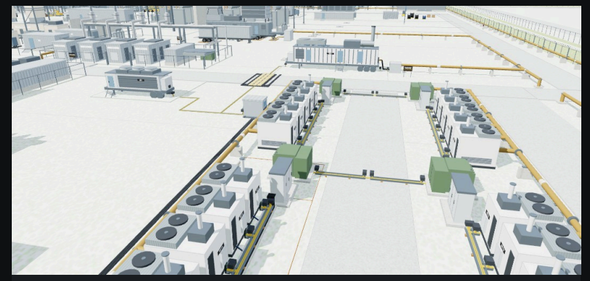
4 VIEWS



5 UNDER THE GROUND AND THROUGH THE WALLS

The ground lifted away, then only the cable

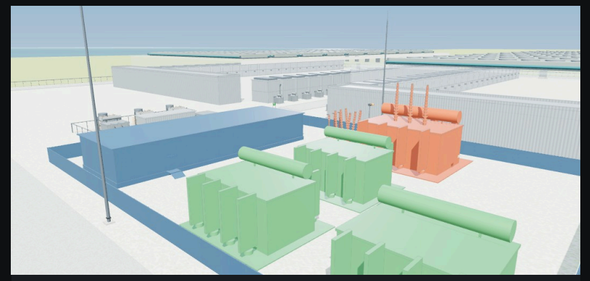
2 VIEWS



6 THE MODULAR ENERGY PARK

Engines, aeroderivatives, unit-block-loop, the battery

6 VIEWS



7 THE DATA CAMPUS, FUEL AND WHAT COMES NEXT

Behind the meter, the gas yard, the close

4 VIEWS

HOW IT RUNS

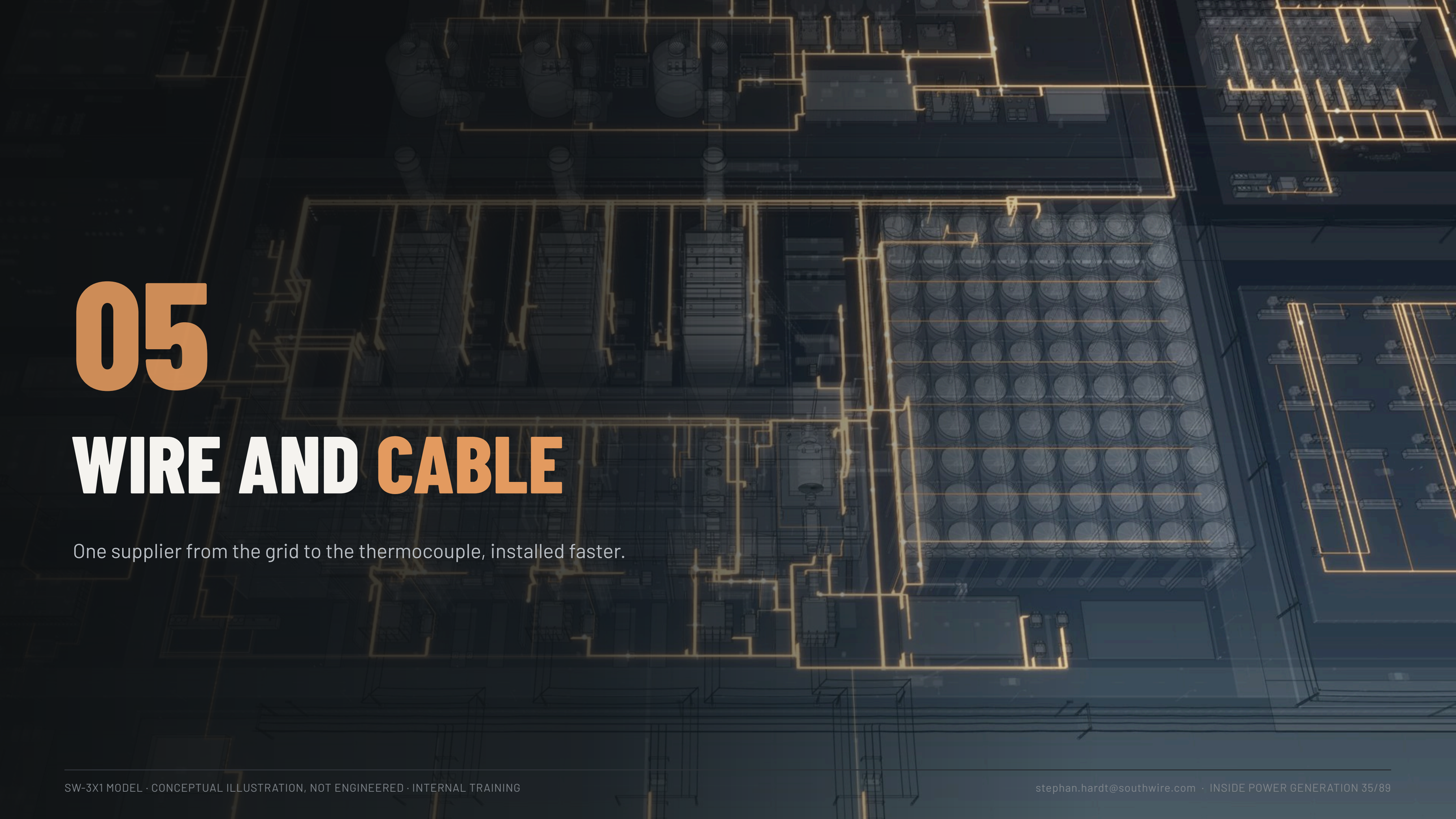
34 views, about 30 s each, in the 3D viewer: ▸ **Nov 11 auto** plays it by itself, P pauses, keys 1-7 jump to a chapter. The full story (85 views) is on the viewer link afterwards.

THE TOUR, RECORDED

Video goes here

PowerPoint: Insert → Screen Recording of ▸ Nov 11 auto, sized to this frame

Play this slide only if the live model is not available



05

WIRE AND CABLE

One supplier from the grid to the thermocouple, installed faster.

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THE WHOLE CAMPUS IN ONE NUMBER

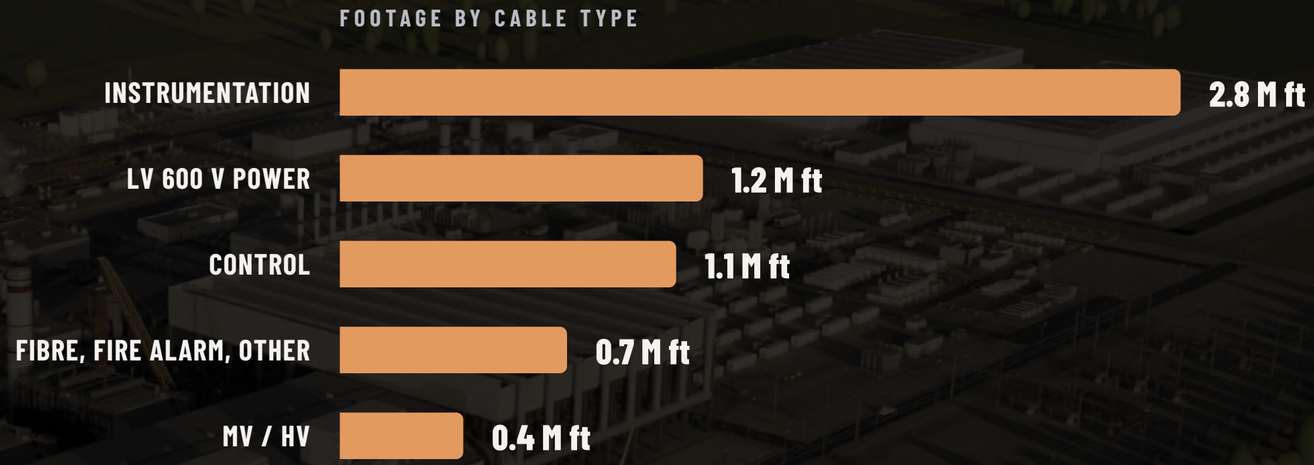
6.2M

FEET OF CABLE · ABOUT 1,174 MILES

6,492
CABLE RUNS

22,012
CABLES

From the model's cable schedule: every circuit on the campus, from the 230 kV switchyard to the last thermocouple.



Most of the footage is small cable: instrument and control. MV is the high-value part, but it isn't the volume.

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EVERY CABLE HAS ONE OF FIVE JOBS

1 POWER IN AND OUT

Generator leads, collector feeders, gen-ties to the grid, incoming supply

MV-105 leads & collectors · 230 kV XLPE · overhead: ACSR, ACSS, AAC, OPGW

E.G. I-MV-0039

MV-105 15 kV 750 kcmil: RICE engine-generator 1 → power control module (generator leads)

2 RUN THE MACHINES

Every motor, pump, fan and heater

MV-105 · TC-ER power · Machine Flex VFD · 2 kV DC · portable

E.G. B-P-0001

Machine Flex VFD 350 kcmil: R4 drive → ACC fan motor F01

3 CONTROL THE MACHINES

Start, stop, open, close, trip

TC-ER control

E.G. A-C-0004

TC-ER control 14 AWG: R2A e-house → boiler feed pump station

4 MEASURE EVERYTHING

Temperatures, pressures, flows, vibration to the control room

PLTC / ITC instrumentation · thermocouple extension

E.G. A-I-0233

Thermocouple ext. 16 AWG: gas turbine 1 probes → junction box

5 KEEP IT SAFE AND CONNECTED

Grounding, gas areas, fire alarm, buildings, data

Bare copper · ARMOR-X MC-HL · fire alarm · SIMpull THHN & MC · fibre

E.G. C-G-0002

Bare Cu 4/0: ground grid → under the whole plant

Share of the 1,174 miles in the model's cable schedule. Job 1 is few miles but the highest value per foot; job 4 is nearly half of all the footage.

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CORE CABLE CATEGORIES IN A GAS POWER PLANT

HIGH VOLTAGE & OVERHEAD — 10 mi

230 kV XLPE cable · ACSR, ACSS, AAC, OPGW

Switchyard, line exits and the H-MOD line from the modular yard; 230 kV underground ties to carbon capture and hydrogen

230 kV · XLPE, Cu wire shield, Al laminate, HDPE · ACSS rated to 250 °C, low sag

MEDIUM-VOLTAGE POWER — 72 mi

MV-105, 5 to 35 kV

Generator leads and collector feeders (modular yard); 13.8 kV from the unit transformers to switchgear, feed pumps, fans, compressors

EPR 133 % insulation · Cu tape shield · Cu or Al (35 kV collectors) · UL 1072

LOW-VOLTAGE POWER & DRIVES — 215 mi

TC-ER power · XHHW-2 feeders · Machine Flex VFD · 2 kV DC

Every 480 V motor, MCC and heater; drive output to the 80 ACC fans; battery strings and exciter DC

600 V XHHW-2 · Type TC-ER exposed run, UL 1277 · VFD: 3 symmetrical grounds, overall shield

CONTROL — 179 mi

TC-ER control cable

Start, stop, open, close and trip: breakers, motor stations, valves; CT / VT circuits to the relays

600 V · #14 AWG 7 / 12 conductor, #10 AWG for CT / VT · UL 1277, tray and direct burial

INSTRUMENTATION — 522 mi

PLTC / ITC pairs and triads · thermocouple extension · RTD cable

Transmitters, vibration, analyzers, CEMS; turbine and HRSG temperatures back to the e-houses

300 V · #18 shielded pairs / triads, overall Al-mylar shield (stops noise from MV machinery) · Type K extension

HIGH HEAT — 4 mi

Silicone rubber 200 °C · RHH/RHW-2 · high-temperature thermocouple wire

Inside the gas-turbine enclosure and exhaust plenum, the steam-turbine casing, the HRSG hot side and duct burners

Silicone 200 °C, glass braid · RHH/RHW-2 XLPE, CPE jacket · glass / mica TC wire · few miles, high value

HAZARDOUS AREAS — 54 mi

ARMOR-X MC-HL (power, control, 15 kV)

Gas yard, fuel modules, LNG and hydrogen, compressors: wherever gas can be present

Class I Div 1 / 2 · continuous corrugated Al armour, PVC jacket · UL 2225

GROUNDING, FIRE & BUILDINGS — 118 mi

Bare copper · FPLR fire alarm · fibre, Cat 6A · SIMpull THHN & MC

Ground grid under everything; fire and gas detection; the plant network; offices, lighting, receptacles

4/0 bare Cu, 19 strand · FPLR, NEC 760 · single-mode armored fibre · THHN UL 83

Miles from the model's cable schedule (1,174 mi). Products and specs typical, conceptual; confirm against the project specification.

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WHO OWNS IT DECIDES THE RULEBOOK

UTILITY-OWNED PLANT

NESC (IEEE C2)

NEC exempts generation under a utility's exclusive control (NEC 90.2)

UL LISTING
Not legally required; the owner's spec decides, often asks for it

WHO SETS THE SPEC
Owner's spec book: IEEE, ICEA, AEIC

GRID SIDE
Transmission owner standards · NERC protection and cyber

MERCHANT / IPP PLANT & MODULAR YARD

NEC (NFPA 70)

Enforced by the local / state inspector (AHJ) in most states

UL LISTING
Required in practice: the inspector accepts listed products

WHO SETS THE SPEC
Owner's engineer and the EPC, within the NEC

GRID SIDE
Interconnection agreement · transmission owner · NERC if large

BEHIND THE METER (DATA CENTRE, INDUSTRIAL HOST)

NEC in full

Premises wiring: every foot inspected

UL LISTING
Required

WHO SETS THE SPEC
EPC and inspector; the host's standards on top

GRID SIDE
Utility interconnection · NEC 705 · IEEE 1547 at distribution

THE CODES THAT BUILD THE PLANT

FIRE & LIFE SAFETY	NFPA 850 generating plants · NFPA 72 fire alarm · NFPA 70E, OSHA 1910.269
COMBUSTION & FUEL	NFPA 85 HRSG / duct burners · NFPA 37 engines, turbines · NFPA 54 / 56 gas · NFPA 59A LNG · NFPA 2 H2 · 49 CFR 192
HAZARDOUS AREAS	NEC 500 / 505 classified locations · API RP 500 / 505: why ARMOR-X MC-HL in the gas areas
PRESSURE & PIPING	ASME Section I (HRSG) · B31.1 power piping · B31.3 / B31.8
ELECTRICAL EQUIPMENT	IEEE C57 transformers · C37 switchgear · IEEE 80 grounding · IEEE 1185 cable installation, 835 ampacity
MODULAR & BTM PACKAGES	UL 2200 gensets · UL 1558 / 891 gear · UL 845 MCC · UL 508A panels · UL 1741 inverters · UL 9540, NFPA 855 batteries · NEC 445, 706, 705, 700-702
ENVIRONMENTAL	EPA air permit · CEMS, 40 CFR 60 / 75

CABLE: UL LISTING → INDUSTRY STANDARD

HV 230 kV XLPE	not UL-listed	AEIC CS9, ICEA S-108-720
MV-105	UL 1072	ICEA S-93-639, AEIC CS8
TC-ER power & control	UL 1277 (ER: out of tray, NEC 336)	ICEA S-95-658 / S-73-532
THHN · XHHW-2, RHH/RHW-2	UL 83 · UL 44	NEC 310 building wire
MC · MC-HL (ARMOR-X)	UL 1569 · UL 2225	NEC 330, Class I
PLTC · ITC	UL 13 · UL 2250	NEC 725 / 727
Fire alarm FPLR · circuit integrity	UL 1424 · UL 2196	NFPA 72, NEC 760
Fibre · Cat 6A · PV wire	UL 1651 · UL 444 · UL 4703	NEC 770 / 800 / 690
ACSR · ACSS · AAC · OPGW	not UL-listed	ASTM B232 / B856 / B231, IEEE 1138
Every tray cable	flame test UL 1685 / IEEE 1202	called out in every plant spec

US practice, typical; the AHJ, the owner's specification and the interconnection agreement govern each project. Code editions and adoption vary by state.

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EVERY ZONE, EVERY SOUTHWIRE PRODUCT

	HV / EHV	OVERHEAD ACSR·ACSS AAC·OPGW	MV-105	ARMOR-X MC-HL	MACHINE FLEX VFD	TC-ER POWER	TC-ER CONTROL	PLTC / ITC INSTRUM.	HIGH-HEAT SILICONE RHH	THERMO-COUPLE EXT.	2 KV DC POWER	PORTABLE SHD-GC	SIMPULL THHN & MC	BARE CU GROUNDING	FIRE ALARM & DATA	
POWER BLOCK			24	26		76	67	161		20					20	415 mi
AIR-COOLED CONDENSER							10	22								52 mi
SWITCHYARD & GRID						18	43	48								123 mi
PLANT GAS YARD				13												24 mi
LNG FUEL SUPPLY				11				27								53 mi
WATER & UTILITIES						26	19	15								67 mi
ADMIN & SERVICES																28 mi
CARBON CAPTURE						20	13	90								144 mi
BATTERY STORAGE								16			11					51 mi
MODULAR POWER YARD			20			27	15	86		11					15	195 mi
GT INLET CHILLING								16								22 mi
TOTAL MILES	3	6	72	54	16	187	179	490	4	32	12	<1	38	30	50	1,174 mi

Dot area = miles of cable in the zone (number shown from 10 mi). From the model's cable schedule: 6,492 runs, 22,012 cables; grounding spread over the zones by their equipment. Conceptual; product families typical.

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ONE SUPPLIER, GRID TO THERMOCOUPLE

GRID CONNECTION	HV XLPE transmission cable (230 kV) · bare copper grounding
POWER ISLAND: MV	MV-105 EPR 133% (5–35 kV) · ARMOR-X MC-HL 15 kV
POWER ISLAND: LV AND DRIVES	Type TC-ER power · ARMOR-X MC-HL 600 V · VFD-rated cable
CONTROL AND INSTRUMENTATION	TC-ER control · PLTC / ITC shielded pairs · thermocouple extension
HAZARDOUS AREAS	ARMOR-X MC-HL, Class I Div 2: gas skids, fuel yards, burner fronts
BATTERY STORAGE	PV / RHW-2 DC conductor (2 kV) · MV-105 collection (35 kV)
MODULAR AND TEMPORARY POWER	Type SHD-GC portable MV · Type W · cam-lock sets
INSTALLATION	SIMpull reels and SIMpull truck · cut-to-length · pre-terminated harness kits

Plus the services: Spec support: sizing, derating, spec reviews · Cut-to-length against the cable schedule · Harness kits, pre-terminated and labelled · Project logistics: reel tracking, sequenced deliveries · Field support: pulling calcs, terminations, tests.

Typical Southwire families for each application; confirm constructions, sizes and listings against current Southwire specifications.

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THE REEL YARD: SIMPULL REELS STAGED FOR THE PULLS

WIRE AND CABLE · INSTALLED FASTER

ACCELERATING TIME TO POWER

Cable is decided in the field, on the critical path. Every reel set-up and every termination saved is schedule saved.

SIMPULL REELS

Several conductors paralleled on one reel: one pull per feeder instead of one per conductor.

SIMPULL TRUCK

Delivery, payoff and reel return in one vehicle.

CUT-TO-LENGTH

Reels cut and labelled to the cable schedule, pull by pull: less waste, no field measuring.

PRE-TERMINATED

Factory-made, tested harnesses for skids and panels: on the critical path you test instead of terminate.

KNOWLEDGE CHECK · ZOOM POLL

THREE QUESTIONS, ONE MINUTE

1 WHICH CABLE TYPE HAS THE MOST MILES IN THE POWER BLOCK?

- A MV / HV POWER
- B LV POWER
- C INSTRUMENTATION
- D FIBRE

2 WHO USUALLY PLACES THE CABLE ORDER?

- A THE OWNER
- B THE OWNER'S ENGINEER
- C THE EPC OR PACKAGE BUILDER
- D THE UTILITY

3 IN A COMBINED CYCLE, THE STEAM TURBINE ADDS ABOUT HOW MUCH MORE POWER, WITH NO EXTRA FUEL?

- A 5%
- B ONE-THIRD
- C DOUBLE
- D NONE

THREE THINGS TO REMEMBER

TAKE AWAY

1 **GAS IS GROWING AGAIN**

driven by data centres and the need for firm power.

2 **MOST OF A PLANT'S CABLE IS SMALL**

instrument and control; the big MV and power cable is the smaller share.

3 **SOUTHWIRE COVERS THE WHOLE PLANT**

one supplier, staged on SIMpull reels, spec-compliant from MV down to the thermocouple.

Southwire®
POWER GENERATION SOLUTIONS

Q&A

Questions from the chat. The recording and the 3D viewer link will be shared after the session.

Stephan Hardt
stephan.hardt@southwire.com

AFTER TODAY

RESOURCES AND WHO TO CALL

RESOURCES

THE 3D PLANT MODEL

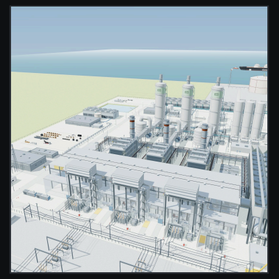
The interactive viewer you saw today, with the Nov 11 tour and the cable X-ray. Link sent with the recording.

THE BROCHURES

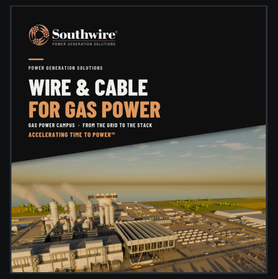
Wire & cable for gas power · EPC solutions · boiler solutions.

THE RECORDING AND THESE SLIDES

Shared after the session, with a glossary of the terms used today.

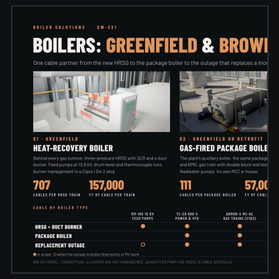


3D VIEWER



WIRE & CABLE BROCHURE

EPC SOLUTIONS



BOILER SOLUTIONS

POWER GENERATION SOLUTIONS TEAM

STEPHAN HARDT	PowerGen strategy, the plant model, owners and OEMs
PATRICK RITCHIE	EPC accounts
GREYSON	EPC outreach to more gas-plant builders
PHIL METZ · BOB BENISH	OEMs and package builders: e-houses, switchgear, modular power

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B

BACKUP SLIDES

Extra detail if questions go there, and the sources.

WHAT THE CABLE DOES HERE · AREA 1 OF 10

TURBINE HALL

180 MILES OF CABLE

2 **RUN THE MACHINES** — **22 mi**

MV-105 5 kV feeds the static starters that spin each gas turbine up to start; TC-ER runs the lube-oil, hydraulic and turning-gear motors.

3 **CONTROL** — **21 mi**

TC-ER control runs from the R2 e-houses to the exciters, fuel valves and trip circuits, so operators can start, load and trip each unit.

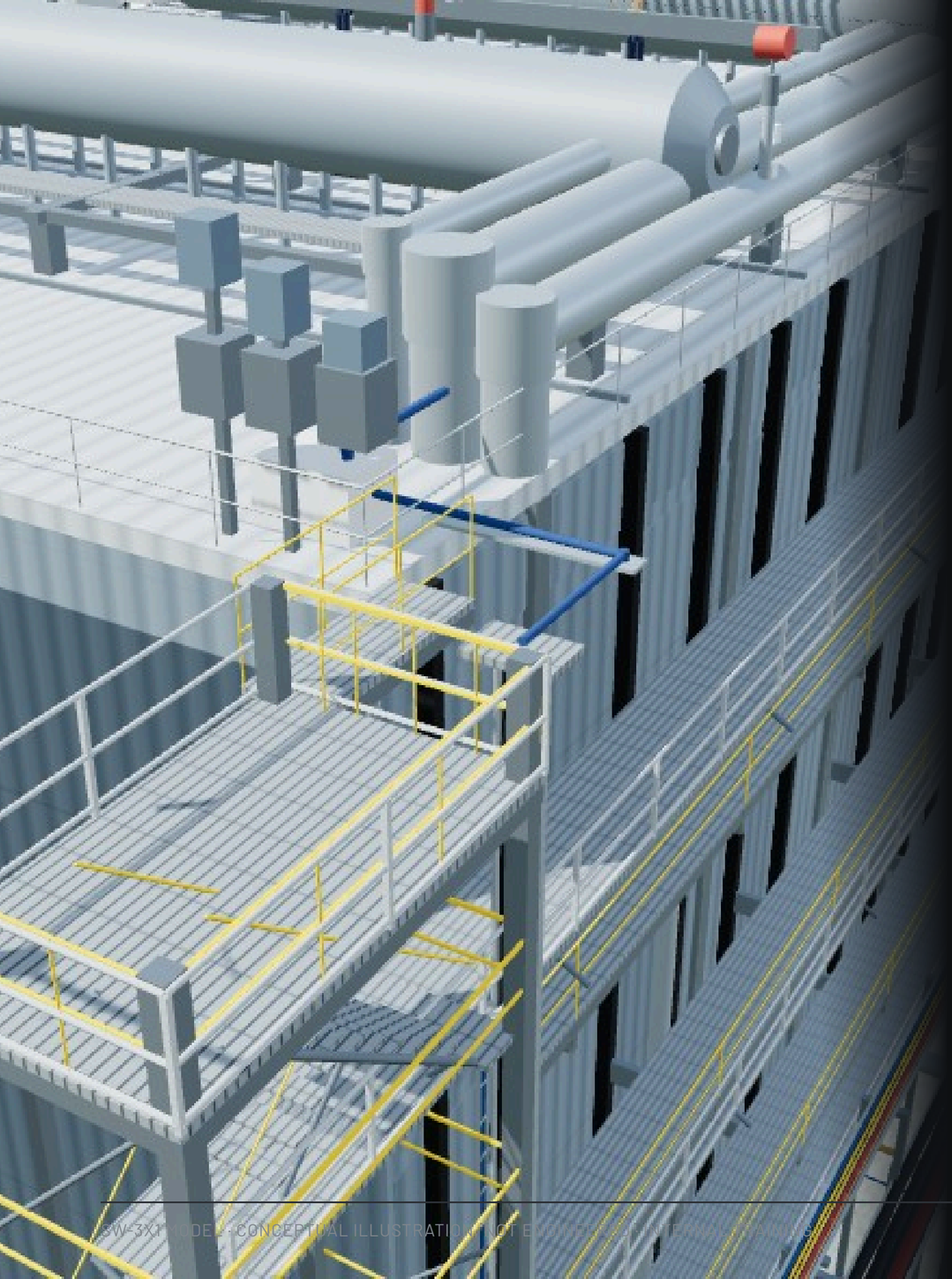
4 **MEASURE** — **96 mi**

PLTC and ITC bring vibration, pressures and flows to the R2 e-houses; thermocouple extension the turbine inlet and exhaust temperatures.

5 **KEEP IT SAFE** — **42 mi**

ARMOR-X MC-HL on the gas-fuel skids (hazardous area); fire and gas loops and DCS fibre; every machine bonded to the ground grid.

ALSO HERE FPLR fire alarm 9 mi · fibre optic 5 mi · RTD cable 5 mi · XHHW-2 feeder sets 4 mi · bare copper 4/0 2 mi · silicone 200 C 1 mi



WHAT THE CABLE DOES HERE · AREA 2 OF 10

HRSGS AND STACKS

112 MILES OF CABLE

2 **RUN THE MACHINES** — **16 mi**

MV-105 15 kV runs from the R1 switchgear to the boiler feed pumps; TC-ER powers the valve actuators, drains and chemical feed.

3 **CONTROL** — **17 mi**

TC-ER control from the R2 e-houses to each motor and valve station, so the boiler can be filled, drained and fired safely.

4 **MEASURE** — **72 mi**

PLTC carries drum level, steam pressure and the stack emissions monitors (CEMS) back to the control system; thermocouple extension the tube temperatures.

5 **KEEP IT SAFE** — **7 mi**

Bare copper grounds each boiler and stack and its lightning down-conductors; fire and gas detection.

ALSO HERE ITC 4 mi · RTD cable 3 mi · fibre optic 2 mi · SIMpull THHN 2 mi · RHH/RHW-2 1 mi

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POWER-BLOCK ELECTRICAL

123 MILES OF CABLE

1 POWER IN AND OUT — 9 mi

Power in: MV-105 15 kV from the unit auxiliary and station transformers into the R1 switchgear, so the plant runs on its own power. (Generator leads to the step-ups are bus duct here.)

2 RUN THE MACHINES — 55 mi

XHHW-2 feeder sets carry 480 V from the unit transformers to the MCCs; MV-105 15 kV runs from R1 out to every transformer: the plant's power spine.

3 CONTROL — 31 mi

TC-ER control carries protection (CT / VT) and transformer cooling and tap-changer control between R1 and each transformer.

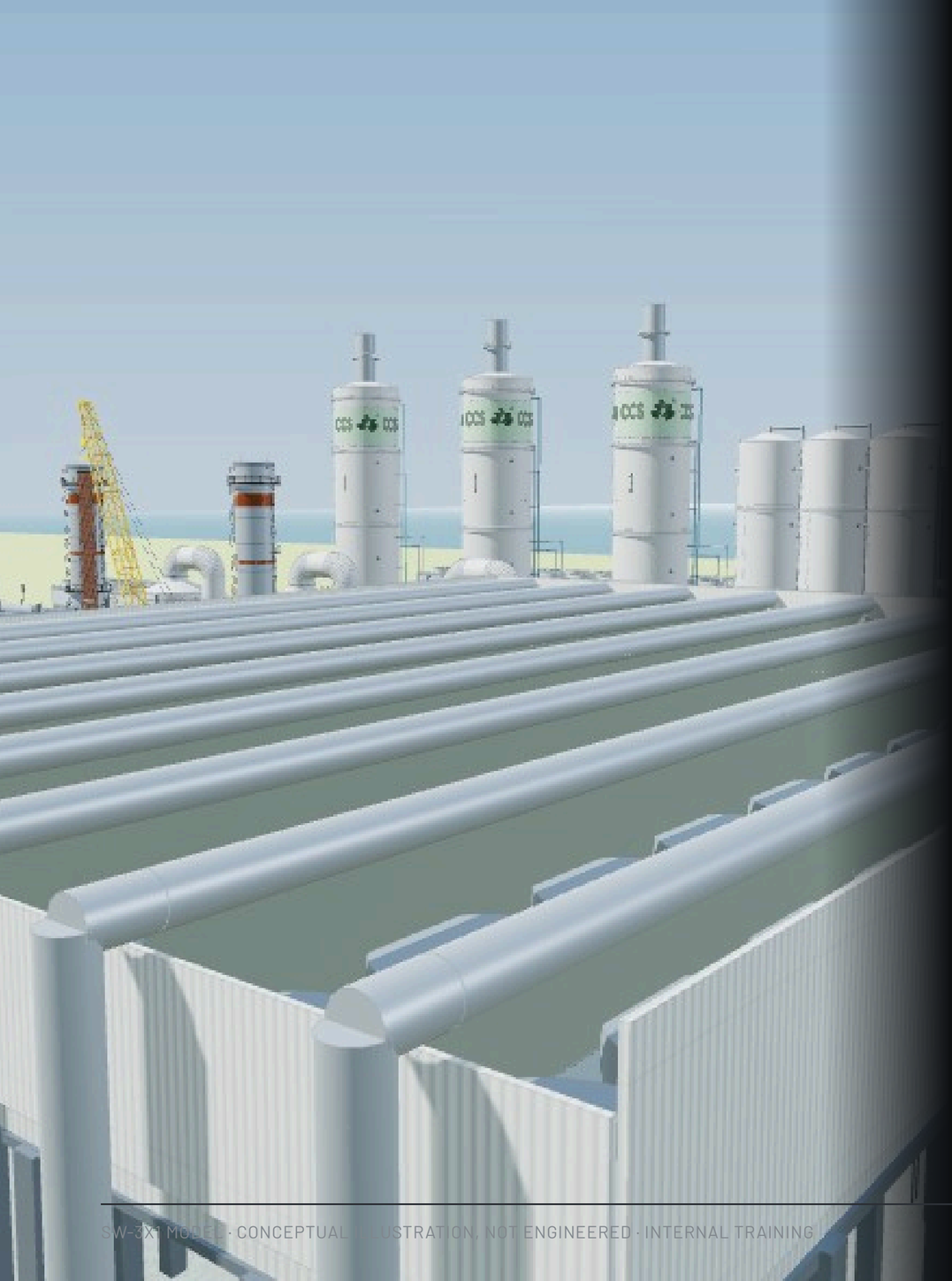
4 MEASURE — 15 mi

PLTC brings transformer temperatures and gas-in-oil monitors back to R1.

5 KEEP IT SAFE — 14 mi

Bare copper ground grid under the transformers; fire detection in R1 and R3; heat-trace feeders.

ALSO HERE TC-ER power 16 mi · SIMpull THHN 7 mi · FPLR fire alarm 4 mi · MV-105 5 kV 2 mi



AIR-COOLED CONDENSER

52 MILES OF CABLE

2 **RUN THE MACHINES** — 15 mi

Machine Flex VFD cable runs from the R4 drives to all 80 fan motors, so each fan's speed follows the steam load; MV-105 to the condensate pumps.

3 **CONTROL** — 10 mi

TC-ER control to the fan and gearbox local stations.

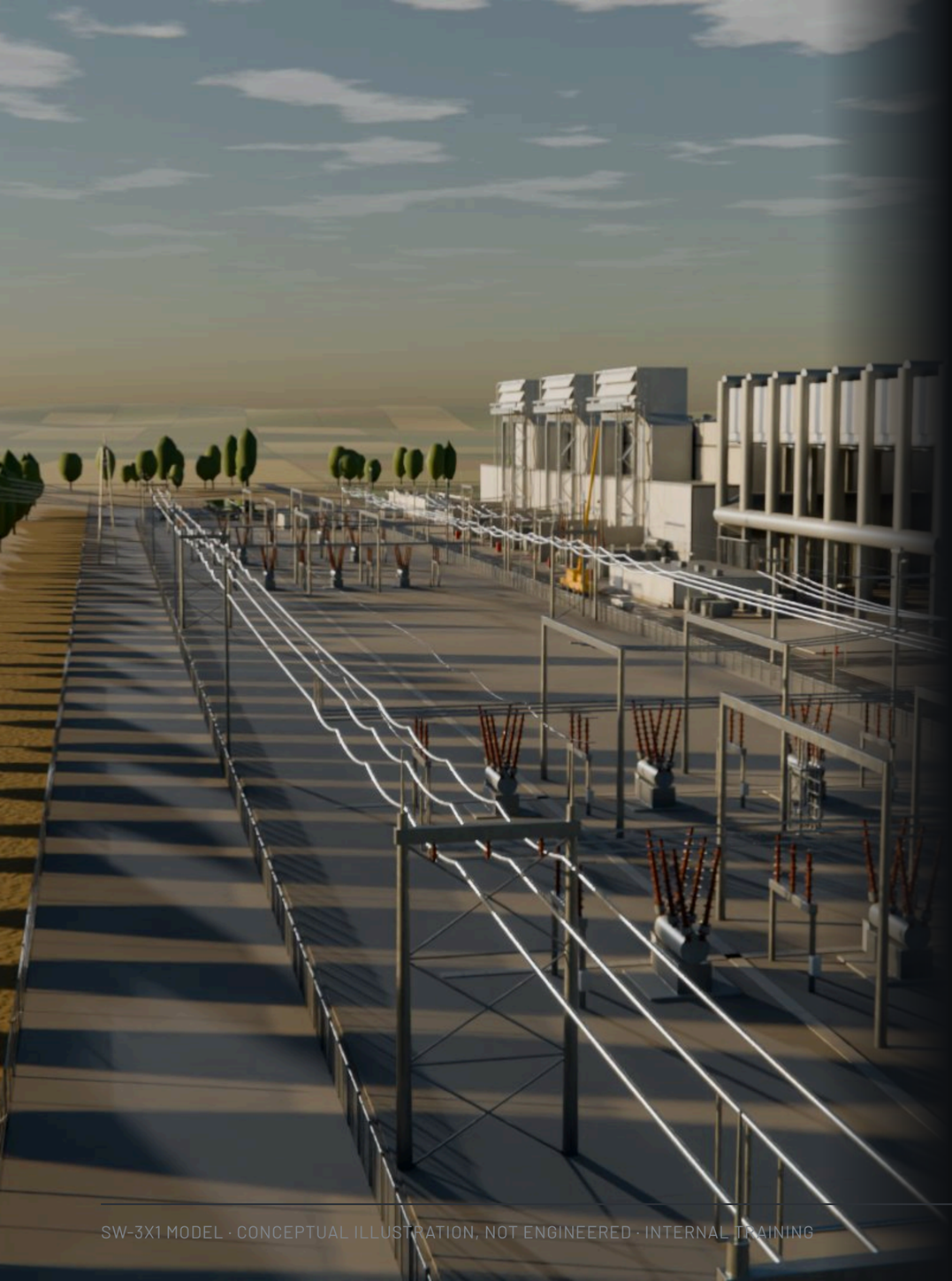
4 **MEASURE** — 22 mi

PLTC carries fan vibration, condensate temperature and level back to R4.

5 **KEEP IT SAFE** — 4 mi

Bare copper grounding of the steel structure and every fan.

ALSO HERE MV-105 15 kV 4 mi · TC-ER power 3 mi · SIMpull THHN 2 mi



SWITCHYARD AND THE GRID

120 MILES OF CABLE

1 **POWER IN AND OUT** — 6 mi
ACSR from the step-ups into the yard, AAC strain bus, ACSS line exits to the POI and the utility's towers, OPGW shield wire with fibre to the relay house.

2 **RUN THE MACHINES** — 18 mi
TC-ER feeds the yard lighting and station service from the relay house.

3 **CONTROL** — 43 mi
TC-ER control carries trip / close and CT / VT circuits from the relay house to every breaker and the revenue metering at the POI, so a fault is cleared in milliseconds.

4 **MEASURE** — 48 mi
PLTC brings breaker and disconnect status and gas-density alarms to the relays.

5 **KEEP IT SAFE** — 4 mi
The station ground grid, rods and risers: it keeps step and touch voltage safe during a fault.

ALSO HERE RTD cable 4 mi · bare copper 4/0 2 mi · SIMpull THHN 1 mi

FUEL: GAS YARD AND LNG

79 MILES OF CABLE

1 POWER IN AND OUT — 1 mi

Power in: a 230 kV XLPE underground tie from the switchyard feeds the hydrogen electrolyzers.

2 RUN THE MACHINES — 10 mi

MV-105 15 kV runs to the LNG send-out pumps and boil-off compressor; TC-ER to heaters and cathodic protection.

3 CONTROL — 1 mi

TC-ER control for the emergency shutdown valves.

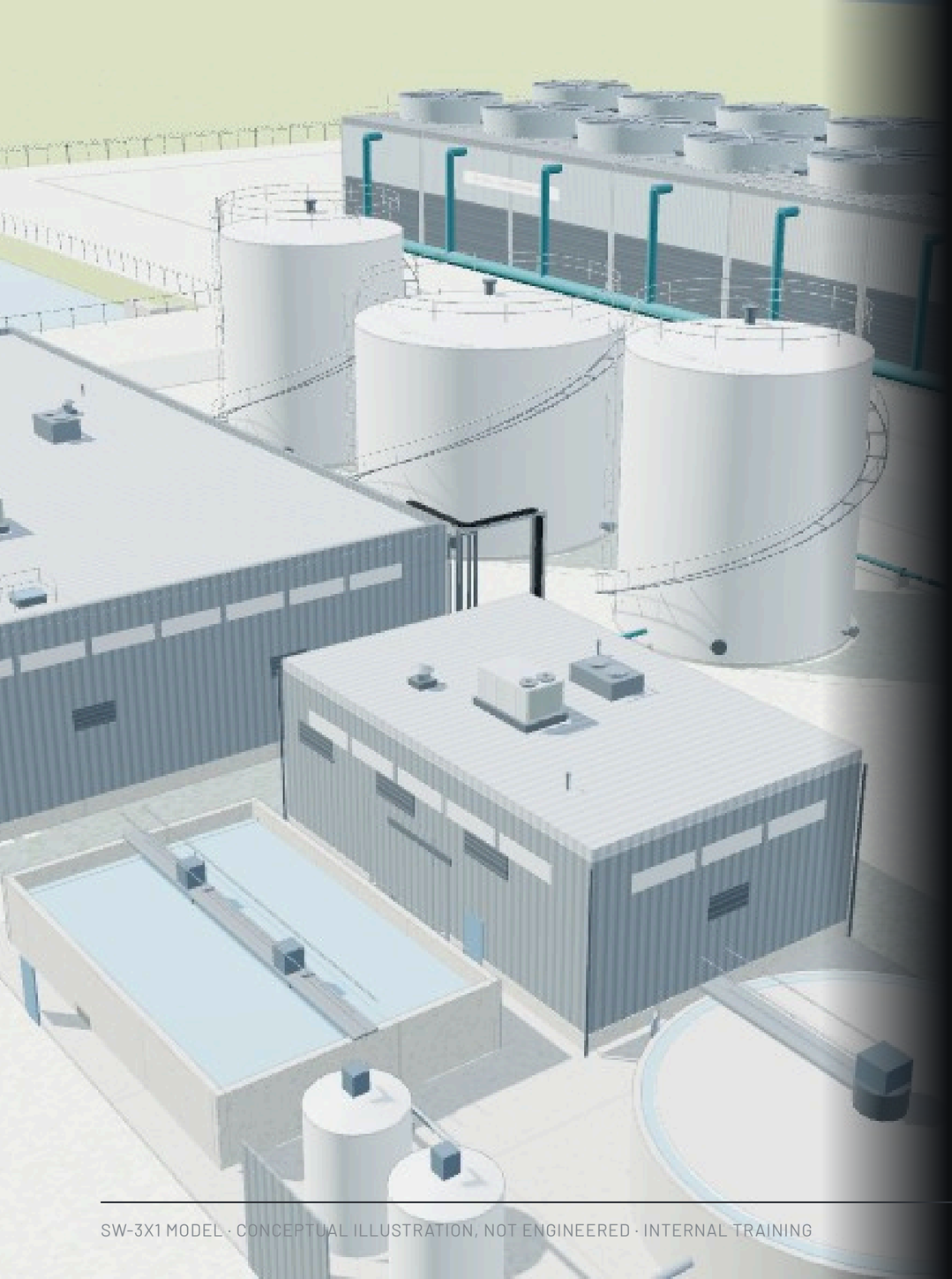
4 MEASURE — 34 mi

ITC carries gas metering, pressure regulation and LNG tank level and temperature.

5 KEEP IT SAFE — 33 mi

ARMOR-X MC-HL carries power and control through the gas areas (Class I), sealed so a spark can't reach the gas; gas detection.

ALSO HERE bare copper 4/0 4 mi · fibre optic 3 mi · ARMOR-X MC-HL 15 kV 2 mi



WHAT THE CABLE DOES HERE · AREA 7 OF 10

WATER, UTILITIES, CHILLING

88 MILES OF CABLE

2 **RUN THE MACHINES** — **29 mi**

TC-ER to the pumps, treatment and fire pumps; Machine Flex VFD to the chiller and tower fans; MV-105 to the big pumps and chillers.

3 **CONTROL** — **21 mi**

TC-ER control from the MCCs to each pump and valve.

4 **MEASURE** — **32 mi**

PLTC carries flow, level and water-quality analyzer signals.

5 **KEEP IT SAFE** — **7 mi**

SIMpull THHN lights and powers the buildings; bare copper grounding.

WHAT THE CABLE DOES HERE · AREA 8 OF 10

CARBON CAPTURE

146 MILES OF CABLE

1 POWER IN AND OUT — 5 mi

Power in: 230 kV XLPE underground from the switchyard to the CCS transformers, then MV-105 into the CCS switchgear.

2 RUN THE MACHINES — 27 mi

MV-105 from the CCS switchgear to the 16 MW booster fans, CO2 compressors and big pumps; TC-ER to the smaller motors.

3 CONTROL — 13 mi

TC-ER control from the CCS switchgear and control building to every valve and motor.

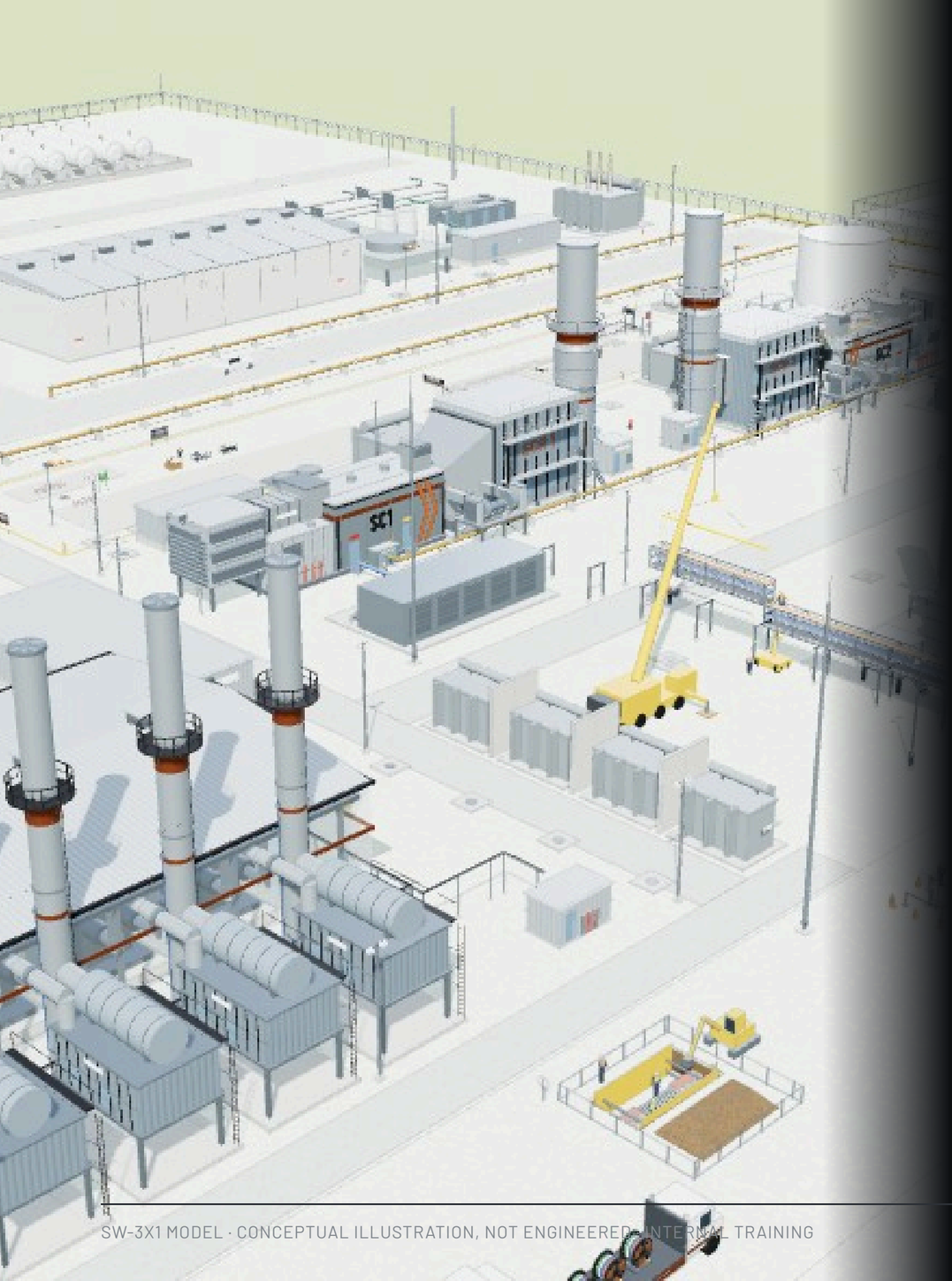
4 MEASURE — 90 mi

PLTC carries the absorber and stripper transmitters and the CO2 analyzers: the largest job in this area.

5 KEEP IT SAFE — 11 mi

Grounding, fire and gas detection and the rack lighting.

ALSO HERE MV-105 15 kV 7 mi · RTD cable 6 mi · SIMpull THHN 6 mi · ITC 3 mi · bare copper 4/0 2 mi · Machine Flex VFD 2 mi



WHAT THE CABLE DOES HERE · AREA 9 OF 10

MODULAR POWER AND BATTERIES

246 MILES OF CABLE

1 **POWER IN AND OUT** — 27 mi

Power out: MV-105 generator leads from each engine and turbine to its step-up, then MV-105 35 kV (AI) collector feeders to the park switchgear and the 230 kV transformers.

2 **RUN THE MACHINES** — 49 mi

TC-ER for package auxiliaries, 2 kV DC cable for the battery strings, Machine Flex VFD to the engine radiator fans.

3 **CONTROL** — 21 mi

TC-ER control from each package and the battery PCS to its breaker.

4 **MEASURE** — 113 mi

Thermocouple extension and PLTC carry engine and turbine temperatures and package instruments: the largest job here.

5 **KEEP IT SAFE** — 37 mi

Ground grid, fire detection in the containers, fibre to the control room.

ALSO HERE fibre optic 12 mi · bare copper 4/0 11 mi · MV-105 15 kV 11 mi · XHHW-2 feeder sets 10 mi · FPLR fire alarm 6 mi · SIMpull THHN 4 mi

ADMIN AND SERVICES

28 MILES OF CABLE

2 **RUN THE MACHINES** — **7 mi**
TC-ER feeds the building services, stormwater pumps and the workshop's welding feeder.

3 **CONTROL** — **3 mi**
TC-ER control for HVAC, the gates and access control.

4 **MEASURE** — **2 mi**
Building and gate signals.

5 **KEEP IT SAFE** — **16 mi**
SIMpull THHN and MC for offices, lighting and receptacles; fibre and Cat 6A to the control room; fire alarm.

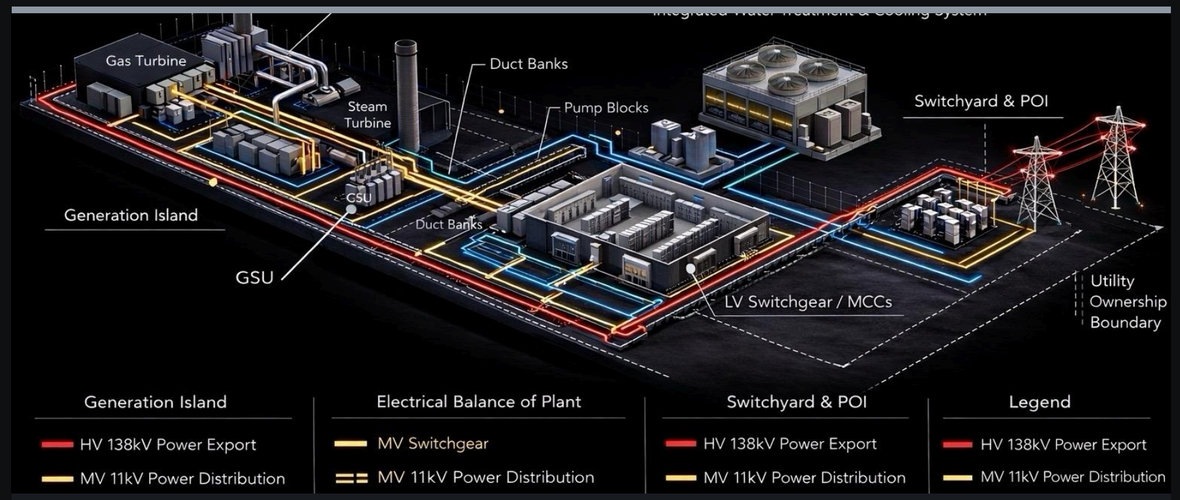
ALSO HERE bare copper 4/0 2 mi · PLTC 1 mi · MC cable 1 mi

ONE REAL CABLE FOR EACH JOB

Say it in one sentence: **[product]** carries **[what]** from **[here]** to **[there]**, so that **[why]**. Every cable below is in the model; click it in the viewer.

JOB	CABLE NO.	SOUTHWIRE PRODUCT	FROM → TO	SO THAT
1 POWER IN AND OUT	I-MV-0039	MV-105 15 kV, 750 kcmil, 2 sets	RICE engine-generator 1 → Power control module	Generator leads: carry the engine's power to its step-up and out.
1 POWER IN AND OUT	C-OH-0003	ACSS 1590 kcmil overhead	230 kV terminal tower → Line entrance	The gen-tie: carries the plant's power out to the grid.
2 RUN THE MACHINES	A-MV-0001	MV-105 15 kV, 250 kcmil, ~1,080 ft	R1 13.8 kV switchgear → Boiler feed pump 1A	Feeds the pump that pushes water into the boiler.
2 RUN THE MACHINES	B-P-0001	Machine Flex VFD, 350 kcmil, ~390 ft	R4 drive (VFD) → ACC fan motor F01	Variable-speed power; ordinary cable fails early on drive output.
3 CONTROL THE MACHINES	A-C-0004	TC-ER control, 14 AWG	R2A e-house → Boiler feed pump local station	Lets operators start, stop and trip the pump.
4 MEASURE EVERYTHING	A-I-0233	Thermocouple extension, 16 AWG pairs	20 probes on gas turbine 1 → Turbine junction box	Tells the control room how hot the turbine runs.
5 KEEP IT SAFE	A-C-0001	ARMOR-X MC-HL control, 14 AWG	R3 e-house → Air compressors	Through a hazardous area, where regular tray cable is not allowed.
5 KEEP IT SAFE	C-G-0002	Bare copper 4/0	Ground grid → Under the whole plant	Sends a fault to earth instead of through a person.

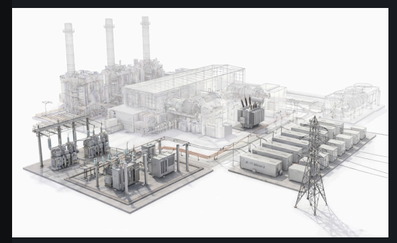
IT STARTED WITH PICTURES THAT LOOKED RIGHT

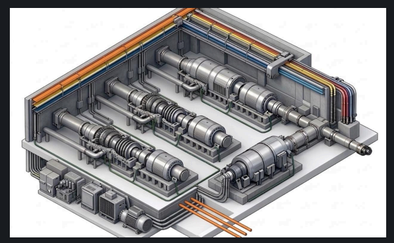


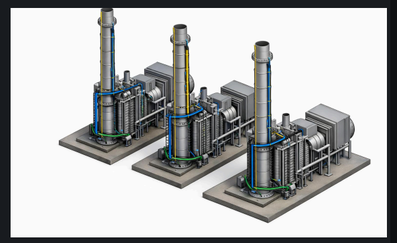
V0 FIRST SKETCHES

MARCH

Cable routes colour-coded to tell the story. But generic: non-US voltages (11 kV, 138 kV), invented equipment, no cable counts.





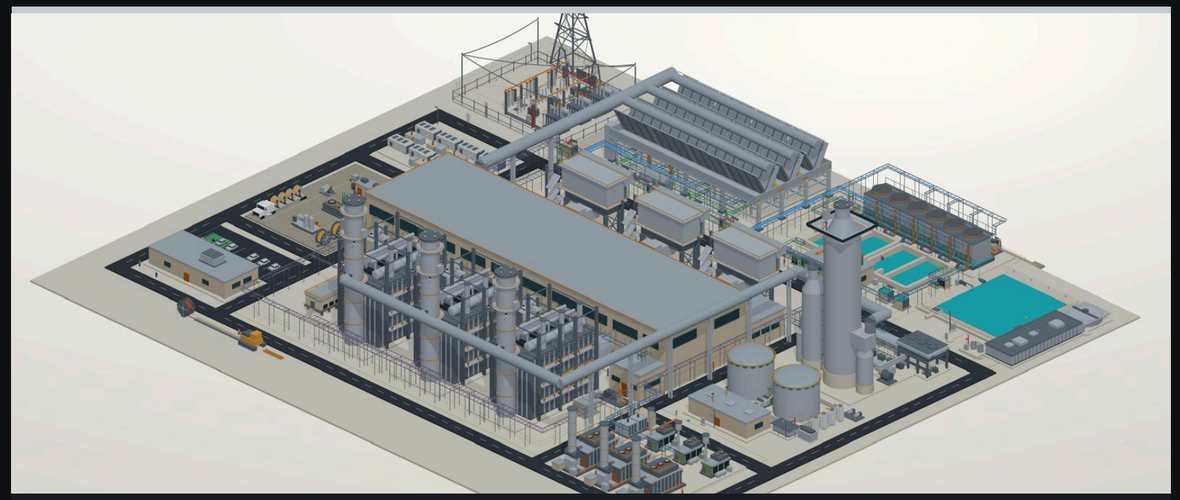


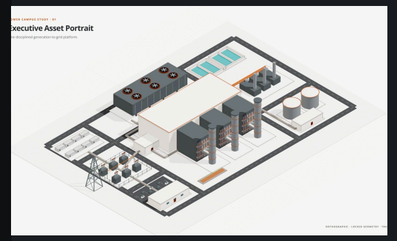
V1 A BRANDED CONCEPT

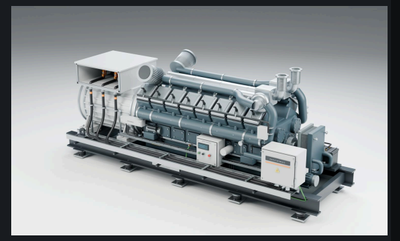
MARCH

A Southwire-branded plant, the turbine hall and HRSGs drawn in detail. Good for a cover, but still invented: nothing you can measure.

THEN A PLANT YOU CAN MEASURE



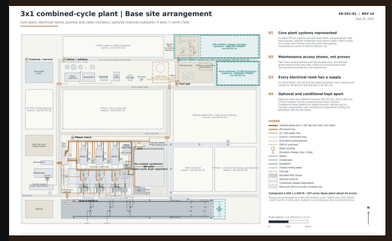


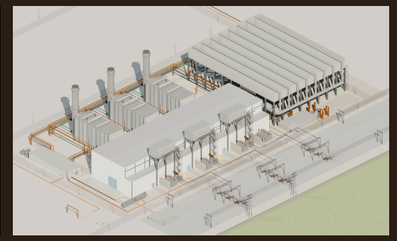


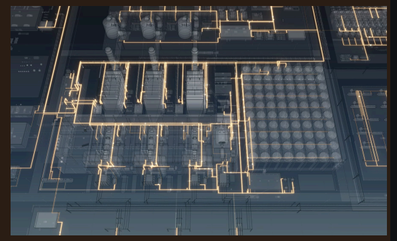
V2 3D STUDIES

SPRING - SUMMER

Real plant elements in 3D: the hall, gas turbines, HRSGs, engines, the plant zones. Closer, but not to scale and not routed: still no cable to count.







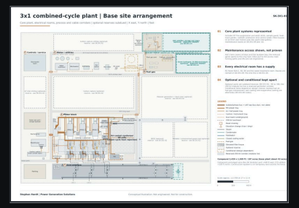
V3 A PLANT YOU CAN MEASURE

SEPTEMBER - TODAY

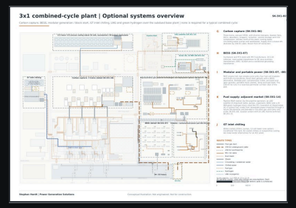
Drafted first: the SK-3X1 drawing set (16 sheets, 230 / 13.8 / 4.16 kV, 480 V), then the model built from it. LOD 3, a 6,492-run cable schedule, the X-ray, the tour.

WHY WE BUILT IT: SEE THE WHOLE PLANT · MEASURE, DON'T GUESS · ONE SOURCE FOR BROCHURES, DECKS, VIEWER AND SCHEDULE · SPEAK THE BUYER'S LANGUAGE

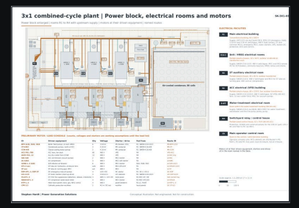
DRAWN FIRST, NOW AT REV 14



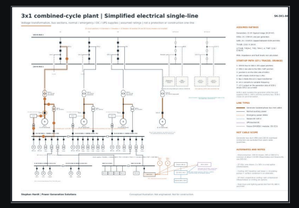
SK-3X1-01
Base site arrangement



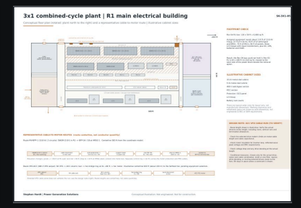
SK-3X1-02
Optional systems



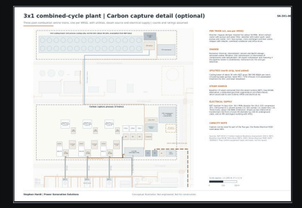
SK-3X1-03
Power block, electrical rooms, motors



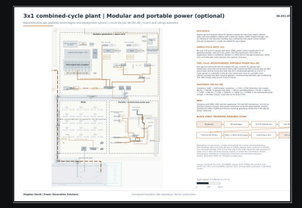
SK-3X1-04
Electrical single-line



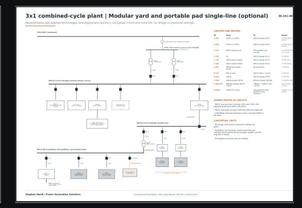
SK-3X1-05
R1 main electrical building



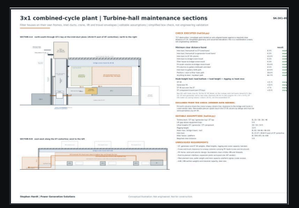
SK-3X1-06
Carbon capture



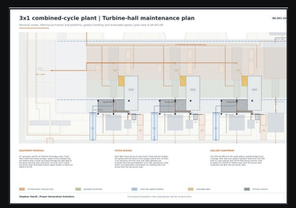
SK-3X1-07
Modular and portable power



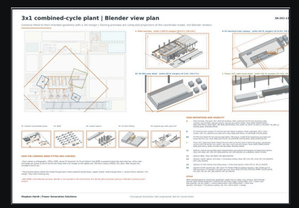
SK-3X1-08
Modular yard single-line



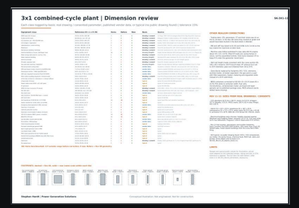
SK-3X1-09
Turbine-hall sections



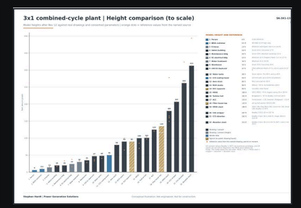
SK-3X1-10
Turbine-hall plan



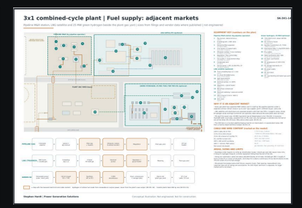
SK-3X1-11
3D view plan



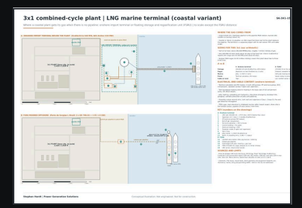
SK-3X1-12
Dimension review



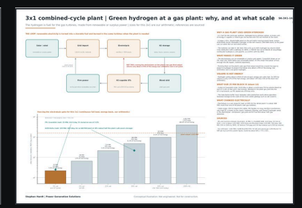
SK-3X1-13
Height comparison



SK-3X1-14
Fuel supply



SK-3X1-15
LNG marine terminal



SK-3X1-16
Green hydrogen

Rev 14

of the SK-3X1 drawing set: 16 sheets, revised as the model found problems. The model is checked back against it on every build.

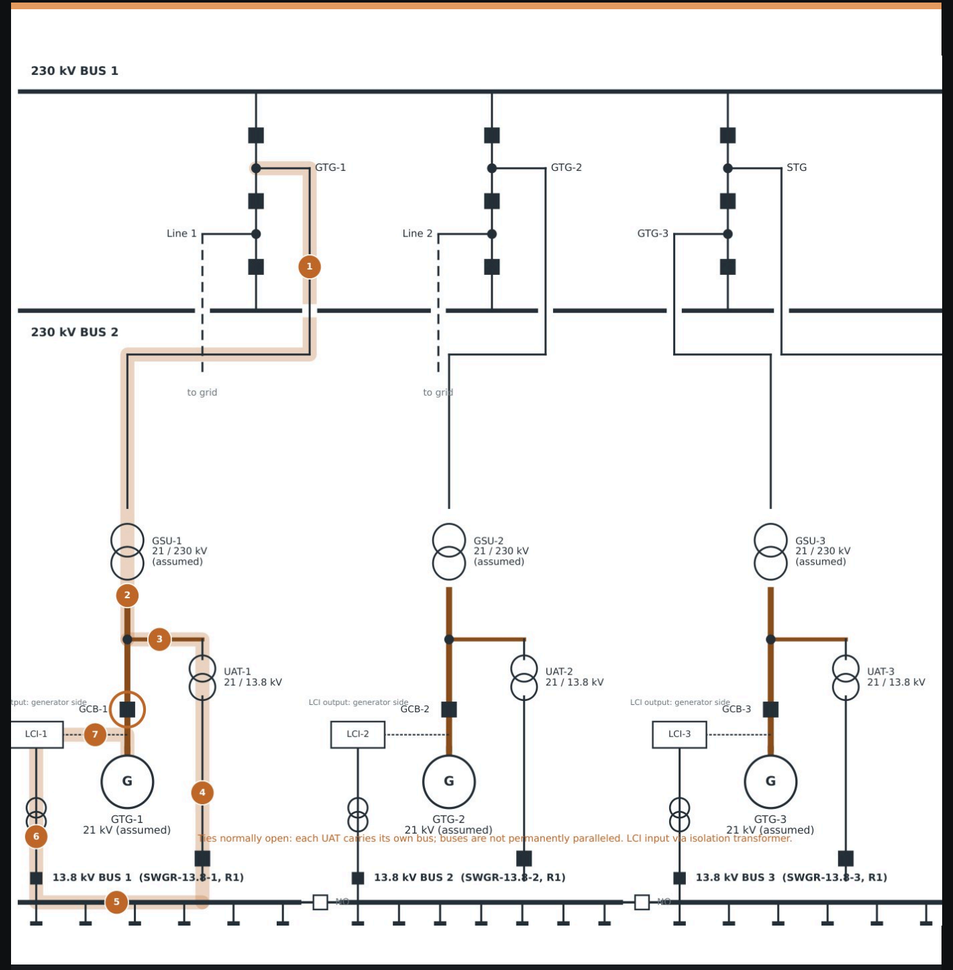
700+

pages of real combined-cycle reference drawings reviewed: site layouts, one-lines, equipment and load lists. They set the equipment, voltages and electrical systems.

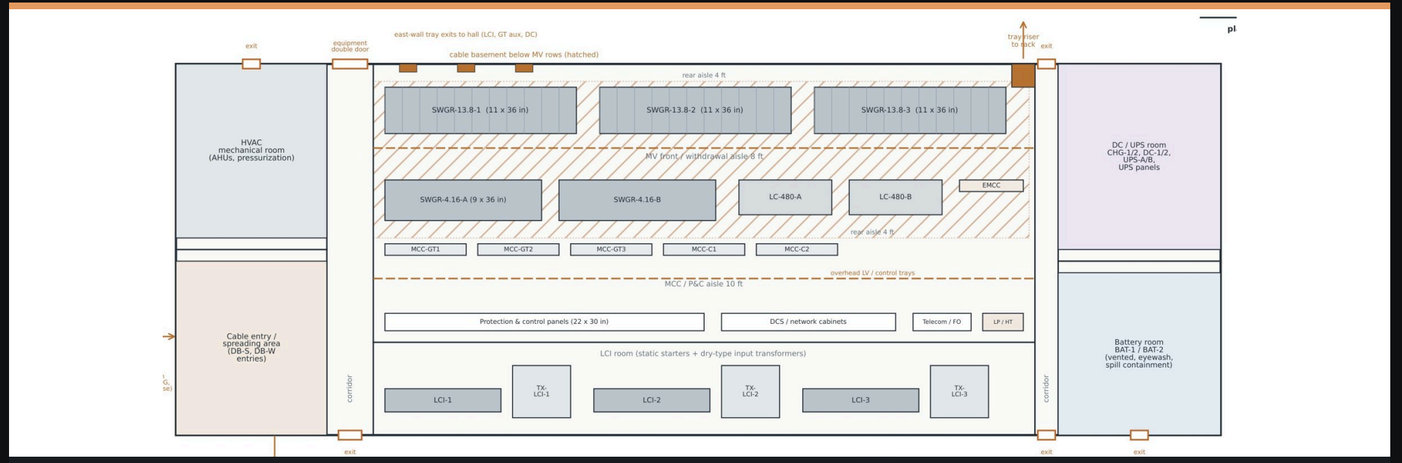
Tagged

everything beyond Rev 14 is marked as a design change and can be switched off: the modular expansion, the data-centre campus, Boiler Row.

REAL ENGINEERING DETAIL, SHEET BY SHEET



SK-3X1-04 Single-line: GT1's start-up path traced, 230 kV down to 13.8 kV



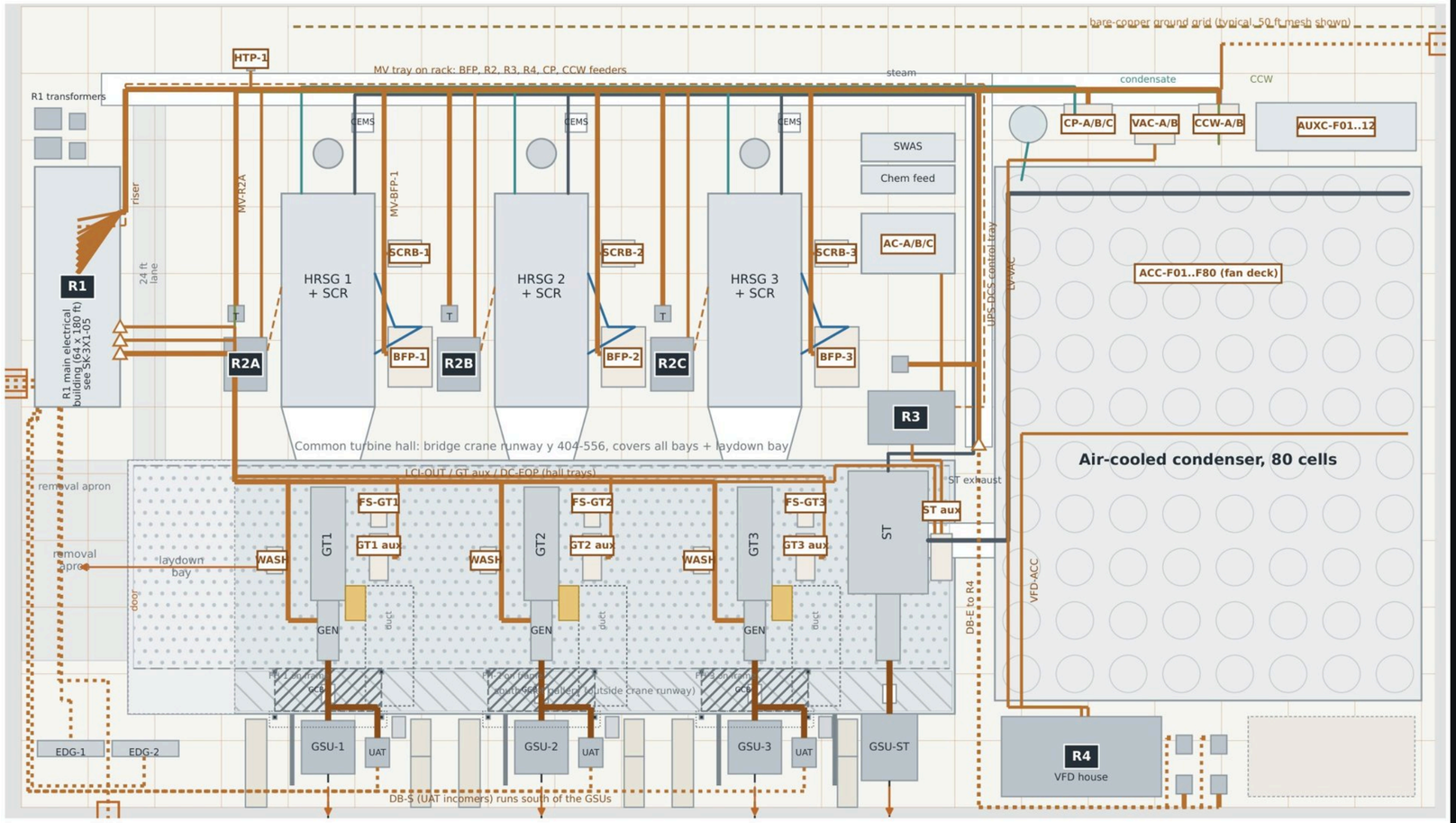
SK-3X1-05 R1 building: Rev 04 could not hold the switchgear; Rev 05 grew it to 64 x 180 ft

PRELIMINARY MOTOR / LOAD SCHEDULE (counts, voltages and starters are working assumptions until the load list)

ID	Driven equipment	Qty	Voltage	Starter / drive	Fed from	Route ID
BFP-1A/B, 2A/B, 3A/B	Boiler feed pumps, at each HRSG	6	13.8 kV	MV breaker, DOL	R1 SWGR-13.8-1/2/3	MV-BFP-1/2/3
CP-A/B/C	Condensate pumps, north of ACC	3	4.16 kV	MV contactor	R1 SWGR-4.16-A/B	MV-CP
CCW-A/B	Closed cooling water pumps	2	4.16 kV	MV contactor	R1 SWGR-4.16-A/B	MV-CCW
ACC-F01..F80	ACC fans, fan deck	80	480 V	VFD	R4	VFD-ACC
AUXC-F01..12	Aux dry cooler fans (CCW)	12	480 V	VFD	R4	LV-VAC tray
VAC-A/B	ACC air-removal vacuum pumps	2	480 V	MCC starter	R4	LV-VAC
AC-A/B/C	Air compressors	3	480 V	MCC soft starter	R3	LV-AIRC
SCRB-1..3 (A/B)	SCR dilution air blowers	6	480 V	MCC starter	R2A / R2B / R2C	LV-SCR-1/2/3
GT1-3 aux	GT lube oil, hydraulics, enclosure fans	sets	480 V	MCC-GT1/2/3	R1	LV-GTAUX-1/2/3
ST aux	ST lube oil, turning gear, EHC	set	480 V	MCC	R3	LV-ST
EOP-GT1..3, EOP-ST	DC emergency lube-oil pumps	4	125 V DC	DC starter	R1 DC-1 / DC-2	DC-EOP
LCI-1/2/3	GT static starters (start-up drive)	3	13.8 kV in	LCI cabinets	R1 SWGR-13.8-1/2/3	LCI-OUT-1/2/3
FS-GT1..3	GT CO2 fire suppression (detection, release, interlocks)	3	480 V + 24 V DC	local panel	R1 MCC-GT / DC	hall control tray
WASH-GT1..3	GT compressor water-wash skids	3	480 V	MCC starter	R1 MCC-GT1/2/3	LV-GTAUX
HTP-1..4	Heat-trace panels (rack, gas, NH3, tanks)	4	480 V	panel breakers	R1 LC / R-WT	LV-HTP-1..4
CPR-1/2	Cathodic protection rectifiers	2	AC in / DC out	rectifier	local panels	DC-CP-1/2

SK-3X1-03 Motor and load schedule: every motor, voltage, starter and source

THE POWER BLOCK, EVERY ROOM AND EVERY ROUTE



SK-3X1-03 Power block: rooms R1 to R4, every motor at its equipment, every MV and LV route named

ELECTRICAL FACILITIES

R1 **Main electrical building**
Conventional building, 64 x 180 ft
Supply: UAT-1/2/3 via duct bank DB-S; EDG-1/2 emergency. Holds 13.8 kV and 4.16 kV switchgear, 480 V load centers, GT and common MCCs, emergency MCC, static starters, UPS, station DC, protection, DCS and comms.

R2A-C **Unit / HRSG electrical rooms**
Prefabricated e-houses, 32 x 40 ft, outdoor 13.8/0.48 kV transformer each
Supply: SWGR-13.8-1/2/3. 480 V switchgear, MCC and DCS remote I/O for SCR blowers, ammonia injection, HRSG valves and drains.

R3 **ST auxiliary electrical room**
Prefabricated e-house, 40 x 65 ft, outdoor transformer
Supply: SWGR-13.8-2. 480 V switchgear and MCC for ST lube oil, turning gear, EHC and air compressors.

R4 **ACC electrical (VFD) building**
Prefabricated e-house, 60 x 120 ft, four outdoor transformers
Supply: SWGR-13.8-1/2/2/3. 480 V switchgear, 92 VFDs (80 ACC fans, 12 aux cooler fans), MCC for vacuum pumps.

R-WT **Water-treatment electrical room**
Room within the water-treatment building (SK-3X1-01)
Supply: SWGR-13.8-2 via DB-W. 480 V MCC for water treatment, wastewater and the fire-pump jockey pump.

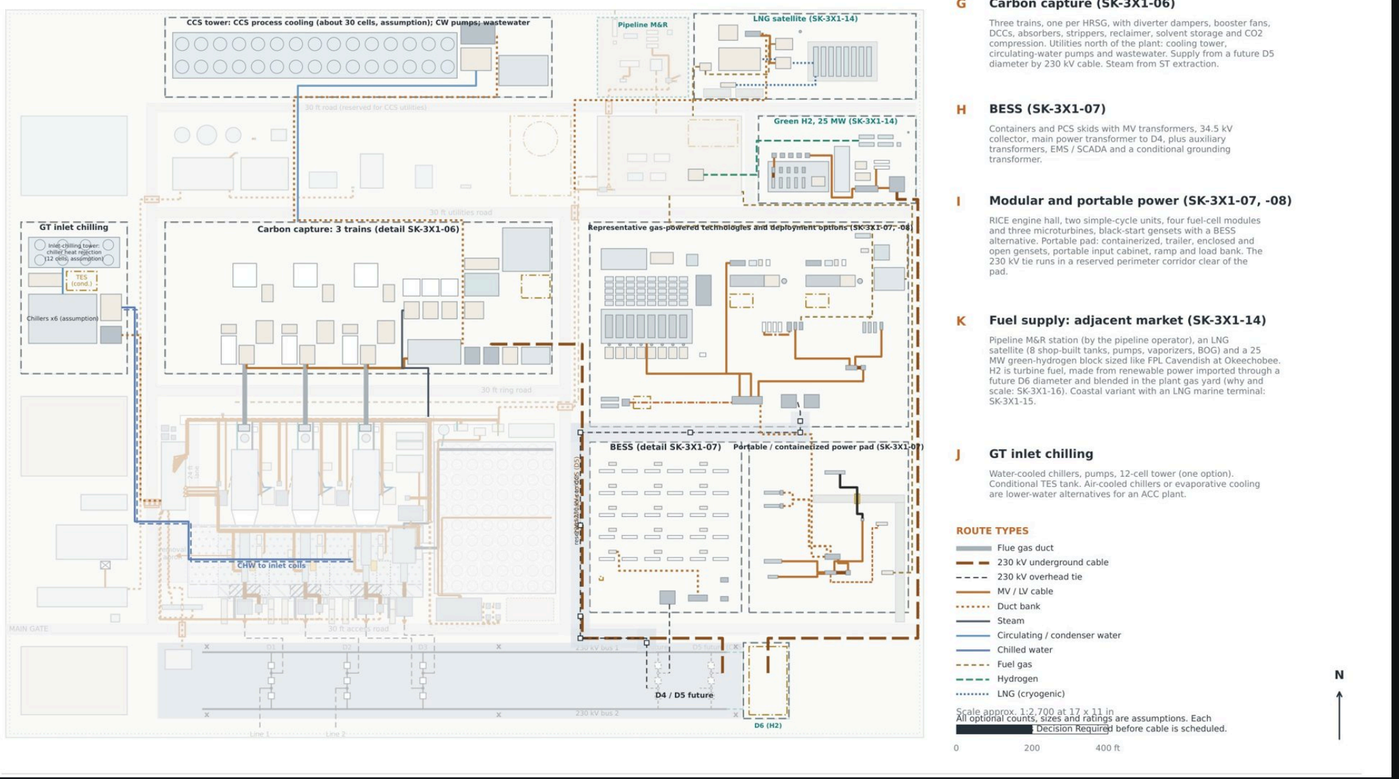
RH **Switchyard relay / control house**
Prefabricated control house, 52 x 70 ft (SK-3X1-01)
Protection, SCADA and communications for the 230 kV yard; 125 V DC and fiber to R1 via DB-S.

CR **Main operator control room**
Room in the control / administration building
Operator consoles only. DCS cabinets are distributed in R1, R2A-C, R3 and R4. Gas yard: local enclosure, not an e-house.

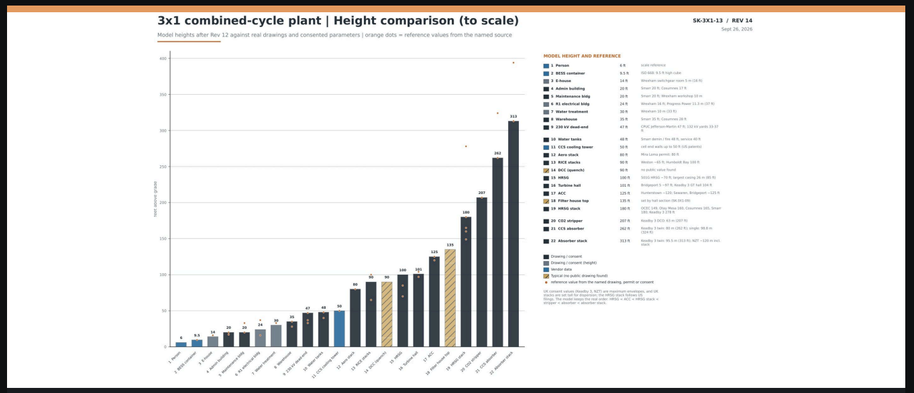
Motors sit at their driven equipment; starters and drives sit in the room named in the table.

Seven electrical rooms: what each one feeds

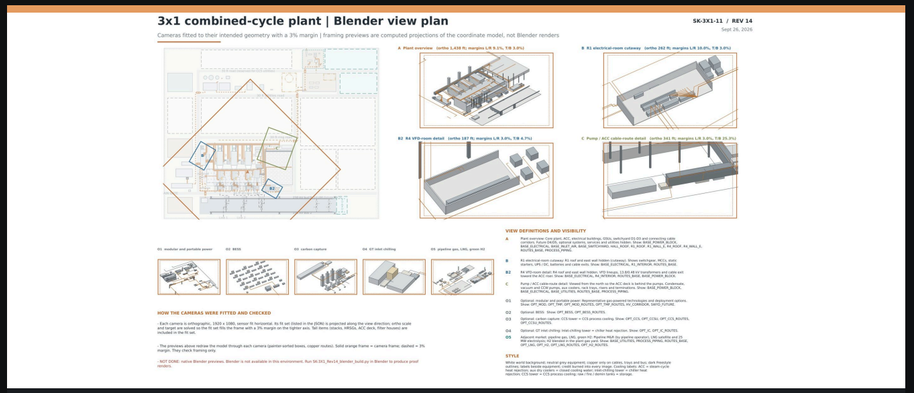
PLANNED BEFORE IT WAS MODELLED



SK-3X1-02 Optional systems: CCS, BESS, modular power, inlet chilling, each with its reserved space



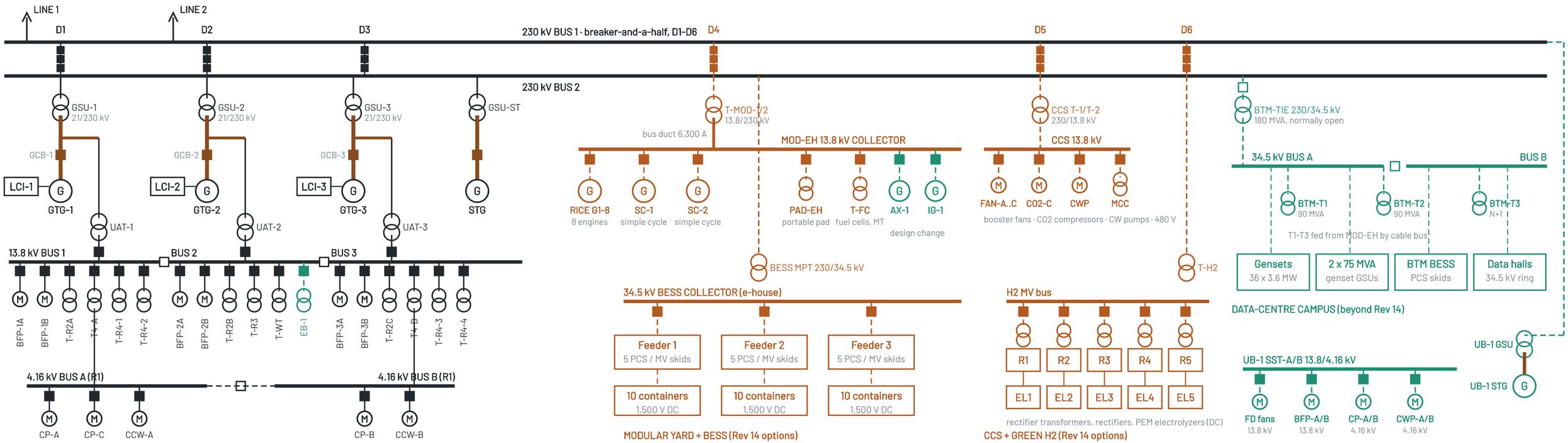
SK-3X1-13 Heights to scale, checked against real references



SK-3X1-11 3D view plan: every view defined on paper first

Checked on every build: the 21 heights on SK-3X1-13 and the 211 footprints on SK-3X1-01 must match the model.

THE FULL SINGLE-LINE, EVERY BUS AND EVERY LOAD



R2A 480 V SCR blowers, MOVs	R2B 480 V HRSG 2 aux	R2C 480 V HRSG 3 aux	R3 480 V ST aux, air comp.	R-WT 480 V water treatment

R4 4 x 480 V · 92 VFDs 80 ACC fans, 12 aux coolers	LC-480-A/B · MCC-GT1..3, MCC-C1/C2 GT aux, lube oil, hydraulics, enclosure fans

EDG-1 / EDG-2
480 V emergency, ATS

EMCC

125 V DC-1/2 · BAT-1/2 · UPS-A/B
breaker trip, DCS, control room, relays

POWER BLOCK (Rev 14 base)

SW-3X1 · MODEL SINGLE-LINE

as built in the 3D model, Oct 2026

- Rev 14 base plant
- Rev 14 options
- design changes beyond Rev 14
- generator isolated-phase bus

Conceptual. Typical ratings. Not for construction.

ITERATED, AUDITED, REWORKED

SINCE THE FIRST 3D MODEL (9 WORKING DAYS)

Rev 14
of the drawing set

98
saved iterations of the model

10
of them fixes, audits and rework

1.6 M
part pairs clash-checked on every build

FIVE LAYERS OF CHECKS, RUN ON EVERY CHANGE

DRAWINGS	211 footprints and 21 heights against the 16 sheets; crane lift clearances in the turbine hall
GEOMETRY	clashes, floating parts, open pipe ends, anything at grade in a road
ELECTRICAL	every motor, panel and transformer has a cable; no cable ends in mid-air or runs through equipment
CABLE SCHEDULE	every run follows a modelled route; sizes from ampacity and voltage drop; totals reconciled
REAL-PLANT LOGIC	gas safety shut-offs before the burners; utilities buy like utilities; US voltages

WHAT THE AUDITS CAUGHT, AND FIXED

- ✓ A gas line routed through two station transformers: rerouted
- ✓ The boiler gas riser upstream of its safety shut-offs: moved
- ✓ Equipment with no cable feeding it, cable ends in mid-air: wired and closed
- ✓ A cooling tower in a different style from its neighbours: rebuilt to match
- ✓ An MV cable tray through the boiler house: rerouted around it
- ✓ Stair flights that never reached the ground: re-seated
- ✓ A duplicate gas-turbine package: removed
- ✓ Cable totals that mixed campus and power block: re-scoped per slide

DESIGN → BUILD → AUDIT → REVIEW → REWORK, EVERY DAY

Every number in this training comes from a model that has to pass all five layers before it is saved.

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OEM ORDER BOOKS

TURBINE BACKLOGS ARE FULL

The three large-frame turbine OEMs together hold **about 220 GW** of gas turbines in backlog and slot reservations. Modular and packaged gas power is growing beside them.

MODULAR

+72%

power-generation sales at the largest engine-generator maker, Q2 2026

BEHIND THE METER

2.3 GW

modular gas fleet for Texas AI data centres, plus 1 GW for another campus

PACKAGED

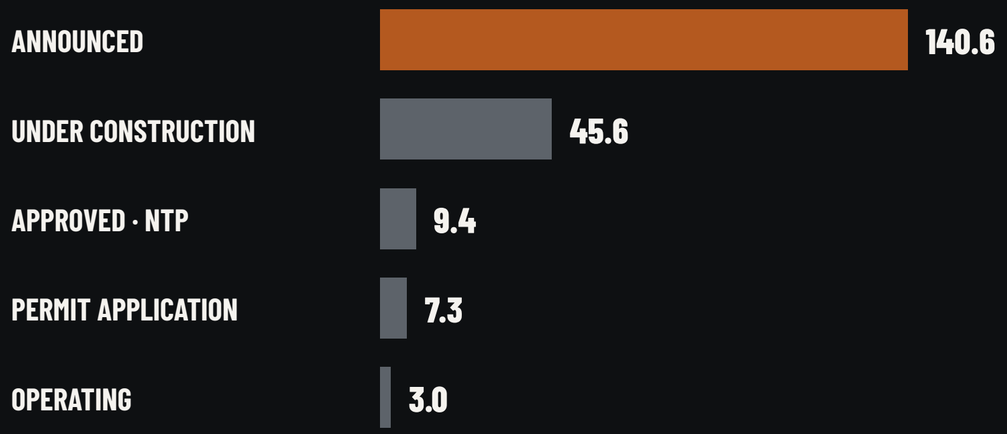
500 MW

standard modular gas-turbine plant, offered as a product

OEM releases, Q2–Q3 2026 (firm backlog plus slot reservations; gas-turbine backlog; large-frame backlog); Utility Dive, Aug 2026; POWER, Oct 2025 and Feb 2026.

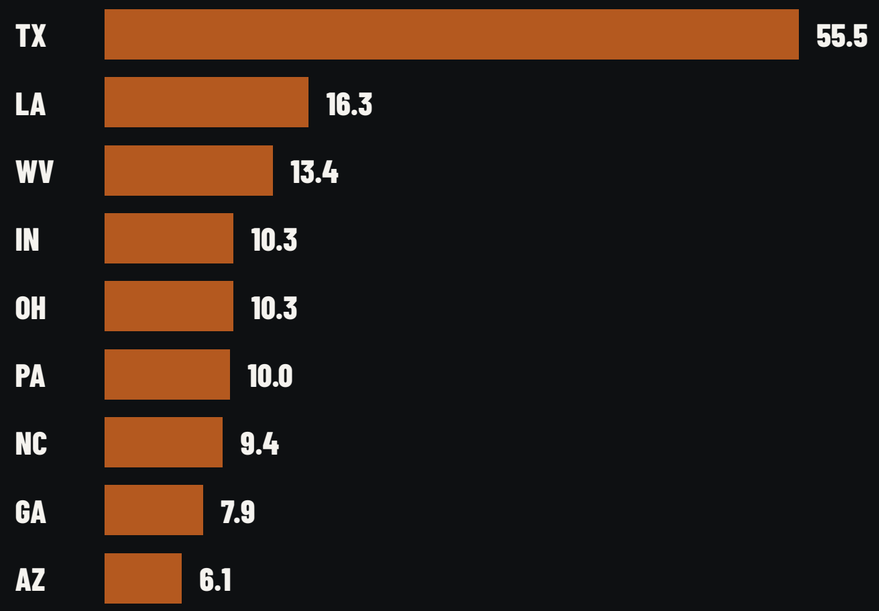
205.8 GW ELIGIBLE, MOST OF IT STILL EARLY

BY STATUS · GROSS GW, COD 2026-2035



68% **of eligible capacity is only announced.** Being on the register does not mean a cable package is open; it means a conversation to start.

TOP STATES · GROSS GW, 386 PROJECTS

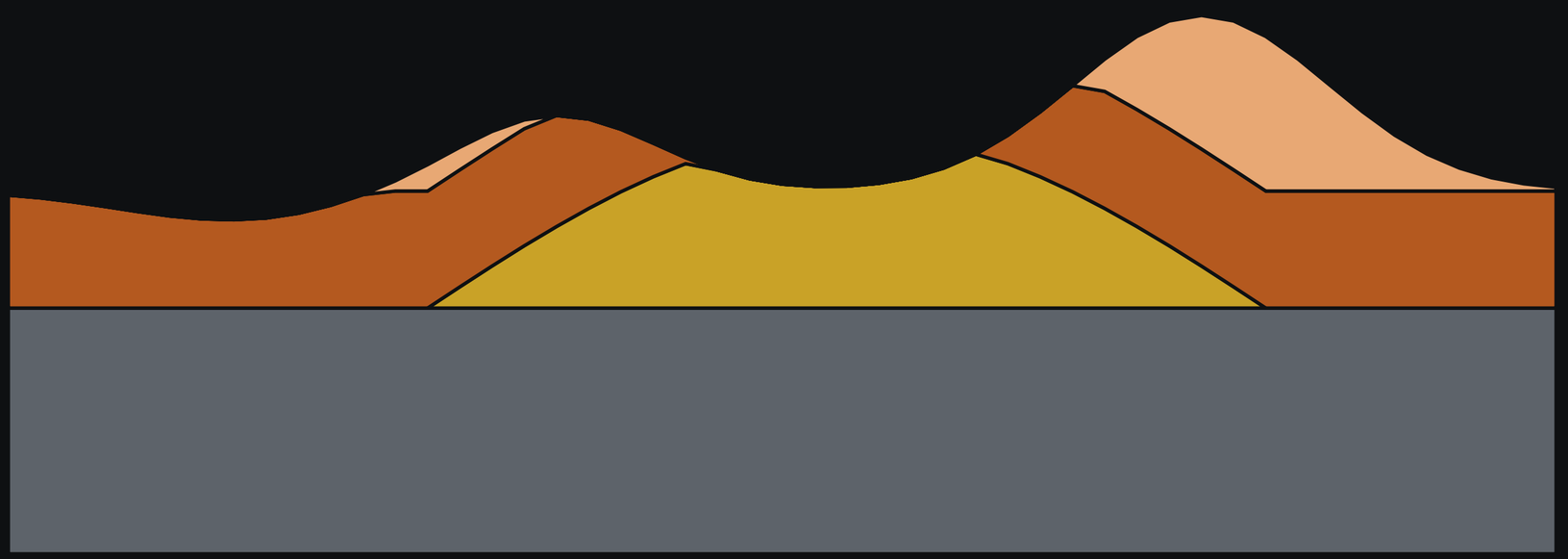


Greenfield 137 GW, brownfield 69 GW. Utilities reuse old sites (coal-site repowers, expansions); independent and behind-the-meter developers mostly build new, often on data-centre campuses.

Southwire PowerGen register, Sep 11, 2026: 386 records across 305 site groups; gross GW, unweighted.

A DAY ON THE GRID: WHO RUNS WHEN

Demand rises in the morning and peaks in the evening. Each kind of plant takes a layer.



- Peaking: simple cycle, engines, batteries
- Mid-merit: combined cycle
- Solar
- Baseload: nuclear, coal, combined cycle

The evening peak arrives as the sun sets: that is when gas and batteries earn their keep.

Illustrative summer weekday, not a real grid's data.

Every start and stop cycles the plant, and its cable: heat, vibration and terminations that move. Peakers may start every day.

FROM THE GENERATOR TO THE LIGHT SWITCH

Power is made at medium voltage, raised to high voltage to travel, then stepped down again near the customer.

1

GENERATOR

13.8–25 kV

21 kV in SW-3X1

CABLE

isolated-phase bus, MV-105

2

STEP-UP (GSU)

→ 115–765 kV

230 kV in SW-3X1

CABLE

HV XLPE cable, switchyard bus

3

TRANSMISSION

115–765 kV

long distance, overhead

CABLE

overhead conductor, OPGW

4

SUBSTATION

→ 4–35 kV

steps it down for the town

CABLE

substation cable, grounding

5

DISTRIBUTION

4–35 kV

feeders along the streets

CABLE

MV URD, overhead distribution

6

CUSTOMER

480 / 240 / 120 V

factory, office, home

CABLE

building wire, MC, service entrance

← INSIDE THE PLANT: TODAY

> **Higher voltage, lower current, smaller losses:** that is why transmission runs at hundreds of kV. Southwire has a product family at every step, from the generator to the outlet.

LEVELS, RATINGS AND THE CABLE THEY DRIVE

One HRSG train from the generator down to the field signals: each voltage level, the equipment and its rating, what it feeds, and the cable that follows.

LEVEL	EQUIPMENT · RATING	FEEDS · CURRENT	CABLE IT DRIVES
230 kV	**GSU-1, 21 / 230 kV step-up**	Plant to the grid, overhead line	**Overhead conductor (outside the HRSG scope)**
21 kV	**GT1 generator, isolated-phase bus; UAT-1 21 / 13.8 kV, 40 MVA**	UAT-1 secondary 1,674 A into SWGR-13.8-1	**4 sets of 3 x 1/C 750 kcmil MV-105**
13.8 kV	**SWGR-13.8-1: metal-clad, 1,200 A, 40 kA, arc-resistant**	Feed pumps BFP-1A / 1B, 5,000 kW each, direct-on-line, 232 A; T-R2A primary, 105 A	**3 x 1/C 250 kcmil MV-105 per pump; 3 x 1/C #2 to T-R2A**
480 V	**T-R2A 2,500 kVA → LC-R2A 4,000 A main, 65 kA → MCC-R2A-1 / -2, 800 A, 65 kA; VFDs**	Secondary 3,007 A; SCR blowers 2 x 110 kW (156 A); 45 HRSG MOVs; feed-pump and chemical-feed auxiliaries; welding receptacles	**10 sets of 4 x 1/C 500 kcmil; 3/0 TC-ER / MC-HL VFD; #10 TC-ER to each MOV**
120 V AC / 125 V DC	**UPS-R2A 10 kVA; 125 V DC battery, 60 cells; DC and UPS panels**	DCS, F&G node, breaker trip and close, burner-management power	**1/C #6 and control TC-ER 14 AWG**
Signals	**4-20 mA / HART, thermocouple millivolts, 24 V DC discretes**	Drum level, pressures, flame scanners, gas-path and tube-metal temperatures	**439 instrument + 84 thermocouple-extension cables, shielded pairs**

750 kW
connected at 480 V on one HRSG train, 78 circuits

2 x 5 MW
feed pumps: the largest motors on the train, at 13.8 kV

3,007 A
on the T-R2A secondary: 40 cables of 500 kcmil

76 %
of the train's cables are instrument or thermocouple pairs (67 % of the footage)

8.4 / 24.6
cal/cm^2 arc flash at 13.8 kV / 480 V (equipment labels)

Ratings are the model's typical values (Load List and nameplates), not engineered. Real plants vary: many put the feed pumps on a 4.16 or 6.9 kV bus.

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WHAT A POWER PLANT DOES TO A CABLE

STRESS	WHERE	THE ANSWER
HEAT	beside turbines, HRSGs and steam lines	high-temperature ratings, route away from hot surfaces
CYCLING	daily starts: heat, vibration, moving terminations	flexible strands, good terminations, support
ELECTRICAL NOISE	variable-frequency drives on fans and pumps	VFD-rated, shielded cable
HAZARDOUS AREAS	gas skids and fuel yards (Class I, Div 2)	ARMOR-X MC-HL, no conduit
FIRE AND SMOKE	halls, control rooms, cable spreading rooms	flame-retardant, low-smoke, circuit integrity
OUTDOORS	sun, oil, water, burial	sunlight- and oil-resistant jackets, direct-burial ratings
SCHEDULE	long pulls on the critical path	SIMpull reels, cut-to-length, pre-terminated

> **The wrong cable fails early, and a cable failure can trip a unit.** That is why the spec is strict, and why it is written for each area of the plant.

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SAME PHYSICS, DIFFERENT OWNERS

The owner decides how the project is structured, and how it gets bought. An independent power producer (IPP) builds each plant as its own company, sells the power on contract or into a market, and funds it with investors.

THE STRUCTURAL EQUATION

THERMAL IPP = **PROJECTCO** (one company per plant) + **ASSET** (conversion plant) + **CONTRACT** (power purchase agreement) + **CUSTOMER** (off-taker / load) + **CAPITAL** (investors, tax equity) + **MARKET** (ERCOT, PJM, MISO)

POWER OWNER ARCHETYPES

IOUS / CO-OPS — REGULATED

- Utility-led, rate-based ownership
- Own and operate generation
- Buy like utilities: standards, cycles

HYBRID IPPS — MIXED MODEL

- Utility-affiliated IPP subsidiaries
- Bridge regulated and competitive
- Two buying behaviours

PURE IPPS — COMPETITIVE

- Privately owned, investor-backed projects
- A direct fit with the project equation
- Speed and cost drive decisions

OUR SALES FOCUS FOR COMPETITIVE THERMAL POWER

GREENFIELD AND BROWNFIELD: FIVE KINDS OF SITE

GREENFIELD · NEW SITES 137 GW in the register

77.0 GW

GREENFIELD, GRID-CONNECTED

A new plant on open land, tied to the grid.

Everything is new: the full cable schedule, the switchyard and an HV gen-tie to the POI.

54.5 GW

DATA-CENTRE / BEHIND-THE-METER CAMPUS

Power built next to the load it serves.

Modular engines, turbines and batteries; MV collection to the campus; fast schedules, bought by package builders.

BROWNFIELD · EXISTING SITES 69 GW in the register

41.3 GW

COAL-SITE REPOWER

New gas units on a retired coal or boiler site.

Reuses the land, water and grid connection; new plant cable tied into an old switchyard.

21.4 GW

EXPANSION AT AN OPERATING PLANT

New units beside ones that keep running.

Work around live equipment; tie-ins wait for outages; shared switchyard and controls.

11.6 GW

INDUSTRIAL CAPTIVE AND SITE REUSE

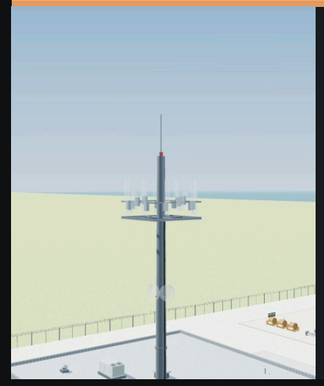
Power, often with steam, for a factory or refinery.

Hazardous areas (MC-HL), tight routes, the host plant's own standards.

And on plants already running: outages, HRSG module replacements, upgrades and conversions. There the outage clock is the cost, so cable is staged, pre-terminated and pulled fast. Utilities mostly reuse old sites; IPPs and behind-the-meter developers mostly build new.

Gross GW by site class, Southwire PowerGen register, Sep 11, 2026 (205.8 GW eligible, COD 2026–2035). Site classes provisional (Rev01).

A MODERN GAS PLANT IS MORE THAN THE POWER BLOCK

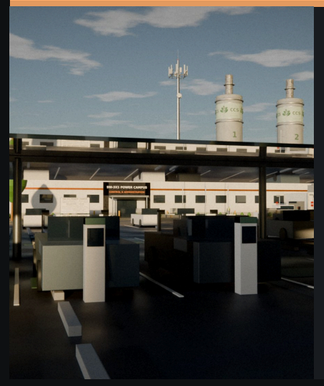


PRIVATE 5G / LTE

A monopole with massive-MIMO antennas: the plant's own wireless network.

CABLE

fibre backhaul, tower power, grounding



EV CHARGING + SOLAR CARPORT

Chargers at every bay, a DC fast charger, PV on the carport roof.

CABLE

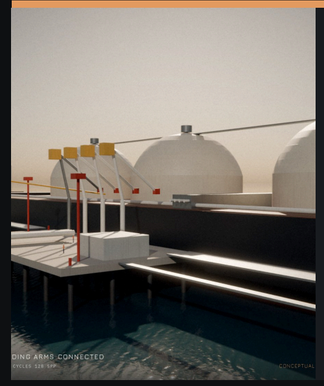
LV feeders, PV wire, a pad-mount transformer

BEHIND-THE-METER DATA CENTRE

A ~150 MW campus fed straight from the plant and the modular yard.

CABLE

34.5 kV collection, cable bus, UPS feeders

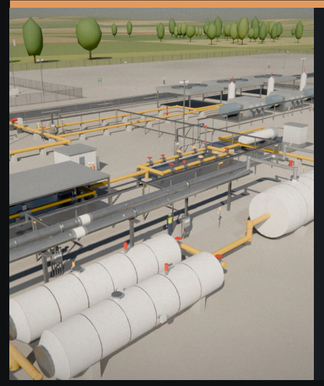


LNG BY SEA

A marine terminal: the carrier at the berth, unloading arms, the tank onshore.

CABLE

hazardous-area cable, long runs to the jetty, MV pumps

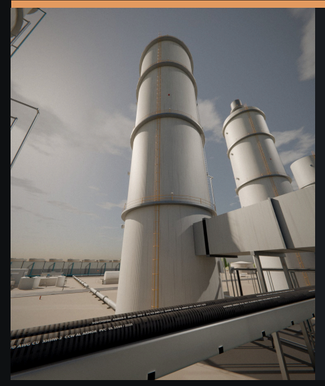


HYDROGEN

Tube-trailer bay and receiving station; an electrolyzer building beside it.

CABLE

Class I Div 2 runs, gas detection, DC cable



CARBON CAPTURE

Absorber, stripper and compressors on the stack exhaust.

CABLE

a second plant's motors, drives and instruments

All six are in the SW-3X1 model. Each one adds its own network of cable on top of the power block: the campus is the customer, not just the turbines.

THE CAMPUS · FROM THE SEA TO THE GRID

ONE SITE, SEVEN STOPS

1 GRID CONNECTION AND SWITCHYARD

2 TURBINE HALL

3 HRSG AND ITS E-HOUSE

4 COOLING AND WATER

5 MODULAR GENERATION

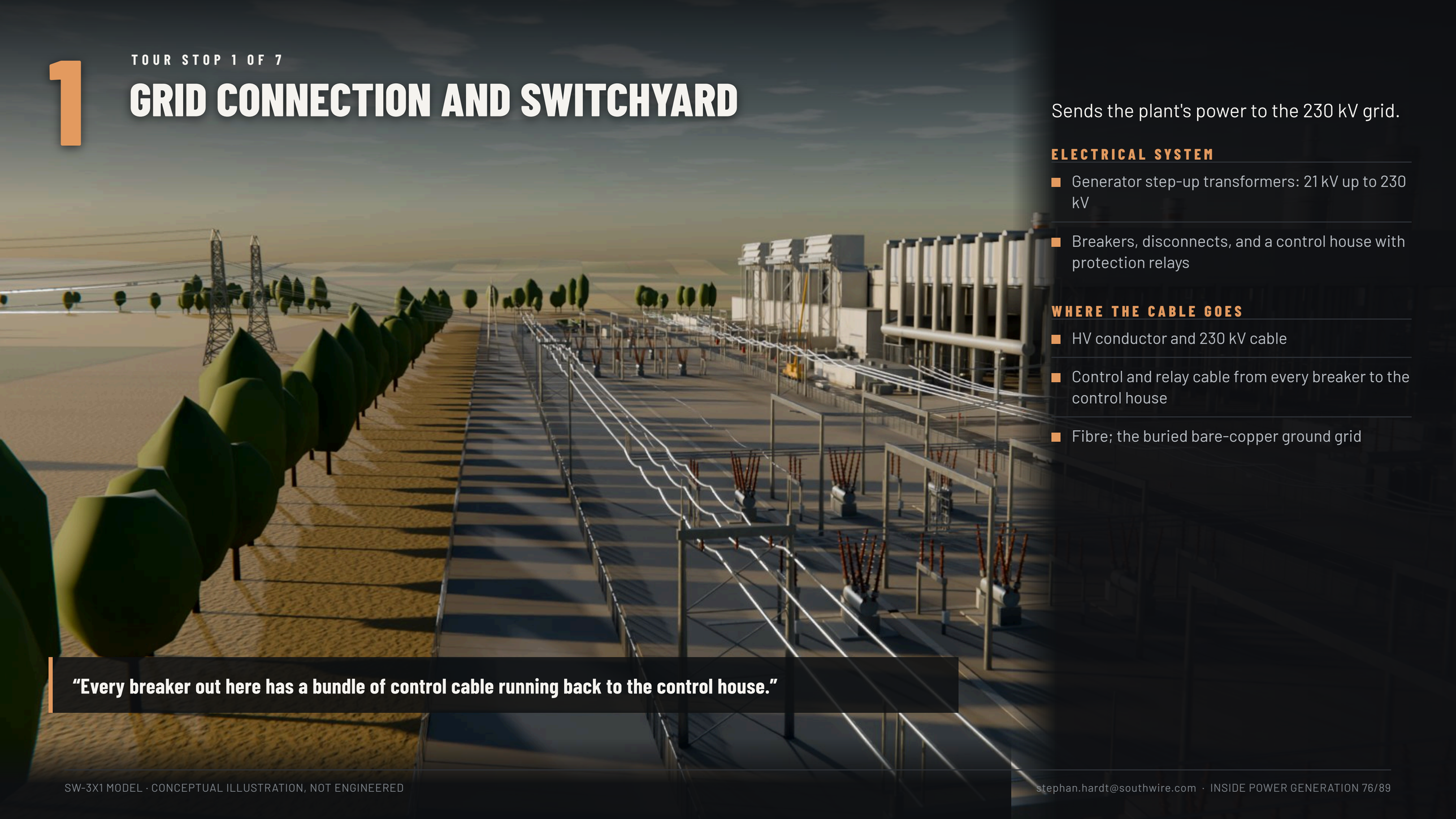
6 BEHIND THE METER: THE DATA CENTRE

7 FUEL: GAS YARD AND LNG

LNG arrives by ship at the marine terminal; the power leaves on 230 kV lines. Everything in between runs on cable.

TOUR STOP 1 OF 7

1 GRID CONNECTION AND SWITCHYARD



Sends the plant's power to the 230 kV grid.

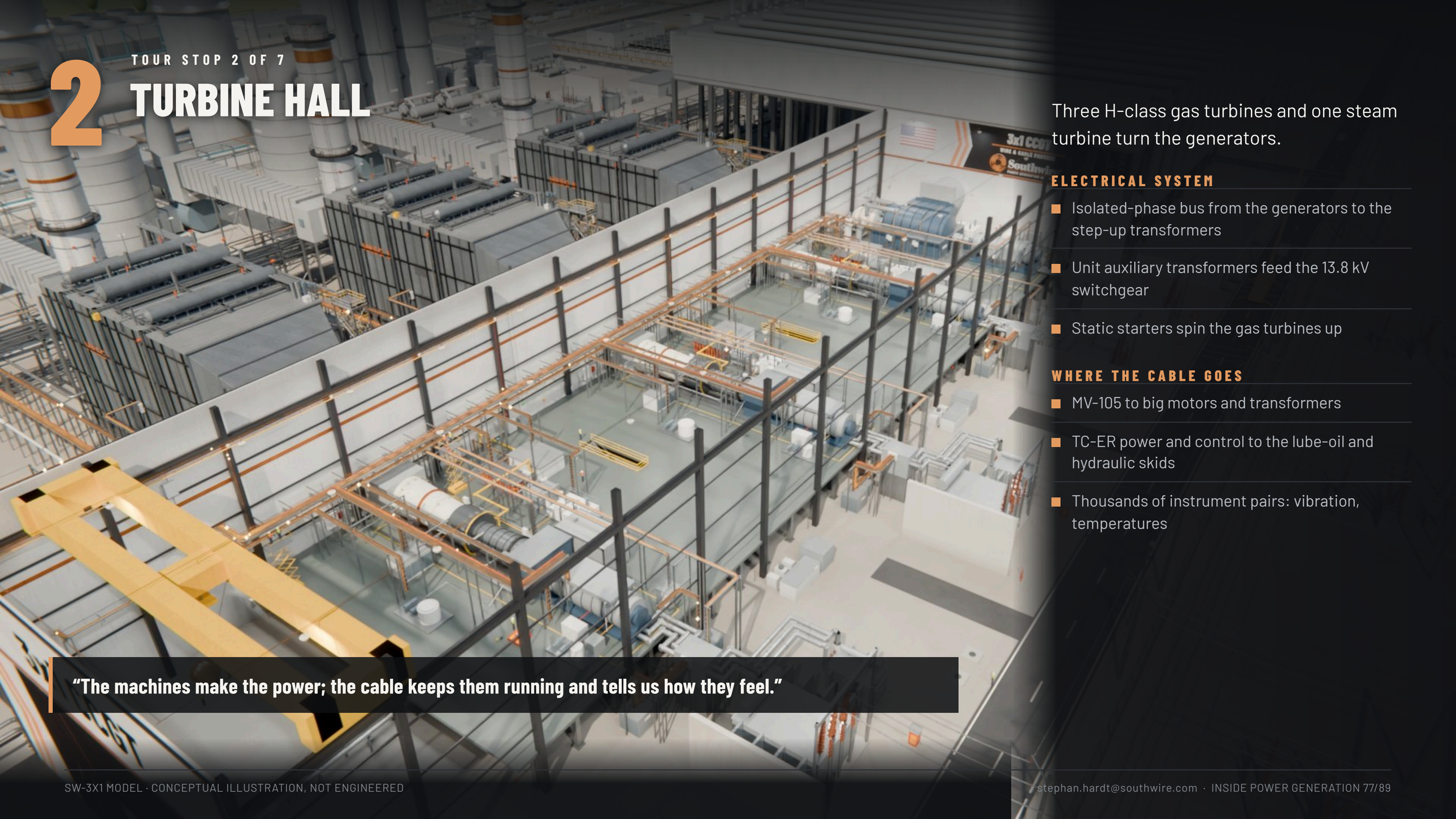
ELECTRICAL SYSTEM

- Generator step-up transformers: 21 kV up to 230 kV
- Breakers, disconnects, and a control house with protection relays

WHERE THE CABLE GOES

- HV conductor and 230 kV cable
- Control and relay cable from every breaker to the control house
- Fibre; the buried bare-copper ground grid

“Every breaker out here has a bundle of control cable running back to the control house.”



TOUR STOP 2 OF 7

2 TURBINE HALL

Three H-class gas turbines and one steam turbine turn the generators.

ELECTRICAL SYSTEM

- Isolated-phase bus from the generators to the step-up transformers
- Unit auxiliary transformers feed the 13.8 kV switchgear
- Static starters spin the gas turbines up

WHERE THE CABLE GOES

- MV-105 to big motors and transformers
- TC-ER power and control to the lube-oil and hydraulic skids
- Thousands of instrument pairs: vibration, temperatures

“The machines make the power; the cable keeps them running and tells us how they feel.”

TOUR STOP 3 OF 7

3 HRSG AND ITS E-HOUSE

Turns turbine exhaust heat into steam; the SCR inside cuts NOx.

ELECTRICAL SYSTEM

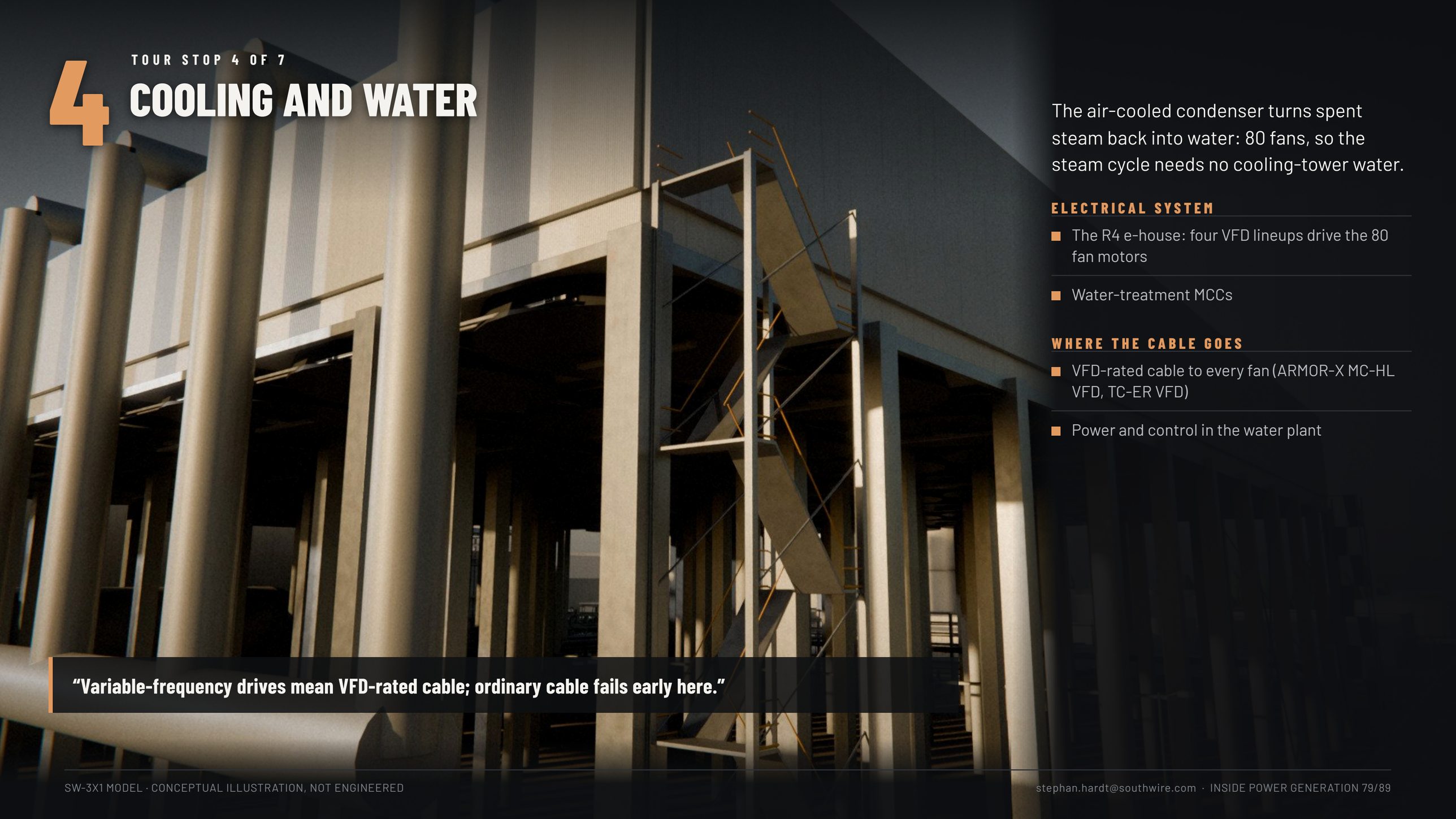
- Two 13.8 kV feed pumps
- A transformer feeds the e-house: 480 V switchgear, MCCs, drives, DCS
- CEMS shelter at the stack

WHERE THE CABLE GOES

- One HRSG train: 795 cables, about 173,000 ft
- About three quarters are instrument and thermocouple pairs
- ARMOR-X MC-HL at the duct-burner gas skid

> **“One boiler train is about 30 miles of cable, and most of it is small instrument cable.”**

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED



TOUR STOP 4 OF 7

4 COOLING AND WATER

The air-cooled condenser turns spent steam back into water: 80 fans, so the steam cycle needs no cooling-tower water.

ELECTRICAL SYSTEM

- The R4 e-house: four VFD lineups drive the 80 fan motors
- Water-treatment MCCs

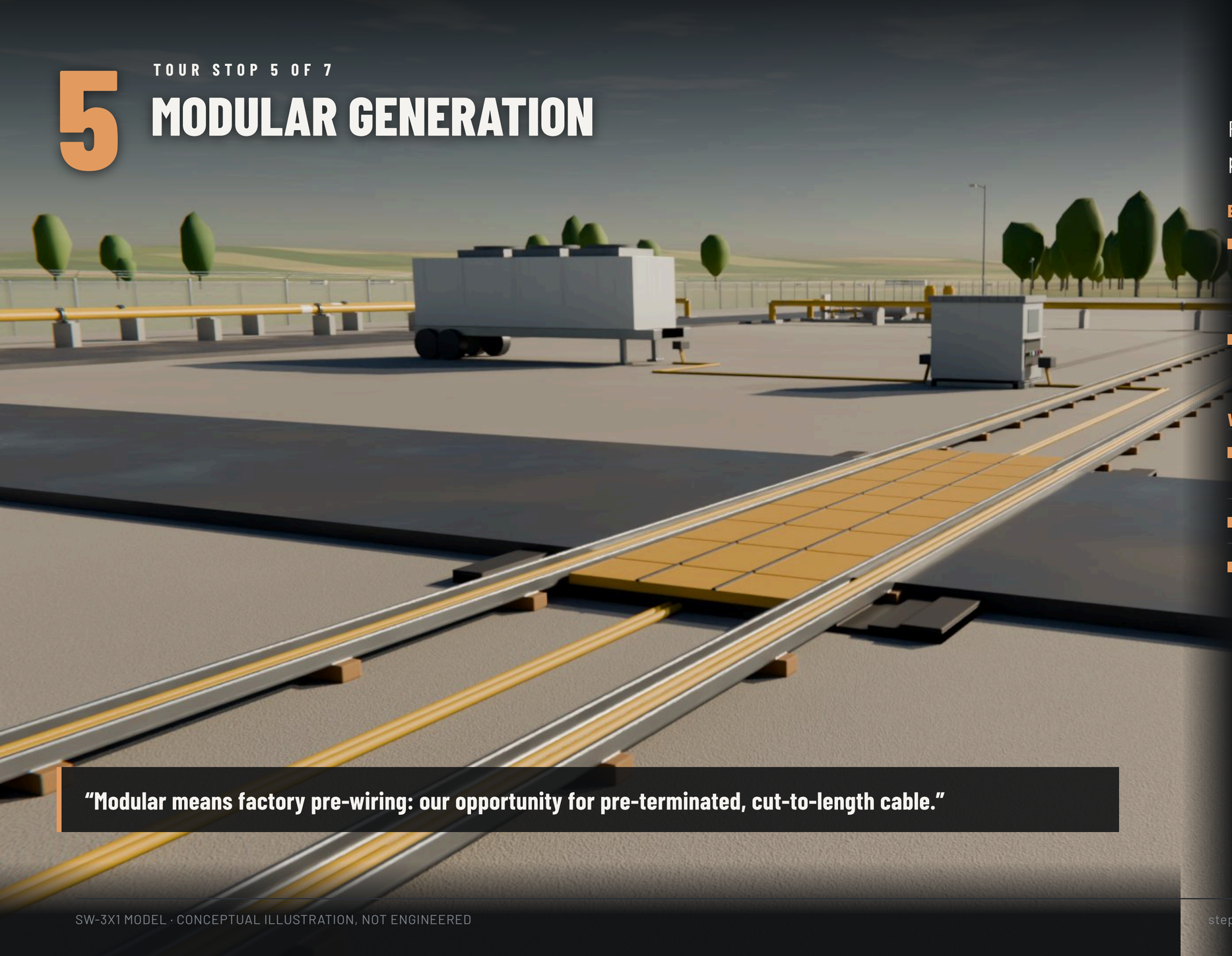
WHERE THE CABLE GOES

- VFD-rated cable to every fan (ARMOR-X MC-HL VFD, TC-ER VFD)
- Power and control in the water plant

“Variable-frequency drives mean VFD-rated cable; ordinary cable fails early here.”

TOUR STOP 5 OF 7

5 MODULAR GENERATION



Fast-start, flexible power beside the big plant.

ELECTRICAL SYSTEM

- Gas engines in a hall, aeroderivative and industrial gas-turbine packages, containers, trailers, fuel cells, microturbines
- Each unit on the 13.8 kV collector; inverters for the battery and fuel cells

WHERE THE CABLE GOES

- MV collection cable; DC cable at the battery and fuel cells
- Factory pre-wired skids and plug-in modules
- Pre-terminated and cut-to-length cable

“Modular means factory pre-wiring: our opportunity for pre-terminated, cut-to-length cable.”

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED

TOUR STOP 6 OF 7

6 BEHIND THE METER: THE DATA CENTRE

A data-centre campus powered directly by the plant's modular units, with the grid as backup.

ELECTRICAL SYSTEM

- 13.8 to 34.5 kV step-up transformers and double-bus switchgear
- A 40 MW battery absorbs load steps
- Two data halls fed 2N through pad-mount transformers and UPS

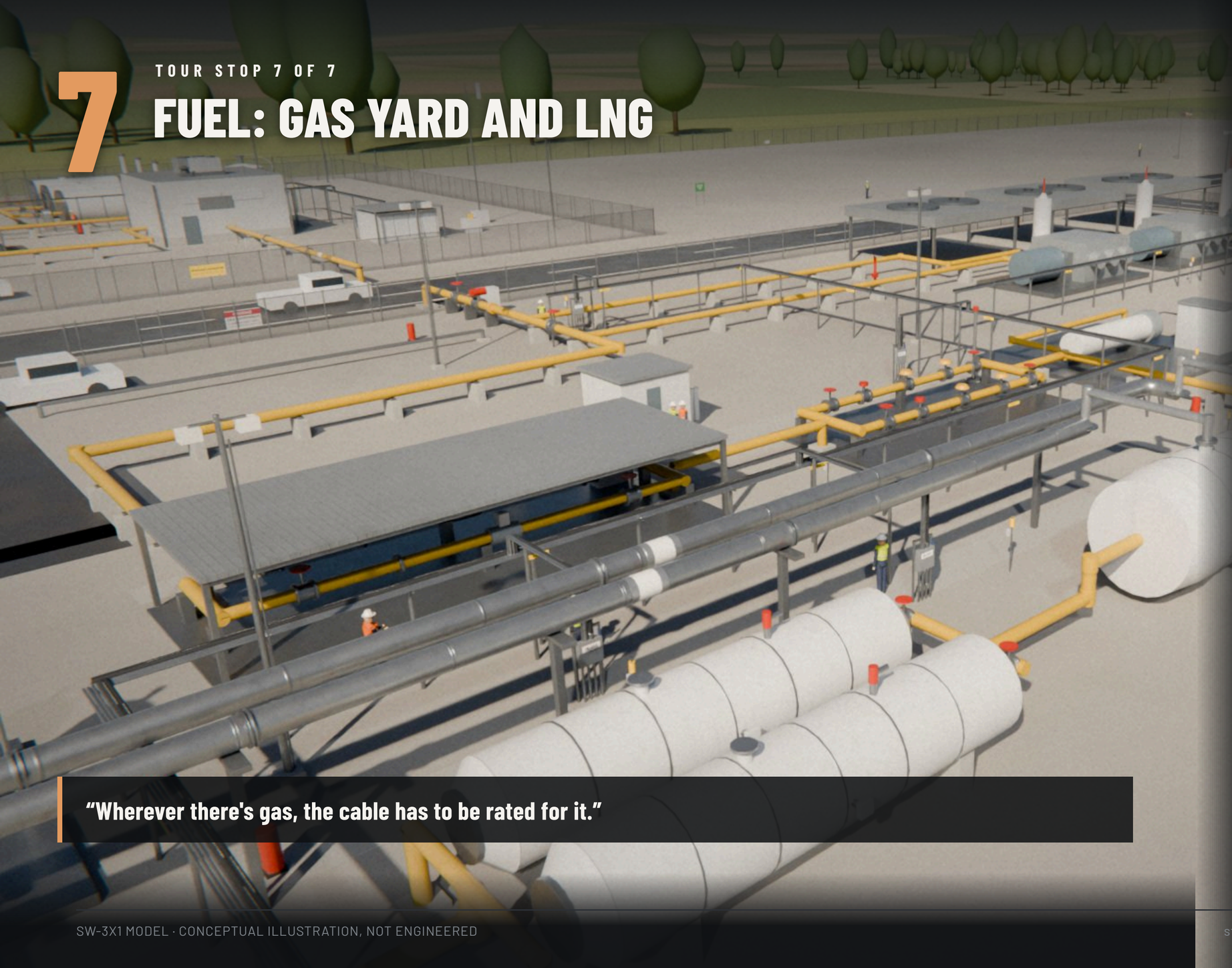
WHERE THE CABLE GOES

- 35 kV MV cable
- Large LV feeders to the UPS skids
- Fibre everywhere

“Data centres are why gas is growing again, and they are cable-dense on both the A and the B side.”

TOUR STOP 7 OF 7

7 FUEL: GAS YARD AND LNG



Brings the fuel in and conditions it; an LNG satellite holds about three hours of backup fuel.

ELECTRICAL SYSTEM

- Metering and regulating station; the LNG e-house (13.8 kV / 480 V)
- Pumps, vaporizers, boil-off compressor

WHERE THE CABLE GOES

- ARMOR-X MC-HL for Class I Div 1 and 2
- TC-ER control, PLTC instrumentation and fire alarm on one strut rack

> **“Wherever there's gas, the cable has to be rated for it.”**

TRACED TO REAL PLANTS, SYSTEM BY SYSTEM

SYSTEM	WHAT REAL PLANTS SHOW	SEEN IN	IN SW-3X1
GRID CONNECTION	230 kV breaker-and-a-half switchyards; one step-up transformer per generator	State siting filings for combined-cycle plants; a 360 MW combined-cycle feasibility study	**SK-04 single-line: bays D1-D6, lines 1 and 2**
GENERATOR LEADS	Generator breaker, isolated-phase bus, a unit auxiliary transformer tapped off the leads	Power-plant distribution course notes; IEEE section one-line examples	**SK-03 / SK-04: GCBs, IPB, UAT-1..3**
PLANT MV BUSES	Auxiliary MV buses fed by unit and station transformers, tied but normally open	A 2025 state utility-commission combined-cycle filing: overall one-line (6.9 kV there, 13.8 kV here)	**13.8 kV buses 1-3, N.O. ties (slide 13)**
480 V AND MCCS	480 V switchgear through 2,500-3,300 kVA transformers to HRSG, turbine and balance-of-plant MCCs	The same filing	**Rooms R2A-C and R3, load centres, MCCs**
BIG MOTORS	Feedwater pumps ~3.5 MW, cooling-water pumps ~2.6 MW, cooling fans ~1.3 MW on a ~640 MW plant	US DOE / NETL combined-cycle baseline studies	**SK-03 motor and load schedule**
HRSG AND BURNERS	Burner safety shut-offs and flame detection; Class I Div 2 fuel skids	A published HRSG purchase specification; city building bulletins; NFPA 85	**Boiler slides; gas valve trains ahead of every burner**
CABLE	MV-105 EPR at 133 %; TC-ER and PLTC; tray flame tests	Owner Division 26 cable specs; ICEA, IEEE 1202, UL 1277 / 1569	**Cable schedule: every run sized and routed**

700+ pages reviewed; these are public examples of the same systems. Every row can be checked: the documents and links are in the reference register. Nothing here is engineered for SW-3X1.

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THE ANSWERS AND WHY

1 WHICH CABLE TYPE HAS THE MOST MILES IN THE POWER BLOCK?

A MV / HV POWER | B LV POWER | **C INSTRUMENTATION** | D FIBRE

Instrumentation: about 200 of the 445 miles. The big cables get the attention; the small ones are most of the miles.

2 WHO USUALLY PLACES THE CABLE ORDER?

A THE OWNER | B THE OWNER'S ENGINEER | **C THE EPC OR PACKAGE BUILDER** | D THE UTILITY

The EPC or package builder buys it; the owner's engineer specifies it. Be in the spec early, be ready when the EPC buys.

3 IN A COMBINED CYCLE, THE STEAM TURBINE ADDS ABOUT HOW MUCH MORE POWER, WITH NO EXTRA FUEL?

A 5% | **B ONE-THIRD** | C DOUBLE | D NONE

About one-third: the HRSGs turn the gas turbines' hot exhaust into steam. Burn the gas once, use the heat twice.

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FOUR WAYS TO BURN GAS FOR POWER

	SIMPLE CYCLE THE PEAKER	COMBINED CYCLE THE WORKHORSE	ENGINES (RICE) THE SPRINTER	AERODERIVATIVE THE JET
UNIT SIZE	**50–400 MW per gas turbine**	**300–1,500+ MW per block**	**2–20 MW per engine, in banks**	**30–60 MW per unit**
EFFICIENCY	**~35–40%**	**~55–64%**	**~40–50%**	**~38–42%**
START TO FULL LOAD	**10–30 min**	**about 1–3 hours**	**about 5 min**	**5–10 min**
RUNS AS	Peaking, mid-merit	Baseload, mid-merit	Peaking, data-centre firming	Peaking, fast backup
IN THE MODEL	A gas turbine alone: exhaust goes straight up the stack	Gas turbines + HRSGs + steam turbine: the SW-3X1 power block	Big gas engines in a hall: the modular yard	A jet engine on the ground: the modular yard

Units that stick: 1 GW = 1,000 MW = 1,000,000 kW. **MW** is power (how hard it runs right now); **MWh** is energy (power × hours). Efficiency is how much of the fuel's energy leaves as electricity; the rest goes up the stack as heat.

Typical ranges for new US units; efficiency on a lower-heating-value basis. Starts vary with design and how hot the unit is.

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BACKUP · FUNDAMENTALS · THE BASICS

HOW AN HRSG WORKS

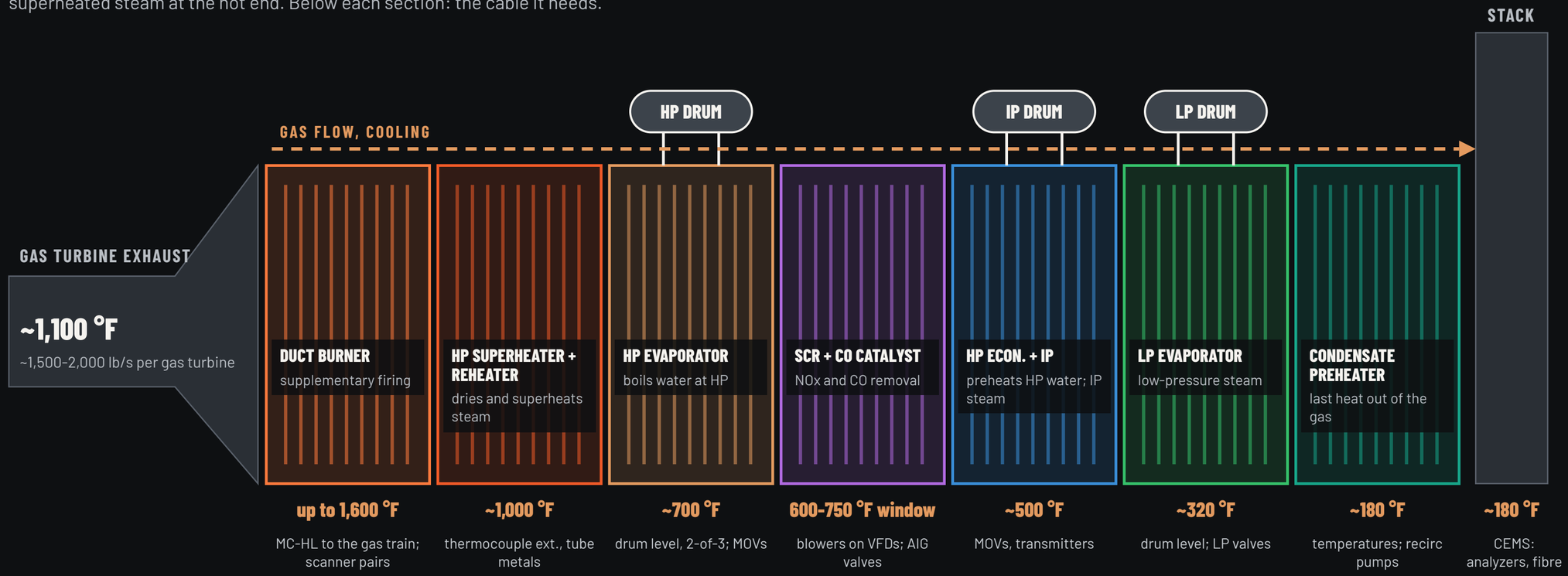
The exhaust crosses rows of finned tubes, hottest sections first; water enters at the cold end and leaves as superheated steam at the hot end. Below each section: the cable it needs.

WHAT IT IS

- **Without it:** ~60 % of the fuel's energy goes up the stack as ~1,100 °F exhaust (~40 % efficient).
- **With it:** a boiler heated by that exhaust; its steam drives the steam turbine.
- **The payoff:** ~1/3 more power from the same fuel (~60 %+ efficient).

FUN FACTS

- Fuel, CO2 and cost per MWh drop by about a third.
- The NOx catalyst (SCR) lives **inside** it, where the gas is just the right temperature.
- Duct burners can fire extra gas for peak demand.
- No HRSG: a gas-turbine plant. With one: combined cycle.



Steam to the steam turbine, typical: HP ~1,800-2,400 psig, ~1,050 °F · reheat / IP ~400-600 psig, ~1,050 °F · LP ~50-80 psig

~1/3
of a combined-cycle plant's output comes from HRSG steam, with no extra fuel (duct firing aside)

~60 %+
combined-cycle efficiency, against ~40 % for the gas turbine alone

3 x 3
three pressures, three drums; each drum level measured three ways, typically voted 2-out-of-3

~1 hour
a hot start: about an hour: every start cycles the tubes, which is why tube-metal thermocouples matter

Typical three-pressure reheat HRSG with SCR behind an H-class gas turbine; section order, temperatures and pressures vary by design.

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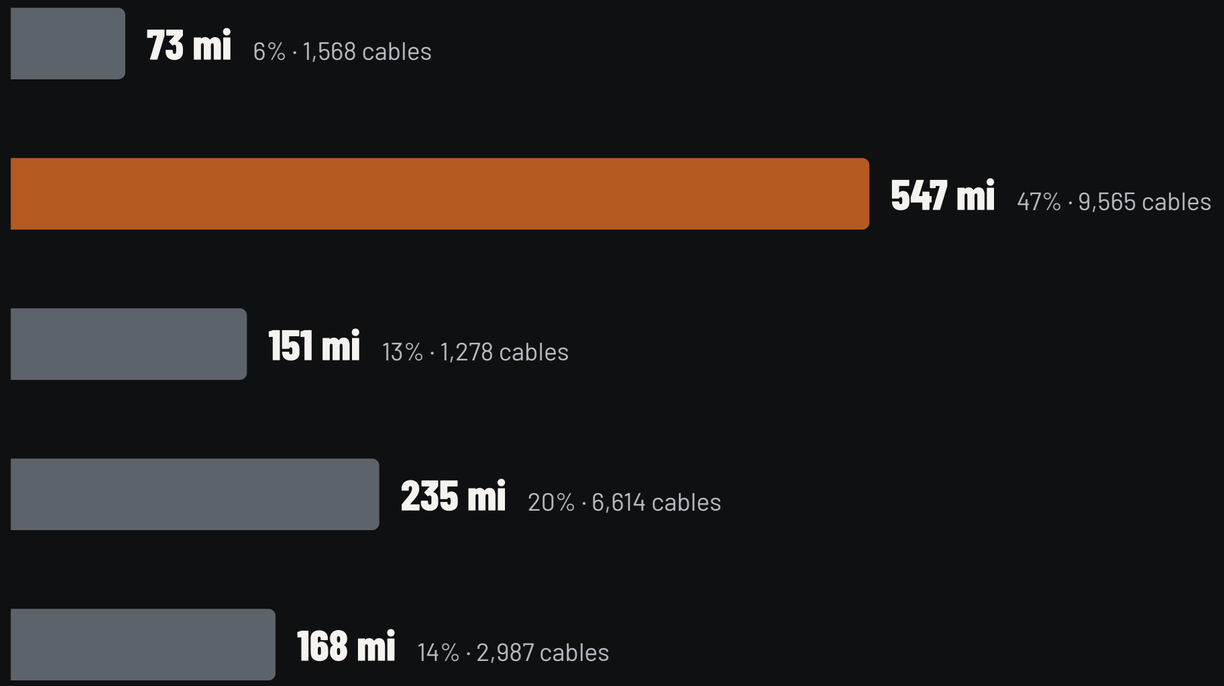
FOLLOW THE POWER: THE CABLE IS EVERYWHERE

THE WHOLE SW-3X1 CAMPUS BY PLANT AREA, FROM THE MODEL'S CABLE SCHEDULE (1,174 MILES)



THE POI

Point of interconnection: where the plant meets the grid. The plant side is the owner's and EPC's scope; past it, the utility or transmission owner builds and buys.

The gen-tie (often HV underground) runs from the switchyard to the POI, and the scope line there is often blurred: a place we help.

Most of the cable is inside the box: thousands of instrument, control and motor circuits in the turbine hall and HRSGs. The grid connection carries the biggest cables; the power block carries the most.

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WHAT THE SPEC CALLS FOR

Every cable application on the HRSG, the requirements an owner's or EPC's cable specification writes for it, and the Southwire product built to meet them.

APPLICATION · WHERE	WHAT THE SPECIFICATION CALLS FOR	SOUTHWIRE
MV POWER, 15 KV FEED-PUMP MOTORS; THE T-R2A PRIMARY	1/C Cu, EPR insulation, **133 % level**, 90 °C, tape shield, PVC jacket. ICEA S-93-639, UL 1072 (MV-105). Terminations IEEE 48; acceptance test VLF (IEEE 400.2) / NETA ATS	**MV-105 15 kV, 220 mil NL-EPR 133 %, SIMpull · SPEC 46302**
480 V FEEDERS AND POWER T-R2A SECONDARY, LC-R2A TO MCCS, MOTORS AND MOVS	Cu XHHW-2, 90 °C wet / dry, sized at **125 %** of full load; voltage drop ≤ 3 %. ICEA S-95-658, UL 44 / UL 1277; 10+ parallel sets: compare with bus duct	**XHHW-2 1/C (SIMpull) · Type TC-ER 3/C + ground · SPEC 45252**
TRAY CABLE: POWER AND CONTROL EVERY TRAY BETWEEN THE E-HOUSE AND THE HRSG	Type TC-ER (UL 1277) so it may leave the tray as an exposed run to the equipment (NEC 336.10); sunlight resistant, direct burial; **IEEE 1202 / UL 1685** 70,000 Btu/h vertical-tray flame test	**Control TC-ER 14 AWG multiconductor · SPEC 45053**
HAZARDOUS AREA: GAS TRAIN, DUCT BURNER DUCT-BURNER FUEL SKID, BMS PANEL TO THE VALVES	Class I Div 2 wiring methods per NEC 501.10(B); MC-HL (UL 1569 + UL 2225) also qualifies for Div 1. Burner management per **NFPA 85**, duct burner UL 295, listed fittings	**ARMOR-X MC-HL, welded armor · SPEC 45352**
VARIABLE-FREQUENCY DRIVES SCR DILUTION-AIR BLOWERS, FD FANS	VFD-rated: symmetrical three-ground construction, overall shield or continuous armor for the common-mode current, XLPE insulation for reflected-wave peaks	**ARMOR-X MC-HL VFD · TC-ER VFD · SPEC 45352**
INSTRUMENTATION, 4-20 MA / HART DRUM LEVEL, PRESSURES, FLAME SCANNERS, ANALYZERS	**Individually shielded** twisted pairs + overall shield, drain wires; 300 V PLTC (UL 13, NEC 725) or ITC; ICEA S-73-532; segregated from power in tray	**Instrumentation PLTC, SPOS · SPEC 43303**
THERMOCOUPLE EXTENSION GAS-PATH, TUBE-METAL AND DUCT TEMPERATURES	Alloy matched to the thermocouple (type K = KX), ANSI MC96.1 / ASTM E230 colour code; **never copper**; shielded pairs; PLTC; 105 °C near the casing	**Thermocouple extension, PLTC multi-pair · SPEC 42345 / 42350**
FIRE ALARM, FIBRE, BUILDING WIRE F&G LOOPS, CEMS DATA, E-HOUSE LIGHTING	FPLR / FPLP (NEC 760, NFPA 72); single-mode fibre, armored in tray; THHN / THWN-2 in conduit (UL 83)	**SIMpull THHN / THWN-2 · fire-alarm and fibre cable**

CHECKED AGAINST A REAL PLANT

A recent US utility combined-cycle filing shows the same structure: **HRSG MCCs A / B** fed from 480 V switchgear through 2,500 / 3,300 kVA transformers off the auxiliary bus (6.9 kV there, 13.8 kV here).

DESIGN AND INSTALL

Raceway and cable design for generating stations: IEEE 422. Grounding: IEEE 80 / 665. Area classification: NEC 500, API RP 500. Field acceptance: NETA ATS.

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APPENDIX

SOURCES AND QUALIFIERS

ELECTRICITY MIX 2025; PLANNED 2026 ADDITIONS	EIA, Electricity explained (2025 utility-scale generation); EIA Today in Energy, Feb 2026
LOAD GROWTH, DATA CENTRES	Grid Strategies, National Load Growth Report, 2025 (forecast)
378 GW PIPELINE, 52 GW BUILDING	Global Energy Monitor, Aug 2026; early-stage proposals will not all be built
TURBINE LEAD TIMES	S&P Global Commodity Insights, May 2025
OEM BACKLOGS AND MODULAR ORDERS	OEM releases, Q2-Q3 2026; Utility Dive, Aug 2026; POWER, Oct 2025 and Feb 2026
CAPACITY ADDED PER YEAR, 1996-2026	EIA and Southwire provisional series, to be verified
CABLE OPPORTUNITY, REGISTER, STATES, SITE CLASSES	Southwire PowerGen planning model and register, Sep 11, 2026: assumed rates and probabilities
PLANT TYPES, GRID VOLTAGES, UNITS	Typical industry ranges; vary by design
CABLE COUNTS AND MILES	SW-3X1 model cable schedule: conceptual, not engineered
PLANT AND YARD RENDERS	SW-3X1 plant model: conceptual and illustrative, not engineering data

VERIFY ALL FIGURES AGAINST PRIMARY SOURCES BEFORE EXTERNAL USE

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